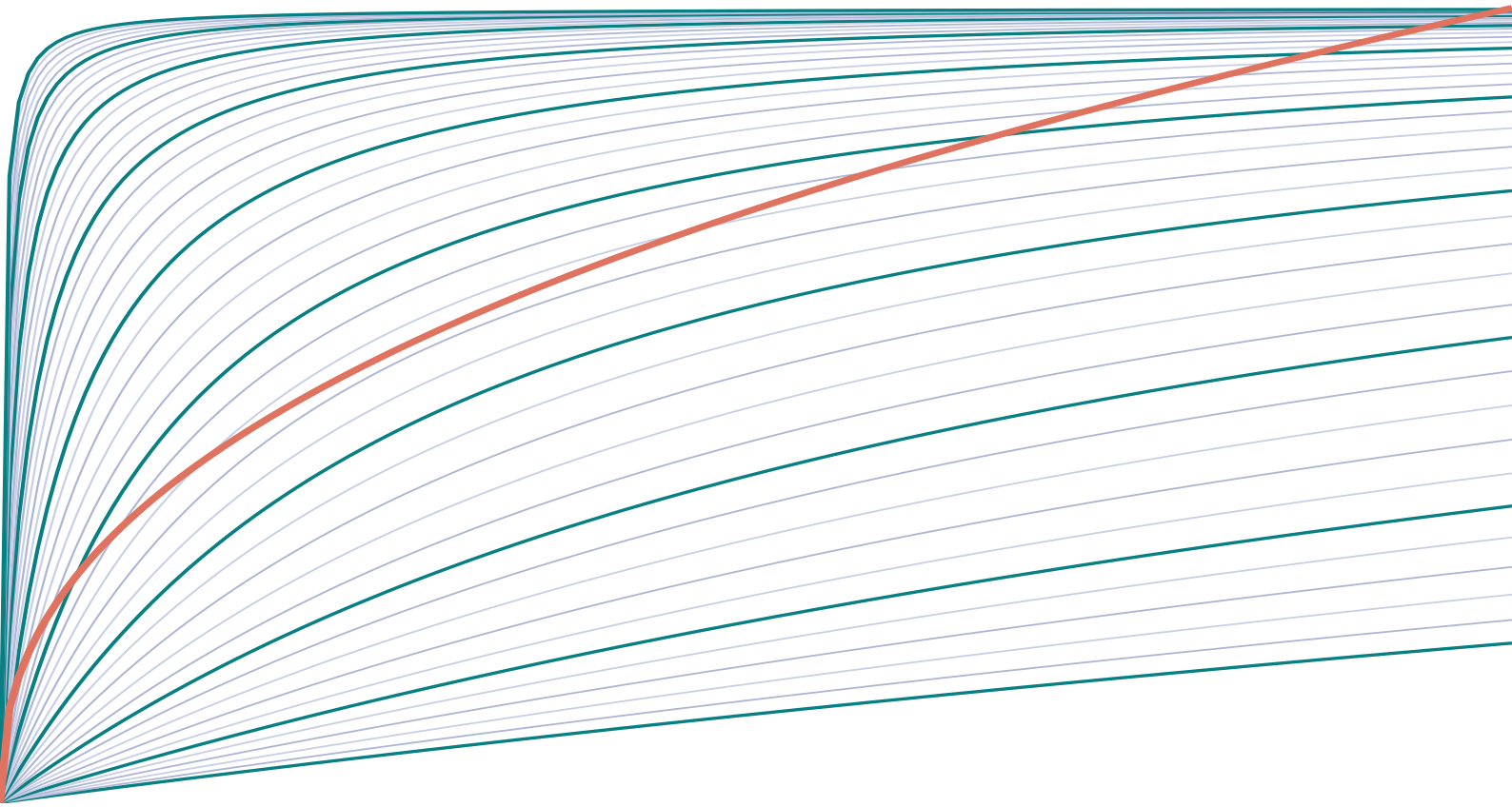
# Isotherm++

Adsorption isotherm models for mass transport simulation: equations, exact derivatives, sensitivities and original sources.



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# Isotherm++ 2.0



SCIENTIFIC SOFTWARE | AUSTRALIAN ENGLISH

**Adsorption isotherm models, mathematical reference and C++20 integration.**

The scientific documentation and C++ API reference are maintained from a common reStructuredText source tree for the HTML and PDF manuals.

Each model provides the adsorbed amount, its exact derivative with respect to the independent variable, validated parameter domains and read and replace access to its parameters. All models share one interface based on C++20 concepts and traits, without inheritance or virtual functions, so the isotherm used by a transport solver can be changed without modifying the solver.

Each model page gives the original publication, the equation implemented, its derivative, the admissible domain, the numerical parameter sensitivities and a compiled C++20 example. The library covers single-component equilibrium; it does not include a parameter-fitting module.

# Scientific model catalogue

The catalogue is grouped **by the number of parameters stored by each C++ model** and then **alphabetically within each parameter group**. Some fixed-activity temperature sections (such as Henderson and Smith) store water activity as a physical condition, not an additional fitted coefficient. Temperature supplied to an evaluation function is not a stored parameter.

NOTE

Some models are the same equation under a different parametrisation. They keep their usual names, and the equivalence is stated on the model pages. The following equivalences are also checked by the tests: Hill, Sips, Koble–Corrigan and Holl–Krich are equivalent to Langmuir–Freundlich; the three-parameter Fritz–Schlünder model is equivalent to Redlich–Peterson; and Khan is equivalent to Radke–Prausnitz I.

The order is by stored parameter count and then by model name; a six-parameter group is necessary for triple-site Langmuir.

01

# Henry isotherm

ONE PARAMETER / K_H / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

William Henry, *Experiments on the Quantity of Gases Absorbed by Water, at Different Temperatures, and under Different Pressures*, *Philosophical Transactions of the Royal Society of London*, **93** (1803), 29–42 and 274–276. DOI: 10.1098/rstl.1803.0004.

Henry's 1803 paper deals with gases dissolved in water. For adsorption onto solids, the same proportional relation describes uptake at low surface coverage. Use the model where uptake is approximately proportional to equilibrium concentration.

For a review of sorption models, see Limousin *et al.* (2007), *Sorption isotherms: A review on physical bases, modeling and measurement*, *Applied Geochemistry*, **22**, 249–275. DOI: 10.1016/j.apgeochem.2006.09.010.

## MATHEMATICAL FORMULATION

$$q_e(C_e) = K_H C_e, \qquad K_H > 0, \quad C_e \geq 0.$$

Here $q_e$ is uptake (for example, mg g$^{-1}$), $C_e$ is aqueous equilibrium concentration (for example, mg L$^{-1}$), and $K_H$ is the proportionality coefficient (for example, L g$^{-1}$). These units are illustrative; applications must supply consistent units. The implementation requires a finite positive coefficient and non-negative finite concentration.

The concentration derivative and low-concentration limit are

$$\frac{\mathrm{d}q_e}{\mathrm{d}C_e} = K_H, \qquad \lim_{C_e \to 0^+} \frac{q_e}{C_e} = K_H.$$

Henry's law has no saturation limit. Do not use it at concentrations where the measured uptake is no longer approximately linear.

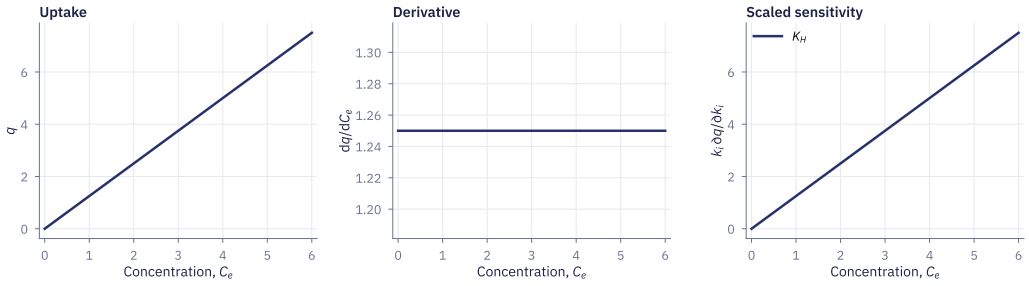



**Figure 1.** (a) Henry uptake for $K_H = 1.25$; (b) Concentration derivative of Henry uptake for $K_H = 1.25$; (c) Scaled sensitivity to $K_H$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Henry model{1.25};
const isotherm::Real uptake = isotherm::evaluate(model, 2.0);   // 2.5
const isotherm::Real slope = isotherm::derivative(model, 2.0);  // 1.25
```

The source browser shows the C++ files with links to their original source.

## CONCENTRATION DERIVATIVE

The independent variable of the implemented Henry model is equilibrium concentration rather than thermodynamic activity. Consequently, the plot shows $\mathrm{d}q_e/\mathrm{d}C_e = K_H = 1.25$, evaluated by the compiled C++20 model. An activity derivative would require an explicitly specified relation between activity and concentration.

### NUMERICAL SENSITIVITY

For the Henry coefficient $K_H$, the scaled sensitivity is

$$\lambda_{K_H}(C_e) = K_H \frac{\partial q_e}{\partial K_H}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

The model requires:

$$K_H > 0, \qquad C_e \geq 0.$$

# 02

# Anti-Langmuir isotherm

TWO PARAMETERS / H, b / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

The Anti-Langmuir isotherm is used to represent unfavourable adsorption behaviour, with an uptake curve that is convex with respect to equilibrium concentration. No single original publication is assigned to this model here. For background on nonlinear single-component adsorption equilibria, see Guiochon, Felinger, Shirazi and Katti, *Fundamentals of Preparative and Nonlinear Chromatography*, 2nd ed., Academic Press (2006), Chapter 3. Publisher record.

Cavazzini *et al.*, *Adsorption equilibria of butyl- and amylbenzene on monolithic silica-based columns*, *Journal of Chromatography A*, **957** (2002), 111–126, reported experimentally measured upward-concave adsorption isotherms fitted using the Anti-Langmuir model. DOI: 10.1016/S0021-9673(02)00320-5.

For the extension to binary chromatography, see Mazzotti, *Local Equilibrium Theory for the Binary Chromatography of Species Subject to a Generalized Langmuir Isotherm*, *Industrial & Engineering Chemistry Research*, **45** (2006), 5332–5350. DOI: 10.1021/ie060297v. The implementation documented here is a single-component model, not the binary generalisation.

### MATHEMATICAL FORMULATION

$$q_e(C_e) = \frac{HC_e}{1 - bC_e}.$$

The admissible parameter and input domain is

$$H > 0, \quad b > 0, \quad 0 \leq C_e < \frac{1}{b}.$$

Here $q_e$ denotes equilibrium uptake and $C_e$ equilibrium concentration. The coefficient $H$ has units of uptake divided by concentration, while $b$ has units of reciprocal concentration. The coefficient $H$ corresponds to $a$ in the alternative notation $q_e = aC_e/(1 - bC_e)$. The independent variable is concentration, not thermodynamic activity. For this equation,

$$\frac{\mathrm{d}q_e}{\mathrm{d}C_e} = \frac{H}{(1 - bC_e)^2}.$$

The admissible concentration range is $0 \leq C_e < 1/b$. The pole at $C_e = 1/b$ is a mathematical limit of the model, not physical saturation. The equation is convex for positive concentration.

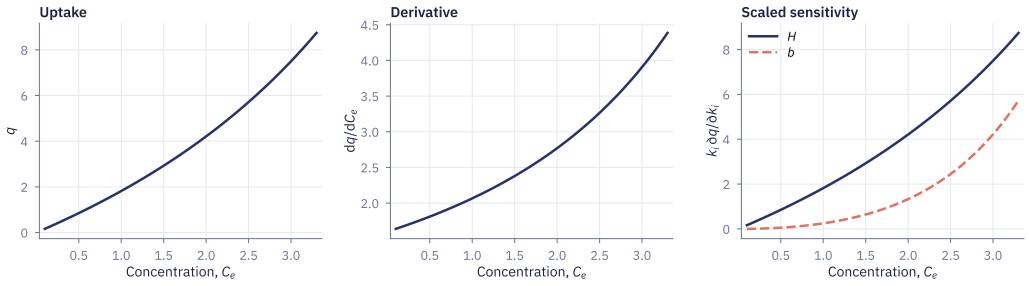



**Figure 2.** (a) Anti-Langmuir uptake for $H = 1.6$, $b = 0.12$; (b) Concentration derivative of Anti-Langmuir uptake for $H = 1.6$, $b = 0.12$; (c) Scaled sensitivities $\lambda_H$, $\lambda_b$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::AntiLangmuir model{1.6, 0.12};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 0.85106383
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 1.8107741
```

The source browser shows the C++ files with links to their original source.

### NUMERICAL SENSITIVITY

For each fitted coefficient $k_i$, define the scaled sensitivity

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$H > 0, \quad b > 0, \quad 0 \le C_e < \frac{1}{b}.$$

# 03

# Dubinin–Radushkevich isotherm

TWO PARAMETERS / q_s, beta / EXPLICIT / TEMPERATURE

### ORIGINAL PUBLICATION AND SCOPE

Dubinin and Radushkevich, *Equation of the Characteristic Curve of Activated Charcoal*, *Proceedings of the Academy of Sciences of the USSR*, **55** (1947), 331–333. The bibliographic details are reproduced in a later review of cooperative sorption.

For a subsequent theoretical treatment, see Dubinin, *The Potential Theory of Adsorption of Gases and Vapors for Adsorbents with Energetically Nonuniform Surfaces*, *Chemical Reviews*, **60** (1960), 235–241. DOI: 10.1021/cr60204a006.

The model describes adsorption in the adsorption-potential (pore-filling) framework. This implementation follows a frequently used liquid-phase concentration-based representation, rather than claiming to reproduce all thermodynamic activity or vapour-pressure conventions of the original work.

### MATHEMATICAL FORMULATION

$$q_e(C_e, T) = q_s \exp(-\beta \varepsilon^2), \qquad \varepsilon = RT \ln\left(1 + \frac{1}{C_e}\right).$$

The admissible parameter and input domain is

$$q_s > 0, \quad \beta > 0, \quad C_e \ge 0, \quad T > 0\ \mathrm{K}, \quad C_e \text{ dimensionless}.$$

Here $q_s$ is a capacity parameter and $\beta$ is an inverse energy-squared coefficient. The independent temperature $T$ **must always be in kelvin (K)**. By default $R = 8.31446261815324$ J mol$^{-1}$ K$^{-1}$, and $\varepsilon$ is the Polanyi potential in J mol$^{-1}$. Consequently, $\beta$ must be expressed in mol$^2$ J$^{-2}$ when these units are used for $R$. **The quantity supplied as $C_e$ must be dimensionless** (for example, a physical concentration divided by a stated reference concentration). Taking the logarithm of a dimensional concentration is not mathematically valid.

For positive concentration, the analytical concentration derivative exposed by the model is

$$\frac{\partial q_e}{\partial C_e} = \frac{2\beta RT\varepsilon}{C_e(1 + C_e)} q_e.$$

At $C_e = 0$, the implementation uses the limits $q_e = 0$ and $\partial q_e/\partial C_e = 0$. The algebraic linearisation of this equation is $\ln q_e = \ln q_s - \beta\varepsilon^2$, for $q_e > 0$.

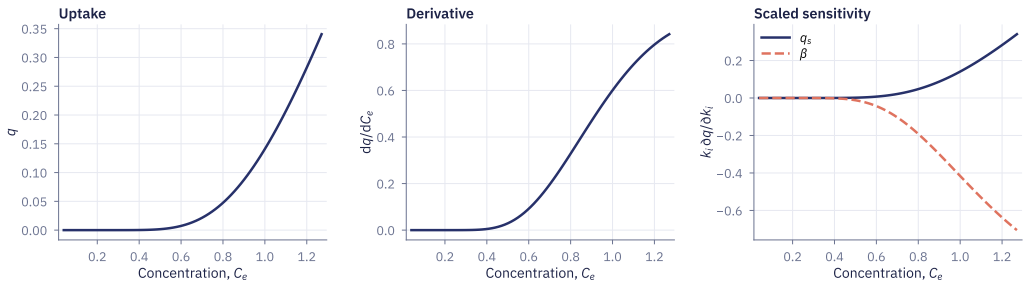



**Figure 3.** (a) Dubinin–Radushkevich uptake for $q_s = 2.7$, $\beta = 0.000001$ at $T = 298.15$ K; (b) Concentration derivative of Dubinin–Radushkevich uptake for $q_s = 2.7$, $\beta = 0.000001$ at $T = 298.15$ K; (c) Scaled sensitivities $\lambda_{q_s}$, $\lambda_\beta$ versus equilibrium concentration at $T = 298.15$ K.

## C++20 USAGE

```
#include <isotherm/isotherm.hpp>

const isotherm::models::DubininRadushkevich model{2.7, 0.000001};
const isotherm::Real temperature = 298.15;
const isotherm::Real uptake = isotherm::evaluate(model, 1.0, temperature);   //
0.1409653
const isotherm::Real slope = isotherm::derivative(model, 1.0, temperature);  //
0.60044838

// Equivalent energy units: beta must be converted together with R.
const isotherm::models::DubininRadushkevich model_kj{
    2.7, 1.0, isotherm::gas_constant_unit::kilojoule_per_mole_kelvin};
const isotherm::Real same_uptake =
    isotherm::evaluate(model_kj, 1.0, temperature);                          //
0.1409653
```

The source browser shows the C++ files with links to their original source.

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the scaled sensitivity is defined by

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_s > 0, \quad \beta > 0, \quad C_e \geq 0, \quad T > 0 \text{ K}, \quad C_e \text{ dimensionless.}$$

04

# Elovich isotherm

TWO PARAMETERS / q_ref, K_E / IMPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Elovich, S. Yu., and Larionov, O. G. (1962), *Theory of adsorption from nonelectrolyte solutions on solid adsorbents, II. Experimental verification of the equation for the adsorption isotherm from solutions*, *Bulletin of the Academy of Sciences of the USSR, Division of Chemical Science*, **11** (2), 198–203. DOI: 10.1007/BF00908017.

Dusart, O., Bouabane, H., and Mazet, M. (1991), *Adsorption sur charbon actif d'acides aminés dans l'eau : détermination de paramètres d'équilibre par différentes équations*, *Journal de Chimie Physique*, **88**, 259–270. DOI: 10.1051/jcp/1991880259. The 1991 study is identified in later literature as an early source for an equation equivalent to the form commonly used in adsorption fitting.

For a detailed historical and mathematical assessment, see Debord, J., Harel, M., Bollinger, J.-C., and Chu, K. H. (2022), *The Elovich isotherm equation: Back to the roots and new developments*, *Chemical Engineering Science*, **262**, 118012. DOI: 10.1016/j.ces.2022.118012.

## MATHEMATICAL FORMULATION

The single-component equilibrium relationship is implicit:

$$\vartheta \exp(\vartheta) = K_E C_e, \qquad q_e = q_{\mathrm{ref}} \vartheta.$$

The admissible parameter and variable domain is

$$q_{\mathrm{ref}} > 0, \qquad K_E > 0, \qquad 0 \le C_e \le \frac{\exp(1)}{K_E}, \qquad 0 \le \vartheta \le 1.$$

The implementation solves for the non-negative coverage fraction $\vartheta$ on the restricted interval $0 \le \vartheta \le 1$, using a bracketed numerical algorithm. The capacity scale $q_{\mathrm{ref}}$ is an uptake parameter and $K_E$ has units of inverse concentration. The concentration derivative evaluated by the C++ implementation for $C_e > 0$ is

$$\frac{\mathrm{d}q_e}{\mathrm{d}C_e} = \frac{q_e}{C_e(1+\vartheta)}.$$

At $C_e = 0$, its limiting slope is $\mathrm{d}q_e/\mathrm{d}C_e = q_{\mathrm{ref}} K_E$.

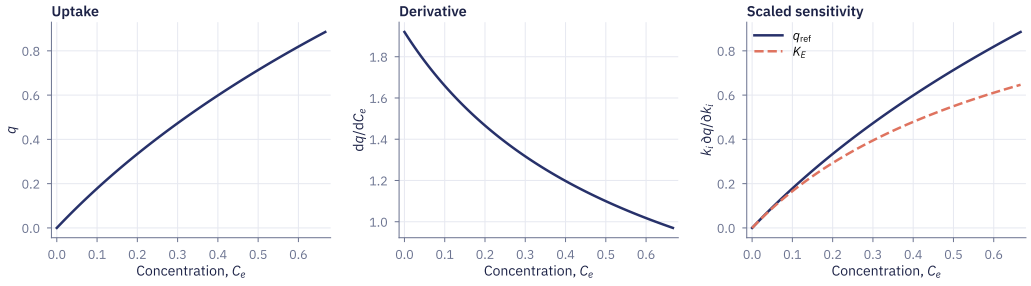



**Figure 4.** (a) Elovich uptake for $q_{\mathrm{ref}} = 2.4$, $K_E = 0.8$; (b) Concentration derivative of Elovich uptake for $q_{\mathrm{ref}} = 2.4$, $K_E = 0.8$; (c) Scaled sensitivities $\lambda_{q_{\mathrm{ref}}}$, $\lambda_{K_E}$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Elovich model{2.4, 0.8};
const isotherm::Real uptake = isotherm::evaluate(model, 1.0);    // 1.1761629
const isotherm::Real slope = isotherm::derivative(model, 1.0);  // 0.7893351
```

NUMERICAL SENSITIVITY

For each coefficient $k_i$, define the scaled parameter sensitivity

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{\mathrm{ref}} > 0, \quad K_E > 0, \quad 0 \le C_e \le \frac{\exp(1)}{K_E}, \quad 0 \le \vartheta \le 1.$$

# 05

# Freundlich isotherm

TWO PARAMETERS / K_F, n / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

Freundlich, H. (1907), *Über die Adsorption in Lösungen*, *Zeitschrift für Physikalische Chemie*, **57U** (1), 385–470. DOI: 10.1515/zpch-1907-5723. The publisher dates the journal article to 1907; a separate 1906 habilitation edition was published by Wilhelm Engelmann.

The Freundlich isotherm is an empirical power-law model used to describe adsorption on heterogeneous surfaces. It does not predict a finite saturation capacity. See also the adsorption-isotherm review (2025).

MATHEMATICAL FORMULATION

$$q_e = K_F C_e^{1/n}.$$

The admissible parameter and input domain is

$$K_F > 0, \quad n > 0, \quad C_e \ge 0.$$

$$\left.\frac{\mathrm{d}q_e}{\mathrm{d}C_e}\right|_{C_e = 0} \text{ is finite only when } 0 < n \le 1.$$

Here $C_e$ denotes equilibrium concentration.

The adjustable parameters are $K_F$, $n$.

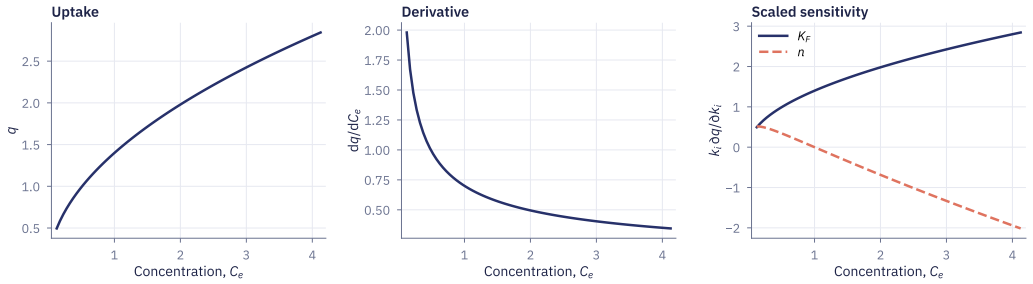



**Figure 5.** (a) Equilibrium uptake evaluated through the C++ model interface; (b) Derivative evaluated through the C++ model interface; (c) Numerical scaled sensitivities versus equilibrium concentration.

C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Freundlich model{1.4, 2.0};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 0.98994949
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 0.98994949
```

NUMERICAL SENSITIVITY

For each parameter $k_i$, the scaled parameter sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e(x)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$K_F > 0, \quad n > 0, \quad C_e \geq 0.$$

$\left.\frac{dq_e}{dC_e}\right|_{C_e = 0}$ is finite only when $0 < n \leq 1$.

# 06

# Halsey isotherm

TWO PARAMETERS / K_H, n / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

Halsey, G., and Taylor, H. S. (1947), *The Adsorption of Hydrogen on Tungsten Powders*, *The Journal of Chemical Physics*, **15** (9), 624–630. DOI: 10.1063/1.1746618. This experimental study preceded Halsey's treatment of adsorption on non-uniform surfaces.

Halsey, G. (1948), *Physical Adsorption on Non-Uniform Surfaces*, *The Journal of Chemical Physics*, **16** (10), 931–937. DOI: 10.1063/1.1746689. The original multilayer relation is expressed through relative gas pressure $p/p_0$ and surface coverage $\vartheta$, in the form $\ln(p/p_0) = -a/\vartheta^r$. It is not a reciprocal-concentration law.

The related historical framework includes Frenkel, J., *Kinetic Theory of Liquids* (Clarendon Press, 1946), and Hill, T. L. (1952), *Theory of Physical Adsorption*, *Advances in Catalysis*, **4**, 211–258. DOI: 10.1016/S0360-0564(08)60615-X. These works are commonly cited in connection with the Frenkel–Halsey–Hill (FHH) treatment of multilayer adsorption.

Chu, K. H., Bashiri, H., Hashim, M. A., Abd Shukor, M. Y., and Bollinger, J.-C. (2023), *The Halsey isotherm for water contaminant adsorption is fake*, *Separation and Purification Technology*, **313**, 123500. DOI: 10.1016/j.seppur.2023.123500. The authors distinguish the original gas-adsorption isotherm from a frequently used liquid-phase power-law form, which they show to be mathematically equivalent to a reparameterised Freundlich relation. Their terminology refers to that specific liquid-phase formulation; it should not be treated as a replacement for comparing actual equations.

The C++ class documented here evaluates $q_e = (K_H/C_e)^{1/n}$. For positive $K_H$ and $n$, this function decreases with increasing $C_e$, diverges as $C_e \to 0^+$, and tends to zero as $C_e \to \infty$. Its mathematical behaviour is therefore distinct from the original Halsey multilayer relation, and the 1948 publication does not establish the physical validity of this particular expression.

The implementation describes its concentration argument as dimensionless. A reference concentration for converting a dimensional $C_e$ to this input has not been established here. Its physical interpretation therefore requires an explicit normalisation convention.

MATHEMATICAL FORMULATION

$$q_e = \left(\frac{K_H}{C_e}\right)^{1/n}.$$

The admissible parameter and input domain is

$$K_H > 0, \quad n > 0, \quad C_e > 0.$$

Here $C_e$ is required by the C++ class to be positive and dimensionless. No reference concentration has been specified for a normalisation $c^* = C_e/C_{\mathrm{ref}}$; hence physical concentration data cannot be converted into model inputs without an additional, explicit definition of $C_{\mathrm{ref}}$.

The adjustable parameters are $K_H$, $n$.

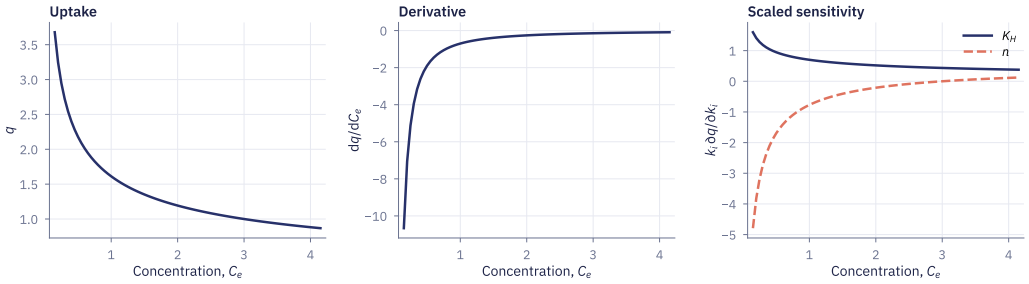



**Figure 6.** (a) Halsey uptake for $K_H = 3.0$, $n = 2.3$; (b) Concentration derivative of Halsey uptake for $K_H = 3.0$, $n = 2.3$; (c) Scaled sensitivities $\lambda_{K_H}$, $\lambda_n$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Halsey model{3.0, 2.3};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 2.1793482
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // -1.8950854
```

### NUMERICAL SENSITIVITY

For each parameter $k_i$, the scaled parameter sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e(x)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$K_H > 0, \quad n > 0, \quad C_e > 0.$$

# 07

# Harkins–Jura isotherm

TWO PARAMETERS / A, B / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Jura and Harkins (1943) introduced an adsorption isotherm for gas–solid systems over a wide range of pressures. Subsequent work by Harkins and Jura (1944) developed the connection between adsorbed films, changes in surface free energy and the determination of solid surface area. These publications establish the historical basis of the Harkins–Jura isotherm.

The present model uses a two-parameter, base-10 logarithmic relation expressed in terms of equilibrium concentration. This concentration-based form should not be assumed to be identical to the original gas-pressure formulation. Chu and Hashim (2023) critically examined the use of the Harkins–Jura equation for aqueous-phase contaminant adsorption, including the limitations of linearised fits. The C++ identifier for this model is `HarkinJura`.

**Primary and related references**

- Jura, G., and Harkins, W. D. (1943). *A New Adsorption Isotherm Which Is Valid Over a Very Wide Range of Pressure. The Journal of Chemical Physics*, **11** (9), 430–431. DOI: 10.1063/1.1723870.

- Harkins, W. D., and Jura, G. (1944). *The Decrease (π) of Free Surface Energy (γ) as a Basis for the Development of Equations for Adsorption Isotherms; and the Existence of Two Condensed Phases in Films on Solids*. *The Journal of Chemical Physics*, **12** (3), 112–113. DOI: 10.1063/1.1723913.
- Harkins, W. D., and Jura, G. (1944). *Surfaces of Solids. XIII. A Vapor Adsorption Method for the Determination of the Area of a Solid without the Assumption of a Molecular Area, and the Areas Occupied by Nitrogen and Other Molecules on the Surface of a Solid*. *Journal of the American Chemical Society*, **66** (8), 1366–1373. DOI: 10.1021/ja01236a048.
- Chu, K. H., and Hashim, M. A. (2023). *Modeling of aqueous phase adsorption: Is it time to bid adieu to the Harkins–Jura isotherm?* *Journal of Molecular Liquids*, **371**, 121122. DOI: 10.1016/j.molliq.2022.121122.


## MATHEMATICAL FORMULATION

$$\frac{1}{q_e^2} = \frac{B - \log_{10}(C_e)}{A}.$$

The admissible parameter and input domain is

$$A > 0, \quad B > 0, \quad 0 \le C_e < 10^B.$$

The concentration derivative requires $C_e > 0$.

Here $C_e$ denotes equilibrium concentration.

The adjustable parameters are $A$, $B$.

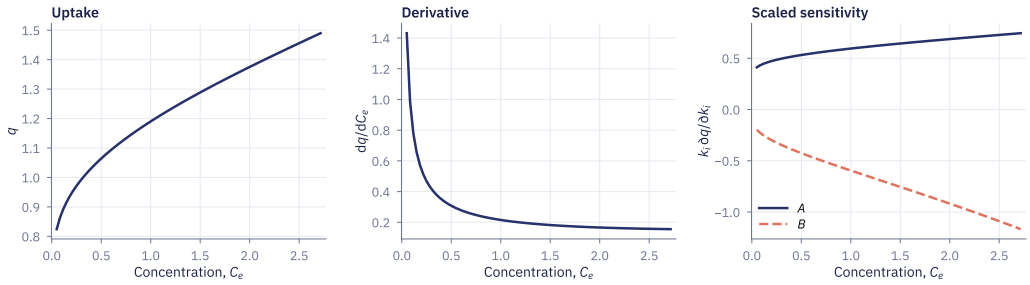



**Figure 7.** (a) Harkins–Jura uptake for $A = 1.7$, $B = 1.2$; (b) Concentration derivative of Harkins–Jura uptake for $A = 1.7$, $B = 1.2$; (c) Scaled sensitivities $\lambda_A$, $\lambda_B$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::HarkinJura model{1.7, 1.2};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 1.064216
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 0.30791065
```

## NUMERICAL SENSITIVITY

For each parameter $k_i$, the scaled parameter sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e(x)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$A > 0, \quad B > 0, \quad 0 \le C_e < 10^B.$$

The concentration derivative requires $C_e > 0$.

08

# Jovanovic isotherm

TWO PARAMETERS / q_sat, K / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Jovanović (1969) derived monolayer and multilayer isotherms for the physical adsorption of gases. The monolayer model approaches a finite saturation capacity through an exponential dependence on the gas-phase pressure variable. The concentration-based expression used here adopts the same saturating functional form, with $C_e$ as its independent variable. Its application to adsorption from solutions is a model adaptation, rather than a direct reproduction of the gas-phase experimental conditions. Chu et al. (2023) distinguish this rising, saturating form from incorrect exponential parametrisations found in the aqueous-adsorption literature.

The primary sources and subsequent developments are:

- Jovanović, D. S. (1969). *Physical adsorption of gases. I: Isotherms for monolayer and multilayer adsorption*. *Kolloid-Zeitschrift und Zeitschrift für Polymere*, **235**, 1203–1213. DOI: 10.1007/BF01542530.
- Jovanović, D. S. (1969). *Physical adsorption of gases. II: Practical application of derived isotherms for monolayer and multilayer adsorption*. *Kolloid-Zeitschrift und Zeitschrift für Polymere*, **235**, 1214–1225. DOI: 10.1007/BF01542531.
- Misra, D. N. (1973). *Jovanovich adsorption isotherm for heterogeneous surfaces*. *Journal of Colloid and Interface Science*, **43**, 85–88. DOI: 10.1016/0021-9797(73)90350-0.
- Quiñones, I., and Guiochon, G. (1996). *Derivation and application of a Jovanovic–Freundlich isotherm model for single-component adsorption on heterogeneous surfaces*. *Journal of Colloid and Interface Science*, **183**, 57–67. DOI: 10.1006/jcis.1996.0518.
- Chu, K. H., Hashim, M. A., Debord, J., Harel, M., Salvestrini, S., and Bollinger, J.-C. (2023). *The Jovanović adsorption isotherm in water contaminant research: Unmasking spurious versions and spotlighting the real thing*. *Chemical Engineering Science*, **281**, 119127. DOI: 10.1016/j.ces.2023.119127.

## MATHEMATICAL FORMULATION

$$q_e = q_{\mathrm{sat}}[1 - \exp(-KC_e)].$$

The admissible parameter and input domain is

$$q_{\mathrm{sat}} > 0, \quad K > 0, \quad C_e \geq 0.$$

Here $C_e$ denotes equilibrium concentration. The capacity $q_{\mathrm{sat}}$ has the same units as $q_e$, while $K$ has the reciprocal units of $C_e$. The model satisfies $q_e(0) = 0$ and approaches $q_{\mathrm{sat}}$ as $C_e \to \infty$.

The adjustable parameters are $q_{\mathrm{sat}}$ and $K$.

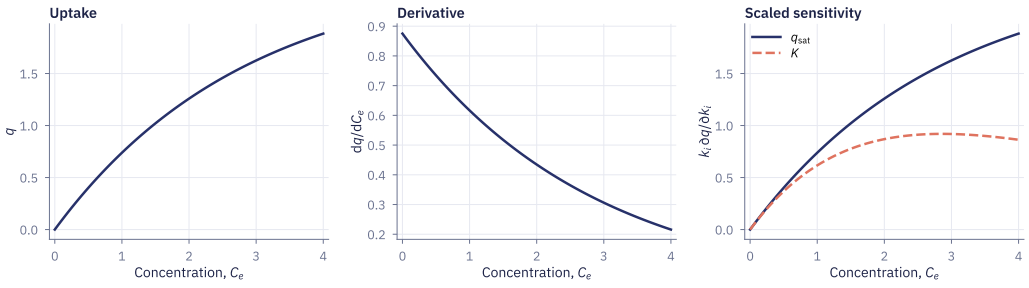



**Figure 8.** (a) Jovanovic uptake for $q_{\mathrm{sat}} = 2.5$, $K = 0.35$; (b) Concentration derivative of Jovanovic uptake for $q_{\mathrm{sat}} = 2.5$, $K = 0.35$; (c) Scaled sensitivities $\lambda_{q_{\mathrm{sat}}}$, $\lambda_K$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Jovanovic model{2.5, 0.35};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);   // 0.40135745
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 0.73452489
```

### NUMERICAL SENSITIVITY

For each parameter $k_i$, the scaled parameter sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e(x)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{\text{sat}} > 0, \quad K > 0, \quad C_e \geq 0.$$

# 09

# Langmuir isotherm

TWO PARAMETERS / q_max, K / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Langmuir, I. (1918), *The Adsorption of Gases on Plane Surfaces of Glass, Mica and Platinum*, *Journal of the American Chemical Society*, **40** (9), 1361–1403. DOI: 10.1021/ja02242a004. This is the primary source for the classical adsorption model.

The development was preceded by Langmuir, I. (1916), *The Constitution and Fundamental Properties of Solids and Liquids. Part I. Solids*, *Journal of the American Chemical Society*, **38** (11), 2221–2295. DOI: 10.1021/ja02268a002, and Langmuir, I. (1917), *The Constitution and Fundamental Properties of Solids and Liquids. II. Liquids*, *Journal of the American Chemical Society*, **39** (9), 1848–1906. DOI: 10.1021/ja02254a006.

The model assumes adsorption on a fixed number of energetically equivalent sites, with one adsorbate molecule per site, a monolayer at saturation and no lateral interactions between adsorbed molecules. Adsorption and desorption are in dynamic equilibrium. In the original gas-adsorption setting, the fractional surface coverage can be expressed as $\vartheta = KP/(1 + KP)$, where $P$ is gas pressure. The concentration form implemented here uses $C_e$ instead of pressure, with $KC_e$ required to be dimensionless. The saturation capacity is $q_{\text{max}}$.

### MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\text{max}} K C_e}{1 + K C_e}.$$

The admissible parameter and input domain is

$$q_{\text{max}} > 0, \quad K > 0, \quad C_e \geq 0.$$

Here $C_e$ denotes equilibrium concentration.

The adjustable parameters are $q_{\text{max}}$, $K$.

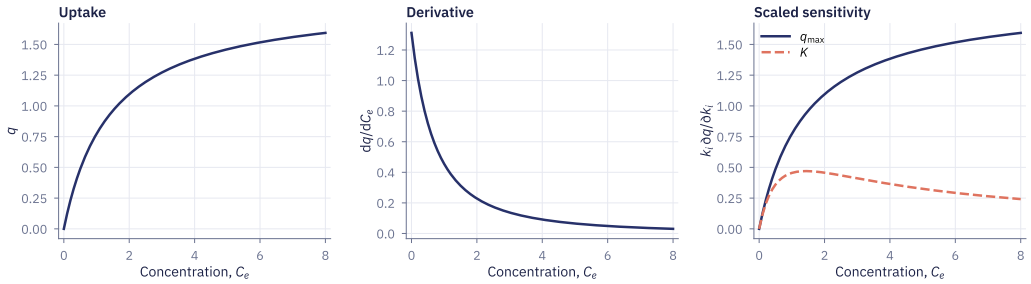



**Figure 9.** (a) Equilibrium uptake evaluated through the C++ model interface; (b) Derivative evaluated through the C++ model interface; (c) Numerical scaled sensitivities versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Langmuir model{1.87902, 0.698076};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);
const isotherm::Real slope = isotherm::derivative(model, 0.5);
```

### NUMERICAL SENSITIVITY

For each parameter $k_i$, the scaled parameter sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e(x)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad C_e \geq 0.$$

# 10 Temkin isotherm

TWO PARAMETERS / K_T, b_T / EXPLICIT / TEMPERATURE

### ORIGINAL PUBLICATION AND SCOPE

The Temkin isotherm is associated with the treatment of adsorption on energetically non-uniform surfaces and interactions between adsorbed molecules. The frequently cited Temkin–Pyzhev paper (1940) examines ammonia-synthesis kinetics, whereas Temkin (1941) directly addresses adsorption equilibrium. The two publications should not be treated as interchangeable sources for a particular isotherm equation.

Temkin, M. I., and Pyzhev, V. (1940), *Kinetics of ammonia synthesis on promoted iron catalysts*, *Acta Physicochimica URSS*, **12**, 327–356. Bibliographic record.

Temkin, M. I. (1941), *Adsorption equilibrium and the kinetics of processes on non-uniform surfaces and in the interaction between adsorbed molecules*, *Zhurnal Fizicheskoi Khimii*, **15**, 296–332. Bibliographic record.

Johnson, R. D., and Arnold, F. H. (1995), *The Temkin isotherm describes heterogeneous protein adsorption*, *Biochimica et Biophysica Acta*, **1247** (2), 293–297. DOI: 10.1016/0167-4838(95)00006-G. This is an example of a subsequent application to protein adsorption.

Chu, K. H. (2021), *Revisiting the Temkin Isotherm: Dimensional Inconsistency and Approximate Forms*, *Industrial & Engineering Chemistry Research*, **60** (35), 13140–13147. DOI: 10.1021/acs.iecr.1c01788. Chu examines the inconsistent dimensions and approximate nature of Temkin expressions commonly used for aqueous adsorption, including their limiting behaviour.

Salvestrini, S., Ambrosone, L., and Kopinke, F.-D. (2022), *Some mistakes and misinterpretations in the analysis of thermodynamic adsorption data*, *Journal of Molecular Liquids*, **352**, 118762. DOI: 10.1016/j.molliq.2022.118762. This paper discusses thermodynamic and dimensional consistency, uncertainty in fitted parameters and the assumptions underlying adsorption models, including Temkin.

Chu, K. H., Hashim, M. A., Hayder, G., and Bollinger, J.-C. (2024), *Rooting out faulty adsorption models. Comment on "Magnetic Prussian blue nanoshells are controllable (sic) anchored on the surface of molybdenum disulfide nanosheets for efficient separation of radioactive cesium from water"*, *Science of the Total Environment*, **921**, 171118. DOI: 10.1016/j.scitotenv.2024.171118. This correspondence includes the Temkin isotherm among the models considered in its critique of an individual adsorption study.

Mudhoo, A. (2025), *Unveiling new insights: Revised Temkin adsorption isotherm parameters from fresh curve fits in adsorption studies*, *Chemical Engineering Science*, **311**, 121585. DOI: 10.1016/j.ces.2025.121585. This study refits published adsorption data using the alternative nonlinear Temkin expression discussed by Chu (2021) and reports differences in estimated parameters.

The C++ class described below implements the logarithmic expression $q_e = (RT/b_T)\ln(K_T C_e)$, not the alternative form investigated by Chu (2021) and Mudhoo (2025). The logarithm requires $K_T C_e$ to be dimensionless, but this condition alone does not establish the dimensional consistency of the loading: the prefactor $RT/b_T$ must also have units consistent with $q_e$. In particular, if $b_T$ is an energy per mole, $RT/b_T$ is dimensionless. The implemented expression has neither a linear low-concentration limit nor a finite saturation limit. The historical papers supply context and must not be read as verification of the particular parameter convention used in this class.

## MATHEMATICAL FORMULATION

**Temperature must always be supplied in kelvin (K)** as an independent thermodynamic input, not an adjustable adsorption parameter.

$$q_e(C_e, T) = \frac{RT}{b_T}\ln(K_T C_e).$$

The admissible parameter and input domain is

$$K_T > 0, \quad b_T > 0, \quad T > 0\ \mathrm{K}, \quad C_e > 0, \quad K_T C_e \geq 1.$$

The gas constant uses the units selected for the model, and $b_T$ must use compatible energy units. Temperature is always in kelvin.

Thermal two-parameter Temkin model, evaluated at a supplied absolute temperature; temperature does not count as a fitted parameter.

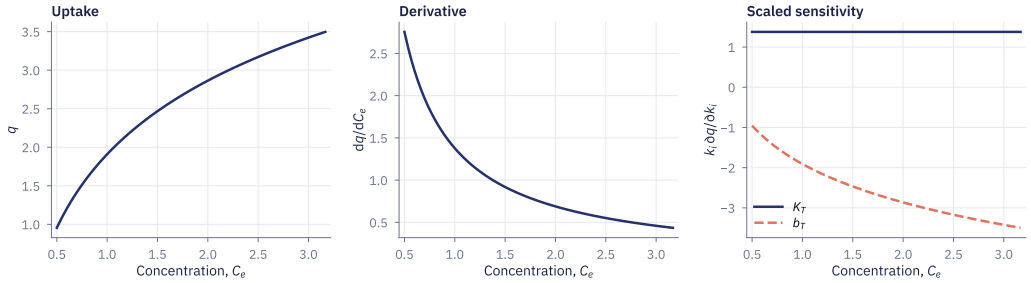



**Figure 10.** (a) Temkin uptake for $K_T = 4.0$, $b_T = 1800.0$ at $T = 298.15\,\mathrm{K}$; (b) Concentration derivative of Temkin uptake for $K_T = 4.0$, $b_T = 1800.0$ at $T = 298.15\,\mathrm{K}$; (c) Scaled sensitivities $\lambda_{K_T}$, $\lambda_{b_T}$ versus equilibrium concentration at $T = 298.15\,\mathrm{K}$.

## ANALYTICAL CONCENTRATION DERIVATIVE

$$\frac{\partial q_e}{\partial C_e} = \frac{RT}{b_T C_e}.$$

C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Temkin model{4.0, 1800.0};
const isotherm::Real temperature_k = 298.15;                                    //
Always in kelvin.
const isotherm::Real uptake = isotherm::evaluate(model, 1.0, temperature_k);    //
1.9092023
const isotherm::Real slope = isotherm::derivative(model, 1.0, temperature_k);   //
1.3771983

// Select kJ/(mol K) for R; convert b_T from J to kJ consistently.
const isotherm::models::Temkin model_kj{
    4.0, 1.8, isotherm::gas_constant_unit::kilojoule_per_mole_kelvin};
const isotherm::Real same_uptake =
    isotherm::evaluate(model_kj, 1.0, temperature_k);                           //
1.9092023
```

LINEARISED RELATIONSHIP

$$q_e = \frac{RT}{b_T}\ln K_T + \frac{RT}{b_T}\ln C_e.$$

NUMERICAL SENSITIVITY

For each model coefficient $k_i$, the scaled sensitivity at fixed temperature is

$$\lambda_i(C_e, T) = k_i \frac{\partial q_e(C_e, T)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$K_T > 0, \quad b_T > 0, \quad T > 0\ \mathrm{K}, \quad C_e > 0, \quad K_T C_e \geq 1.$$

# 11

# Volmer isotherm

TWO PARAMETERS / q_max, K / IMPLICIT

ORIGINAL PUBLICATION AND SCOPE

Volmer, M. (1925), *Thermodynamische Folgerungen aus der Zustandsgleichung für adsorbierte Stoffe*, *Zeitschrift für Physikalische Chemie*, **115U** (1), 253–260. DOI: 10.1515/zpch-1925-11519. This is the principal historical reference for the thermodynamic description of mobile adsorbed films. The associated two-dimensional equation of state is conventionally expressed as $\pi(\Omega - \beta) = RT$, where $\pi$ is spreading pressure, $\Omega$ is molar surface area and $\beta$ represents an excluded-area term. These thermodynamic quantities should not be confused with the two coefficients of the implemented isotherm.

Volmer, M., and Mahnert, P. (1925), *Über die Auflösung fester Körper in Flüssigkeitsoberflächen und die Eigenschaften der dabei entstehenden Schichten*, *Zeitschrift für Physikalische Chemie*, **115U** (1), 239–252. DOI: 10.1515/zpch-1925-11518. This companion paper provides historical context on surface films at liquid interfaces; it is not a separate derivation of the concentration-based equation used below.

The present model uses an implicit relation between equilibrium concentration and fractional surface coverage. Its specific adsorption parametrisation is given in the next section; the thermodynamic surface-film equation of state and the implemented concentration relation should not be presented as identical equations.

MATHEMATICAL FORMULATION

$$KC_e = \frac{\vartheta}{1-\vartheta}\exp\left(\frac{\vartheta}{1-\vartheta}\right), \qquad q_e = q_{\max}\vartheta.$$

The admissible parameter and input domain is

$$q_{max} > 0, \quad K > 0, \quad C_e \geq 0, \quad 0 \leq \vartheta < 1.$$

The implemented derivative requires $C_e > 0$.

Here $C_e$ denotes equilibrium concentration.

The adjustable parameters are $q_{max}$, $K$.

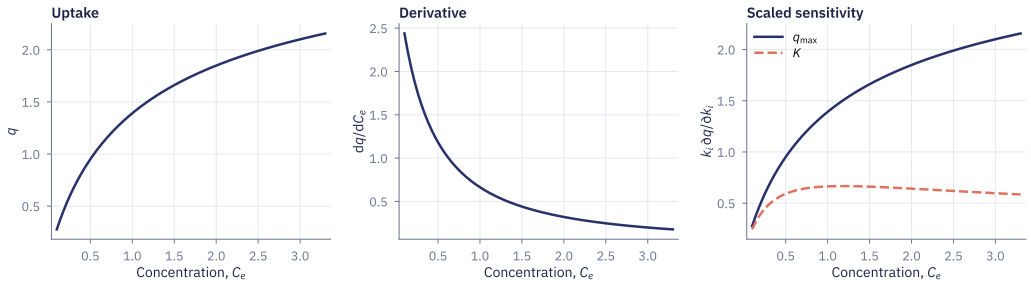



**Figure 11.** (a) Volmer uptake for $q_{max} = 4.5$, $K = 0.7$; (b) Concentration derivative of Volmer uptake for $q_{max} = 4.5$, $K = 0.7$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Volmer model{4.5, 0.7};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 0.95048081
const isotherm::Real slope = isotherm::derivative(model, 0.5);   // 1.1827348
```

### NUMERICAL SENSITIVITY

For each parameter $k_i$, the scaled parameter sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e(x)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad C_e \geq 0, \quad 0 \leq \vartheta < 1.$$

The implemented derivative requires $C_e > 0$.

# 12

# Aranovich isotherm

THREE PARAMETERS / q_max, b, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Aranovich, G. L. (1992), *The theory of polymolecular adsorption*, *Langmuir*, **8** (2), 736–739. DOI: 10.1021/la00038a071. This is the primary reference for the modified multilayer-adsorption treatment. The original Aranovich-type correction is commonly associated with a square-root factor in reduced pressure, whereas the model documented here permits the multilayer exponent $n$ to be adjusted.

The subsequent Aranovich–Donohue research develops related, but not necessarily algebraically identical, models:

- Aranovich, G. L., and Donohue, M. D. (1996), *Multilayer adsorption of slightly soluble organic compounds from aqueous solutions*, *Journal of Colloid and Interface Science*, **178** (2), 764–769. DOI: 10.1006/jcis.1996.0175. This study formulates multilayer adsorption using a reduced concentration and an adjustable divergence exponent.
- Aranovich, G. L., and Donohue, M. D. (1996), *Adsorption of supercritical fluids*, *Journal of Colloid and Interface Science*, **180** (2), 537–541. DOI: 10.1006/jcis.1996.0334. This is a related application of lattice-based adsorption theory, not an experimental validation of this particular three-parameter equation.
- Donohue, M. D., and Aranovich, G. L. (1998), *Classification of Gibbs adsorption isotherms*, *Advances in Colloid and Interface Science*, **76–77**, 137–152. DOI: 10.1016/S0001-8686(98)00044-X. This addresses the classification of adsorption behaviours, including non-monotonic Gibbs excess adsorption.
- Aranovich, G. L., and Donohue, M. D. (2003), *Adsorption compression: An important new aspect of adsorption behavior and capillarity*, *Langmuir*, **19** (7), 2722–2735. DOI: 10.1021/la020739h. This extends the broader physical analysis of adsorbed phases.
- Yılmaz, M., Hashim, M. A., and Chu, K. H. (2022), *Description of sigmoid adsorption isotherms of water pollutants by the Aranovich–Donohue equation*, *Separation Science and Technology*, **57** (18), 2908–2915. DOI: 10.1080/01496395.2022.2097093. This reports applications of an Aranovich–Donohue formulation to atypical liquid-phase adsorption curves; it is not a direct validation of every Aranovich variant.

The implemented three-parameter expression combines a Langmuir-type low-coverage term with the multilayer factor $(1-x)^{-n}$. For gas adsorption, $x = p/p_{\mathrm{sat}}$ is a dimensionless relative pressure; for solution applications, a reduced concentration requires an explicitly defined reference, such as $x = C_e/C_{\mathrm{sat}}$, with $C_{\mathrm{sat}}$ the relevant saturation or solubility concentration. The C++ class receives $x$ directly and does not calculate this normalisation.

## MATHEMATICAL FORMULATION

$$q = \frac{q_{\max} b x}{(1+bx)(1-x)^n}$$

The 3 adjustable parameters are: `q_max`, `b`, `n`. The independent variable is `water activity or relative pressure`.

0 <= relative pressure x < 1; n>0 is the multilayer divergence exponent. This selected A-D-Langmuir convention is not a universal definition of Aranovich models.

At x near saturation the mathematical model diverges; the derivative and uptake do not extend continuously to x=1.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad b > 0, \quad n > 0, \quad 0 \le x < 1.$$

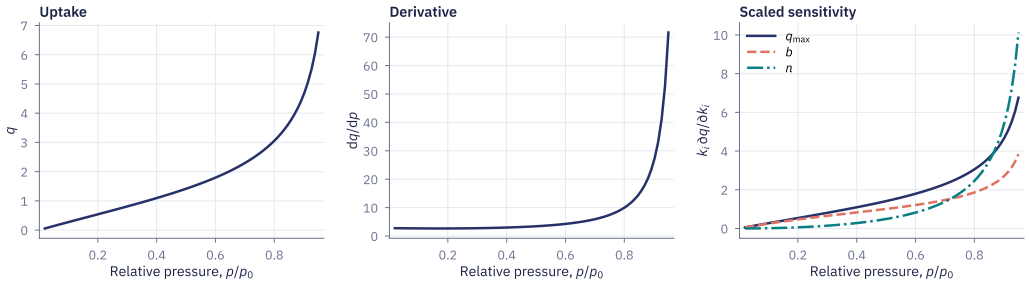



**Figure 12.** (a) Aranovich uptake for $q_{\max} = 3.5$, $b = 0.8$, $n = 0.5$; (b) Relative-pressure derivative of Aranovich uptake for $q_{\max} = 3.5$, $b = 0.8$, $n = 0.5$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_b$, $\lambda_n$ versus relative pressure.

C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Aranovich model{3.5, 0.8, 0.5};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 1.4142136
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 3.4345187
```

NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad b > 0, \quad n > 0, \quad 0 \le x < 1.$$

# 13 Bingel–Walton isotherm

THREE PARAMETERS / q_max, a, b / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

Bingel and Walton (2023) introduced an empirical adsorption isotherm based on Bass's innovation-diffusion model. Their original formulation uses gas pressure and was designed to describe adsorption curves of IUPAC types I, III and V. The authors assessed the model using 31 literature data sets, including stepped isotherms of flexible metal-organic frameworks.

Bingel, L. W., and Walton, K. S. (2023), *Surprising Use of the Business Innovation Bass Diffusion Model To Accurately Describe Adsorption Isotherm Types I, III, and V*, *Langmuir*, **39** (12), 4475–4482. DOI: 10.1021/acs.langmuir.3c00147.

The published model has two fitted coefficients, $a$ and $b$; $q_{\max}$ is obtained from the observed saturation loading (or the largest measured loading when saturation is not reached), rather than fitted as a third coefficient. The C++ class nevertheless accepts all three quantities as constructor arguments. Its concentration-based expression retains the functional form of the original pressure-based equation, but this substitution does not itself demonstrate experimental validity for liquid-phase adsorption.

For methodological background, see Bingel, L. W., Walton, K. S., and Sholl, D. S. (2022), *Experimentally Verified Alkane Adsorption Isotherms in Nanoporous Materials from Literature Meta-Analysis*, *Journal of Chemical & Engineering Data*, **67** (7), 1757–1764. DOI: 10.1021/acs.jced.1c00967. This earlier work concerns adsorption-data reproducibility, not the derivation of the Bingel–Walton equation.

MATHEMATICAL FORMULATION

$$q_e = q_{\max} \frac{1 - \exp[-(a+b)C_e]}{1 + (b/a)\exp[-(a+b)C_e]}$$

The C++ class is constructed with three positive values, $q_{\max}$, $a$ and $b$. In the original publication only $a$ and $b$ are fitted: $q_{\max}$ is supplied independently. The model gives $q_e(0) = 0$ and approaches $q_{\max}$ as $C_e \to \infty$. The initial slope is $aq_{\max}$.

Here $C_e$ denotes equilibrium concentration. When supplied in physical concentration units, both $a$ and $b$ must have the reciprocal units so that $(a+b)C_e$ is dimensionless. The published gas-phase formulation instead uses pressure and coefficients with inverse-pressure units; for relative pressure the independent variable is dimensionless. The class requires $a > 0$ and $b > 0$, which is more restrictive than the original monotonicity condition $a + b > 0$ (with $a > 0$).

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad a > 0, \quad b > 0, \quad C_e \geq 0.$$

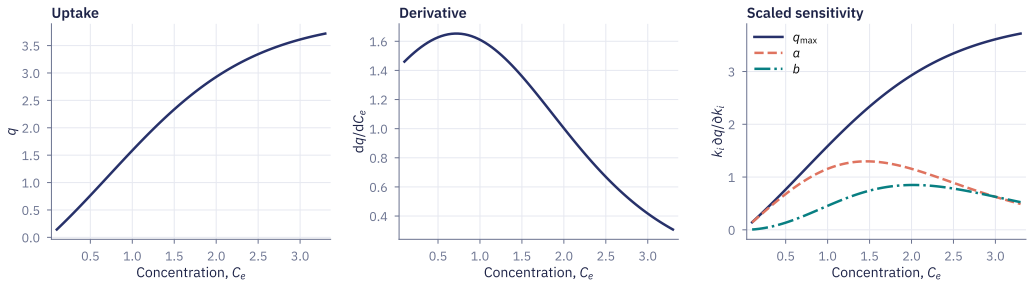



**Figure 13.** (a) Bingel–Walton uptake for $q_{\max} = 4$, $a = 0.35$, $b = 0.8$; (b) Concentration derivative of Bingel–Walton uptake for $q_{\max} = 4$, $a = 0.35$, $b = 0.8$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_a$, $\lambda_b$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::BingelWalton model{4, 0.35, 0.8};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 0.76510973
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 1.6272208
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad a > 0, \quad b > 0, \quad C_e \geq 0.$$

# 14

# Brouers–Sotolongo isotherm

THREE PARAMETERS / q_max, a, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Brouers, Sotolongo, Marquez and Pirard (2005) developed a statistical approach to adsorption on heterogeneous and microporous surfaces, linking power-law behaviour to heavy-tailed Lévy distributions. The Brouers–Sotolongo isotherm is commonly expressed as a saturating Weibull-type function, as used by the present class.

Brouers, F., Sotolongo, O., Marquez, F., and Pirard, J.-P. (2005), *Microporous and heterogeneous surface adsorption isotherms arising from Levy distributions*, *Physica A*, **349** (1–2), 271–282. DOI: 10.1016/j.physa.2004.10.032.

Ncibi, M. C., Altenor, S., Seffen, M., Brouers, F., and Gaspard, S. (2008), *Modelling single compound adsorption onto porous and non-porous sorbents using a deformed Weibull exponential isotherm*, *Chemical Engineering Journal*, **145** (2), 196–202. DOI: 10.1016/j.cej.2008.04.001. This study applies the model to phenol and methylene blue adsorption.

Brouers, F. (2014), *Statistical Foundation of Empirical Isotherms*, *Open Journal of Statistics*, **4** (9), 687–701. DOI: 10.4236/ojs.2014.49064. This paper relates the Brouers–Sotolongo and generalised forms to statistical distribution families and adsorption-energy distributions.

Brouers, F., and Marquez-Montesino, F. (2016), *Dubinin isotherms versus the Brouers–Sotolongo family isotherms: A case study*, *Adsorption Science & Technology*. DOI: 10.1177/0263617416670909. The comparison discusses connections between Dubinin-type models and the generalised Brouers–Sotolongo family; it does not make every family member identical to the three-parameter expression implemented here.

Selmi, T., Seffen, M., Celzard, A., and Fierro, V. (2020), *Effect of the adsorption pH and temperature on the parameters of the Brouers–Sotolongo models*, *Environmental Science and Pollution Research*, **27** (19), 23437–23446. DOI: 10.1007/s11356-018-3835-8. The study examines the dependence of parameters in related equilibrium and kinetic models on experimental conditions.

## MATHEMATICAL FORMULATION

$$q_e = q_{\max}\left[1 - \exp(-aC_e^n)\right]$$

The coefficients $q_{\max}$, $a$ and $n$ are positive. $q_{\max}$ is the limiting saturation loading and $n$ is a dimensionless shape exponent. If $C_e$ is a dimensional equilibrium concentration, $a$ has units of $[C_e]^{-n}$, making $aC_e^n$ dimensionless. If dimensionless concentration is used instead, it must be specified as $c^* = C_e/C_{\mathrm{ref}}$, with a stated positive reference concentration in the same units as $C_e$; the numerical value of $a$ then depends on that choice. The C++ class receives the independent-variable value directly and does not perform concentration normalisation. Its `evaluate` method includes $C_e = 0$; `derivative` requires $C_e > 0$.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad a > 0, \quad n > 0, \quad C_e \geq 0.$$

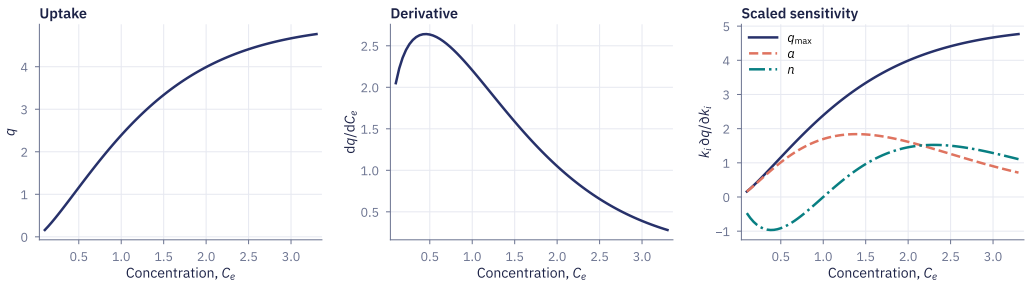



**Figure 14.** (a) Brouers–Sotolongo uptake for $q_{\max} = 5$, $a = 0.65$, $n = 1.3$; (b) Concentration derivative of Brouers–Sotolongo uptake for $q_{\max} = 5$, $a = 0.65$, $n = 1.3$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_a$, $\lambda_n$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::BrouersSotolongo model{5, 0.65, 1.3};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.9256032
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 2.4296495
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad a > 0, \quad n > 0, \quad C_e \geq 0.$$

15

# Brunauer–Emmett–Teller isotherm

THREE PARAMETERS / q_m, C, c_s / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Brunauer, S., Emmett, P. H. and Teller, E. (1938), *Adsorption of Gases in Multimolecular Layers*, *Journal of the American Chemical Society*, **60** (2), 309–319. DOI: 10.1021/ja01269a023. This is the original BET treatment of gas adsorption in successive molecular layers.

Anderson, R. B. (1946), *Modifications of the Brunauer, Emmett and Teller Equation*, *Journal of the American Chemical Society*, **68** (4), 686–691. DOI: 10.1021/ja01208a049. This paper develops an early extension of the BET treatment.

Vyawahare, P., Tun, H., Vaughn, M. W. and Chen, C.-C. (2022), *From Langmuir isotherm to Brunauer–Emmett–Teller isotherm*, *AIChE Journal*, **68** (3), e17523. DOI: 10.1002/aic.17523. The authors develop a thermodynamic, activity-based treatment of multilayer adsorption.

Hu, Q., Lan, R., He, L., Liu, H. and Pei, X. (2023), *A critical review of adsorption isotherm models for aqueous contaminants: Curve characteristics, site energy distribution and common controversies*, *Journal of Environmental Management*, **329**, 117104. DOI: 10.1016/j.jenvman.2022.117104. This review places adsorption models in the context of their assumptions and appropriate use.

Tu, J., Chien, S.-C. and Lin, L.-C. (2026), *Surface area determination of porous materials: the BET approach, its limitations, and alternatives*, *Chemical Society Reviews*, online publication, 12 October 2026. DOI: 10.1039/D6CS00693K. The review examines limitations of BET-based surface-area determination in porous materials.

The implemented three-parameter equation uses $q_m$, $C$ and $c_s$, with the reduced variable $x = C_e/c_s$. Here $C_e$ and $c_s$ must have the same concentration units, whereas $x$ and $C$ are dimensionless. The original gas-phase BET equation is conventionally expressed using relative pressure $p/p_0$. A concentration-based application must therefore specify the meaning of $c_s$; correspondence with a gas-phase saturation pressure or an aqueous activity is not automatic.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_m C c_e}{(c_s - c_e)[1 + (C-1)c_e/c_s]}$$

The three adjustable parameters are `q_m`, `C` and `c_s`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The concentration must remain strictly below the saturation concentration $c_s$; the model diverges as that boundary is approached.

The implementation requires the following admissible parameter and input domain:

$$q_m > 0, \quad C > 0, \quad c_s > 0, \quad 0 \le C_e < c_s.$$

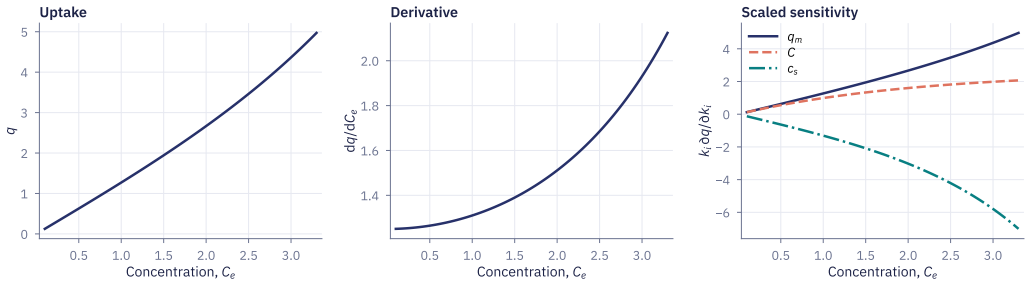



**Figure 15.** (a) Brunauer–Emmett–Teller uptake for $q_m = 5$, $C = 2$, $c_s = 8$; (b) Concentration derivative of Brunauer–Emmett–Teller uptake for $q_m = 5$, $C = 2$, $c_s = 8$; (c) Scaled sensitivities $\lambda_{q_m}$, $\lambda_C$, $\lambda_{c_s}$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::BrunauerEmmettTeller model{5, 2, 8};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.010101
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 1.2881339
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_m > 0, \quad C > 0, \quad c_s > 0, \quad 0 \le C_e < c_s.$$

# 16

# DubininAstakhov isotherm

●●● THREE PARAMETERS / q_max, D, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Dubinin, M. M. and Astakhov, V. A. (1971), *Development of the concepts of volume filling of micropores in the adsorption of gases and vapors by microporous adsorbents*, *Bulletin of the Academy of Sciences of the USSR, Division of Chemical Science*, **20** (1), 3–7. DOI: 10.1007/BF00849307. This work establishes the Dubinin–Astakhov extension of micropore volume-filling theory, generalising the exponent of the Dubinin–Radushkevich model.

Ozawa, S., Kusumi, S. and Ogino, Y. (1976), *Physical adsorption of gases at high pressure. IV. An improvement of the Dubinin–Astakhov adsorption equation*, *Journal of Colloid and Interface Science*, **56** (1), 83–91. DOI: 10.1016/0021-9797(76)90149-1. The study discusses an extension for high-pressure gas adsorption.

Stoeckli, F., Jakubov, T. and Lavanchy, A. (1994), *Water adsorption in active carbons described by the Dubinin–Astakhov equation*, *Journal of the Chemical Society, Faraday Transactions*, **90**, 783–786. DOI: 10.1039/FT9949000783. This work applies the model to water vapour uptake and type-V adsorption behaviour.

Stoeckli, F., López-Ramón, M. V. and Moreno-Castilla, C. (2001), *Adsorption of Phenolic Compounds from Aqueous Solutions, by Activated Carbons, Described by the Dubinin–Astakhov Equation*, *Langmuir*, **17** (11), 3301–3306. DOI: 10.1021/la0014407. This study applies a related potential-based treatment to aqueous adsorption, with its own definition of the adsorption potential.

Brouers, F. and Marquez-Montesino, F. (2016), *Dubinin isotherms versus the Brouers–Sotolongo family isotherms: A case study*, *Adsorption Science & Technology*, **34** (9–10), 552–564. DOI: 10.1177/0263617416670909. The authors compare the relationships and limitations of these model families.

The C++ class uses the reduced form $q_e = q_{\max} \exp[-D(-\ln x)^n]$, where the dimensionless input $x = p/p_0$ is relative vapour pressure ($p_0$ is the saturation vapour pressure at the measurement temperature). For water-vapour sorption, $x$ may be interpreted as water activity under the corresponding vapour–liquid equilibrium convention. The coefficient $D$ is dimensionless and incorporates the adsorption energy and absolute temperature for a fixed-temperature fit, conventionally $D = (RT/E)^n$ when $E$ denotes the characteristic adsorption energy. The class does not take $T$ or $E$ as separate inputs. A liquid-phase concentration must not be passed as $x$ without first defining and justifying its transformation to a dimensionless quantity.

## MATHEMATICAL FORMULATION

$$q = q_{\max} \exp\left[-D(-\ln x)^n\right]$$

The 3 adjustable parameters are: `q_max`, `D`, `n`. The independent variable is `water activity or relative pressure`.

0 < x <= 1, with q(0) defined by limit 0. Exponent n >= 1 to keep the derivative finite at x=1.

D is an effective temperature-specific coefficient, not an independently specified adsorption energy.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad D > 0, \quad n > 0, \quad n \geq 1, \quad 0 \leq x \leq 1.$$

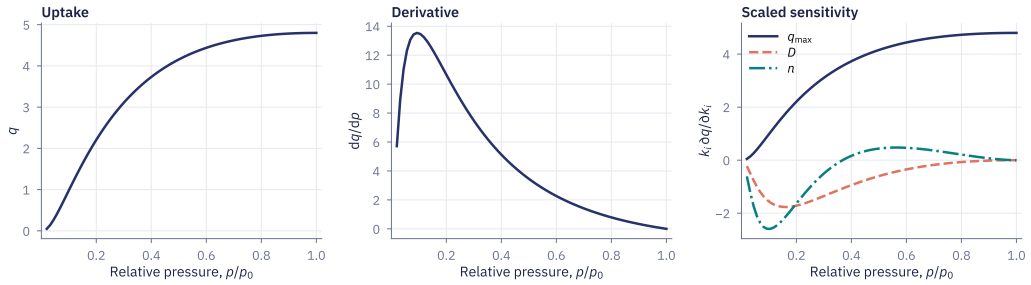



**Figure 16.** (a) Dubinin–Astakhov uptake for $q_{\max} = 4.8$, $D = 0.3$, $n = 2$; (b) Relative-pressure derivative of Dubinin–Astakhov uptake for $q_{\max} = 4.8$, $D = 0.3$, $n = 2$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_D$, $\lambda_n$ versus relative pressure.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::DubininAstakhov model{4.8, 0.3, 2};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 4.1556964
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 3.4566111
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad D > 0, \quad n > 0, \quad n \geq 1, \quad 0 \leq x \leq 1.$$

17

# FloryHuggins isotherm

THREE PARAMETERS / q_max, K, n / IMPLICIT

## ORIGINAL PUBLICATION AND SCOPE

The Flory–Huggins framework originated in the independent statistical-thermodynamic treatments of polymer solutions developed by Huggins and Flory. These papers establish the theoretical background; they should not be presented as original derivations of the particular aqueous-adsorption coverage equation used by this model.

The principal early references are:

- Huggins, M. L. (1941), *Solutions of Long Chain Compounds*, *The Journal of Chemical Physics*, **9** (5), 440. DOI: 10.1063/1.1750930.
- Flory, P. J. (1941), *Thermodynamics of High Polymer Solutions*, *The Journal of Chemical Physics*, **9** (8), 660. DOI: 10.1063/1.1750971.
- Flory, P. J. (1942), *Thermodynamics of High Polymer Solutions*, *The Journal of Chemical Physics*, **10** (1), 51–61. DOI: 10.1063/1.1723621.
- Huggins, M. L. (1942), *Some Properties of Solutions of Long-chain Compounds*, *The Journal of Physical Chemistry*, **46** (1), 151–158. DOI: 10.1021/j150415a018.

The implemented adsorption relation expresses the surface coverage as an implicit function of equilibrium concentration, with three fitted parameters: saturation loading, an equilibrium coefficient and a coverage exponent. It is not the Flory–Huggins form of vacancy solution theory, which is documented separately.

Chu, Hashim, Bashiri, Debord, Harel and Bollinger (2023), *The Flory–Huggins Isotherm and Water Contaminant Adsorption: Debunking Some Modeling Fallacies*, *Industrial & Engineering Chemistry Research*, **62** (2), 1121–1131, DOI: 10.1021/acs.iecr.2c03799, analyse the aqueous-adsorption form and demonstrate that a commonly used linearisation is inconsistent with its nonlinear equation. The C++ model evaluates the implicit nonlinear equation rather than that linearisation.

## MATHEMATICAL FORMULATION

$$\vartheta = Kx(1-\vartheta)^n, \quad q = q_{\max}\vartheta$$

The 3 adjustable parameters are: `q_max`, `K`, `n`. The independent variable is `concentration`.

Positive q_max,K,n. Root theta in [0,1) is unique and found by bisection. Assumes x is the driving concentration (cf. some sources use initial concentration C_i).

This coverage equation should not be confused with Flory–Huggins vacancy solution theory.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

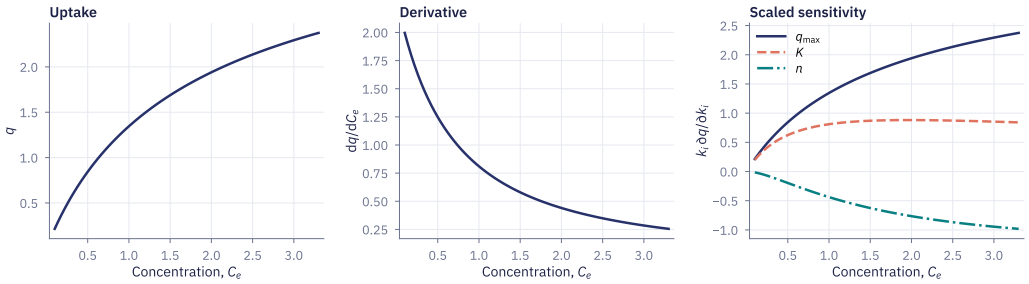



**Figure 17.** (a) Flory–Huggins uptake for $q_{max} = 4.2$, $K = 0.55$, $n = 1.4$; (b) Concentration derivative of Flory–Huggins uptake for $q_{max} = 4.2$, $K = 0.55$, $n = 1.4$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_n$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::FloryHuggins model{4.2, 0.55, 1.4};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 0.84377187
const isotherm::Real slope = isotherm::derivative(model, 0.5);   // 1.248214
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

# 18 FloryHugginsVST isotherm

THREE PARAMETERS / q_max, K_H, alpha / IMPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Vacancy solution theory (VST) treats adsorption equilibrium using an adsorbed phase containing adsorbate molecules and hypothetical vacancies. Suwanayuen and Danner established this approach for gas adsorption. Cochran, Kabel and Danner subsequently introduced the Flory–Huggins activity-coefficient formulation (FH-VST), with three adjustable quantities: a Henry-law constant, a limiting adsorption capacity and a gas–solid interaction parameter.

The principal references are:

- Suwanayuen, S. and Danner, R. P. (1980), *A Gas Adsorption Isotherm Equation Based on Vacancy Solution Theory*, *AIChE Journal*, **26** (1), 68–76. DOI: 10.1002/aic.690260112.
- Cochran, T. W., Kabel, R. L. and Danner, R. P. (1985), *Vacancy Solution Theory of Adsorption Using Flory-Huggins Activity Coefficient Equations*, *AIChE Journal*, **31** (2), 268–277. DOI: 10.1002/aic.690310214.
- Cochran, T. W., Kabel, R. L. and Danner, R. P. (1985), *The Vacancy Solution Model of Adsorption—Improvements and Recommendations*, *AIChE Journal*, **31** (12), 2075–2081. DOI: 10.1002/aic.690311217.

An application to pure and binary adsorption of carbon dioxide, methane and nitrogen on ZSM-5 is reported in *Adsorption of Carbon Dioxide, Methane and Nitrogen: Pure and Binary Mixture Adsorption for ZSM-5 with SiO2/Al2O3 Ratio of 280*, *Separation and Purification Technology*, **33** (2), 199–210 (2003), DOI: 10.1016/S1383-5866(02)00078-3. That study compares FH-VST with other adsorption models; it does not introduce FH-VST.

The original FH-VST formulation uses gas pressure as its driving variable. The C++ class evaluates the corresponding implicit three-parameter expression using the supplied concentration argument. A physical application must use dimensionally consistent units and must not assume that gas pressure and solution concentration are interchangeable without an appropriate thermodynamic relationship. This model is distinct from the surface-coverage FloryHuggins class.

## MATHEMATICAL FORMULATION

$$x = \frac{q_{\max}}{K_H}\frac{\vartheta}{1-\vartheta}\exp\left(\frac{\alpha^2\vartheta}{1+\alpha\vartheta}\right)$$

The 3 adjustable parameters are: `q_max`, `K_H`, `alpha`. The independent variable is `concentration`.

Positive q_max, K_H and alpha. The implicit forward relation is monotone for alpha>0; bisection returns the unique theta in (0,1).

FH-VST is not equivalent to the coverage-only FloryHuggins model; the activity coefficient is part of vacancy solution theory.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K_H > 0, \quad \alpha > 0, \quad C_e \geq 0.$$

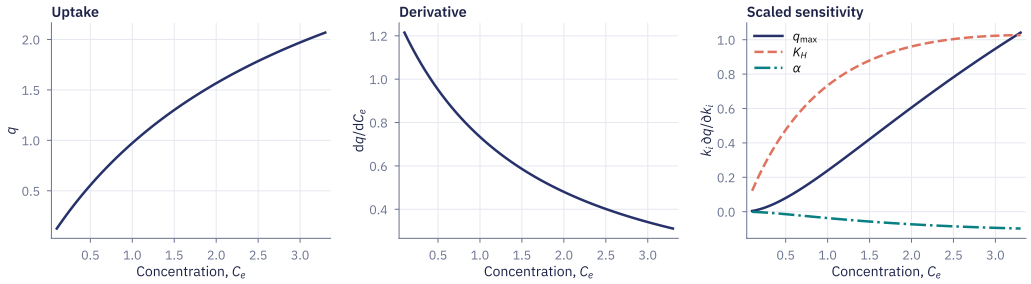



**Figure 18.** (a) Flory–Huggins VST uptake for $q_{\max} = 4.2$, $K_H = 1.3$, $\alpha = 0.35$; (b) Concentration derivative of Flory–Huggins VST uptake for $q_{\max} = 4.2$, $K_H = 1.3$, $\alpha = 0.35$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_{K_H}$, $\lambda_\alpha$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::FloryHugginsVST model{4.2, 1.3, 0.35};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);   // 0.55538254
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 0.95166432
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K_H > 0, \quad \alpha > 0, \quad C_e \geq 0.$$

19

# Fowler–Guggenheim isotherm

THREE PARAMETERS / q_max, K, W / IMPLICIT / TEMPERATURE

## ORIGINAL PUBLICATION AND SCOPE

The Fowler–Guggenheim model extends the Langmuir description of a monolayer by allowing interactions between neighbouring adsorbed molecules. Its statistical-mechanical background is presented by Fowler and Guggenheim (1939), particularly in the discussion of surface layers. Fowler's earlier 1935 paper derived the non-interacting Langmuir isotherm from statistical mechanics and is a theoretical precursor, rather than the source of the lateral-interaction term.

- Fowler, R. H. (1935), *A Statistical Derivation of Langmuir's Adsorption Isotherm*, *Mathematical Proceedings of the Cambridge Philosophical Society*, **31** (2), 260–264. DOI: 10.1017/S0305004100013359.
- Fowler, R. H., and Guggenheim, E. A. (1939), *Statistical Thermodynamics*, Cambridge University Press, discussion of surface layers, pp. 421–451.
- Barbero, G., Evangelista, L. R., and Lelidis, I. (2021), *Effective adsorption energy and generalization of the Frumkin-Fowler-Guggenheim isotherm*, *Journal of Molecular Liquids*, **327**, 114795. DOI: 10.1016/j.molliq.2020.114795.
- Khunpetch, P., Majee, A., Ruixuan, H., and Podgornik, R. (2023), *Curvature effects in interfacial acidity of amphiphilic vesicles*, *Physical Review E*, **108**, 024402. DOI: 10.1103/PhysRevE.108.024402.

The later studies concern generalised lateral-interaction models and charge regulation, respectively; they do not independently validate the precise parameter convention used here. The C++ implementation employs an implicit coverage equation with interaction parameter $W \geq 0$ and a temperature input in kelvin, as specified below. The sign and normalisation of interaction parameters must be checked before comparing fitted coefficients across published formulations.

## MATHEMATICAL FORMULATION

$$Kc_e = \frac{\vartheta}{1-\vartheta}\exp\left(\frac{W\vartheta}{RT}\right), \qquad q_e = q_{\max}\vartheta$$

The three adjustable parameters are `q_max` (capacity), `K` (affinity) and `W` (interaction energy). Capacity and affinity must be strictly positive; interaction energy must be non-negative. All values must be finite. Equilibrium concentration must be non-negative, while the absolute temperature is a separate, strictly positive input **always in kelvin (K)**. By default, the gas constant is 8.31446261815324 J mol$^{-1}$ K$^{-1}$, so the interaction energy uses joules per mole. The constructor can select kJ/(mol K) or L atm/(mol K) provided that the interaction energy is expressed in the corresponding units.

A bracketed, monotonic log-activity solver determines the surface coverage $0 \leq \vartheta < 1$; derivatives use implicit differentiation, rather than numerical differentiation of iterates.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad W \geq 0, \quad C_e \geq 0, \quad T > 0\ \mathrm{K}.$$

The temperature input $T$ **must always be supplied in kelvin (K)**. The interaction energy and selected value of $R$ must use compatible units.

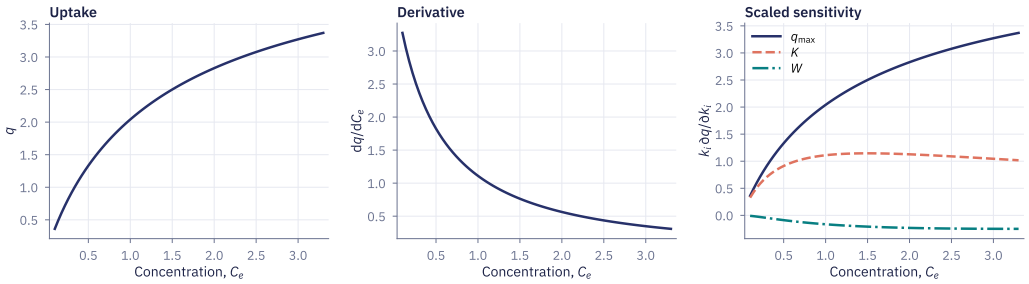



**Figure 19.** (a) Fowler–Guggenheim uptake for $q_{max} = 5$, $K = 0.8$, $W = 900$ at $T = 298.15\,\mathrm{K}$; (b) Concentration derivative of Fowler–Guggenheim uptake for $q_{max} = 5$, $K = 0.8$, $W = 900$ at $T = 298.15\,\mathrm{K}$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_W$ versus equilibrium concentration at $T = 298.15\,\mathrm{K}$.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::FowlerGuggenheim model{5, 0.8, 900};
const isotherm::Real temperature_k = 298.15;                                    //
Always kelvin.
const isotherm::Real uptake = isotherm::evaluate(model, 0.8, temperature_k);    //
1.7982749
const isotherm::Real slope = isotherm::derivative(model, 0.8, temperature_k);   //
1.3283299

// Equivalent energy representation: W=0.9 kJ/mol.
const isotherm::models::FowlerGuggenheim model_kj{
    5, 0.8, 0.9, isotherm::gas_constant_unit::kilojoule_per_mole_kelvin};
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e; T) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad W \geq 0, \quad C_e \geq 0, \quad T > 0\ \mathrm{K}.$$

# 20

# Frenkel–Halsey–Hill isotherm

●●● THREE PARAMETERS / q_m, C_s, s / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

The Frenkel–Halsey–Hill (FHH) theory describes multilayer physical adsorption in terms of an adsorption potential that changes with film thickness. Its classical formulation commonly uses relative vapour pressure. The Isotherm++ class implements the **concentration-based three-parameter convention** specified below; this should not be interpreted as a universal thermodynamic substitution of liquid-phase concentration for relative vapour pressure.

The historical and subsequent sources include:

- Frenkel, J. (1946), *Kinetic Theory of Liquids*, Oxford University Press. A theoretical precursor, not the source of this exact C++ parametrisation.
- Halsey, G. (1948), *Physical Adsorption on Non-Uniform Surfaces*, *The Journal of Chemical Physics*, **16** (10), 931–937. DOI: 10.1063/1.1746689.

- Hill, T. L. (1952), a contribution in *Advances in Catalysis*, volume 4, Academic Press. Specific chapter metadata have not been independently verified for this documentation.
- Pierce, C. (1960), *The Frenkel–Halsey–Hill Adsorption Isotherm and Capillary Condensation*, *The Journal of Physical Chemistry*, **64**, 1184–1187. DOI: 10.1021/j100838a018.
- Pfeifer, P., Obert, M., and Cole, M. W. (1989), *Fractal BET and FHH Theories of Adsorption: A Comparative Study*, *Proceedings of the Royal Society of London A*, **423**, 169–188. DOI: 10.1098/rspa.1989.0049.
- *A Review on the Adsorption Isotherms and Design Calculations for the Optimization of Adsorbent Mass and Contact Time* (2023), *ACS Omega*. DOI: 10.1021/acsomega.2c08155. This review presents the concentration-based FHH relation with an additional coefficient $k$.
- Chu, K. H., and co-authors (2023), *The Halsey Isotherm for Water Contaminant Adsorption Is Fake*, *Separation and Purification Technology*, **313**, 123500. DOI: 10.1016/j.seppur.2023.123500. This paper cautions against uncritical use of certain concentration-based equations marketed as Halsey models. It does not establish the physical validity of every concentration-form FHH parametrisation.


## MATHEMATICAL FORMULATION

The equation implemented by the C++20 class is

$$q_e(C_e) = q_m\left[\ln\left(\frac{C_s}{C_e}\right)\right]^{-1/s}.$$

The parameters are $q_m > 0$ (uptake scaling coefficient), $C_s > 0$ (reference saturation concentration), and $s > 0$ (dimensionless exponent). The independent variable is equilibrium concentration $C_e$, expressed in the **same units** as $C_s$. Temperature is not a separate input to this implementation.

The admissible mathematical domain is

$$q_m > 0, \quad C_s > 0, \quad s > 0, \quad 0 < C_e < C_s.$$

A commonly reported concentration-based FHH relation includes an additional coefficient $k$:

$$-\ln(C_e/C_s) = \frac{k}{(q_e/q_0)^s}, \qquad k > 0.$$

Solving for uptake gives

$$q_e = q_0\, k^{1/s}[\ln(C_s/C_e)]^{-1/s}.$$

The implemented three-parameter form absorbs $k^{1/s}$ into $q_m = q_0 k^{1/s}$. Consequently, the stored $q_m$ is an **effective scale coefficient** and should not be presented as an independently measured monolayer capacity without additional evidence. This convention is not the rational-power Fritz–Schlünder V model.

The analytical concentration derivative is

$$\frac{dq_e}{dC_e} = \frac{q_e(C_e)}{sC_e\ln(C_s/C_e)} > 0.$$

For this parametrisation, uptake tends to zero as $C_e \to 0^+$ and diverges as $C_e \to C_s^-$. It has no finite saturation capacity at the upper boundary. Although $C_s$ can be stored as a fitted coefficient, any physical interpretation as a saturation concentration requires appropriate independent experimental evidence.

The parameters are not affected by the amplitude redundancy of the previous $(q_m, A, B)$ pressure-based implementation, because $C_s$ appears *inside* the logarithm. This does not guarantee practical identifiability from a limited or noisy dataset.

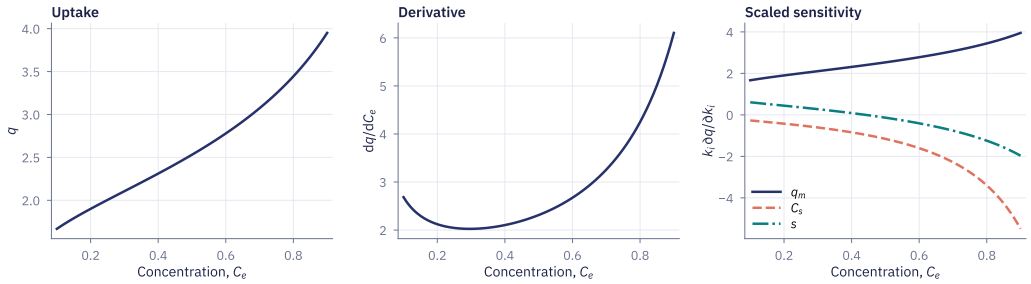



**Figure 20.** (a) Uptake versus $C_e$ for $q_m = 2.4$, $C_s = 1.2$, and $s = 2.5$; (b) Analytical derivative with respect to $C_e$ for the same parameters; (c) Scaled sensitivities $\lambda_{q_m}$, $\lambda_{C_s}$, $\lambda_s$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

// q_m = 2.4, C_s = 1.2, s = 2.5; C_e has the units of C_s.
const isotherm::models::FrenkelHalseyHill model{2.4, 1.2, 2.5};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);   // 2.5311331095
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 2.3129398015
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_m > 0, \quad C_s > 0, \quad s > 0, \quad 0 < C_e < C_s.$$

# 21

# Fritz–Schlunder isotherm

THREE PARAMETERS / Q, K, m / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Fritz–Schlunder models are empirical rational expressions with adjustable powers that provide flexibility for representing adsorption-equilibrium data. Their bibliographical foundation is the work of Fritz and Schlünder (1974) on simultaneous adsorption of organic solutes from dilute aqueous solutions onto activated carbon. That paper establishes the model lineage; the specific three-parameter equation implemented below must be identified from its mathematical expression rather than inferred from the model name alone.


- Fritz, W., and Schlünder, E.-U. (1974), *Simultaneous adsorption equilibria of organic solutes in dilute aqueous solutions on activated carbon*, *Chemical Engineering Science*, **29** (5), 1279–1282. DOI: 10.1016/0009-2509(74)80128-4.
- Kumar, K. V., Monteiro de Castro, M. C., Martinez-Escandell, M., Molina-Sabio, M., and Rodriguez-Reinoso, F. (2010), *Adsorption on Heterogeneous Surfaces: Site Energy Distribution Functions from Fritz–Schlüender Isotherms*, *ChemPhysChem*, **11** (12), 2555–2560. DOI: 10.1002/cphc.201000307.
- Shukor, M. Y., and Sha'arani, S. A. W. (2023), *Adsorption of Sodium Dodecyl Sulfate onto Activated Coconut Shell-Based Adsorbent: Isothermal Remodelling*, *Journal of Environmental Microbiology and Toxicology*, **11** (1), 52–61. DOI: 10.54987/jemat.v11i1.884.

The 2010 study examines distributions of adsorption-site energies associated with Fritz–Schlüender representations. The 2023 comparison includes a three-parameter variant, without establishing that it is preferable for other adsorbates or materials. The implemented form is also algebraically related to Redlich–Peterson-type rational equations; that relationship should not be interpreted as proof of a distinct adsorption mechanism.

## MATHEMATICAL FORMULATION

$$q_e = \frac{QKc_e}{1 + Qc_e^m}$$

The three adjustable parameters are `Q`, `K` and `m`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$Q > 0, \quad K > 0, \quad m > 0, \quad C_e \geq 0.$$

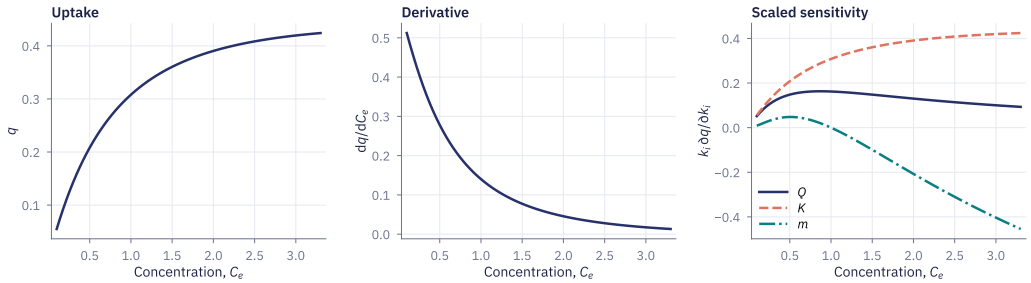



**Figure 21.** (a) Fritz–Schlunder uptake for $Q = 0.9$, $K = 0.65$, $m = 1.15$; (b) Concentration derivative of Fritz–Schlunder uptake for $Q = 0.9$, $K = 0.65$, $m = 1.15$; (c) Scaled sensitivities $\lambda_Q$, $\lambda_K$, $\lambda_m$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::FritzSchlunder model{0.9, 0.65, 1.15};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 0.2758947
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 0.18207201
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$Q > 0, \quad K > 0, \quad m > 0, \quad C_e \geq 0.$$

22

# Frumkin isotherm

THREE PARAMETERS / q_max, K, g / IMPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Frumkin's adsorption treatment accounts for lateral interactions between adsorbed molecules in a surface layer. The historical starting point is Frumkin's 1925 work on fatty-acid films and the equation of state of the surface layer. A second paper published in the same volume addresses adsorption and partitioning; it provides related background rather than an independent derivation of the precise parameter convention implemented here.

- Frumkin, A. (1925), *Die Kapillarkurve der höheren Fettsäuren und die Zustandsgleichung der Oberflächenschicht*, *Zeitschrift für Physikalische Chemie*, **116U** (1), 466–484. DOI: 10.1515/zpch-1925-11629.
- Frumkin, A. (1925), *Einige Bemerkungen zur Theorie der Adsorption und Verteilung*, *Zeitschrift für Physikalische Chemie*, **116U** (1), 501–503. DOI: 10.1515/zpch-1925-11632.
- Mousavi, N. S., Javadi, A., Kovalchuk, V. I., Aksenenko, E. V., Loglio, G., Vollhardt, D., Schneck, E., and Miller, R. (2025), *Frumkin's adsorption model – a successful approach for understanding surfactant adsorption layers*, *Zeitschrift für Physikalische Chemie*, **239** (10), 1705–1724. DOI: 10.1515/zpch-2025-0017.

The 2025 study concerns adsorption at liquid interfaces and illustrates the continuing use of the lateral-interaction description. The C++ model uses an implicit surface-coverage relation and a dimensionless interaction coefficient g; coefficients from other sign or scaling conventions should not be compared without conversion.

## MATHEMATICAL FORMULATION

$$Kx = \frac{\vartheta}{1-\vartheta}\exp(-g\vartheta), \quad q = q_{\max}\vartheta$$

The 3 adjustable parameters are: q_max, K, g. The independent variable is concentration.

Positive q_max,K. Dimensionless g in [0,4); this subcritical range guarantees strictly increasing inverse and a unique equilibrium branch.

g >= 4 may permit multiple occupancy branches and is excluded rather than silently choosing one.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad g > 0, \quad g < 4, \quad C_e \geq 0.$$

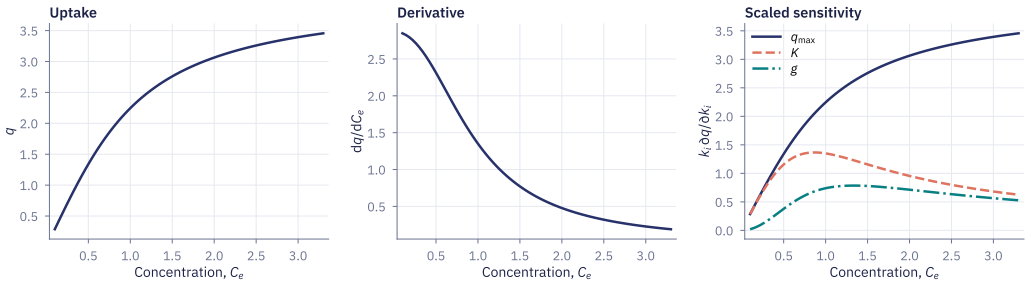



**Figure 22.** (a) Frumkin uptake for $q_{\max} = 4.1$, $K = 0.7$, $g = 1$; (b) Concentration derivative of Frumkin uptake for $q_{\max} = 4.1$, $K = 0.7$, $g = 1$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_K$, $\lambda_g$ versus equilibrium concentration.

C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Frumkin model{4.1, 0.7, 1};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);   // 1.3395007
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 2.3124122
```

NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad g > 0, \quad g < 4, \quad C_e \geq 0.$$

# 23

# GAB isotherm

THREE PARAMETERS / q_m, C, K / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

The Guggenheim–Anderson–de Boer (GAB) isotherm extends the BET multilayer description by introducing an additional factor for the sorption behaviour of layers beyond the first. The name recognises related independent developments by Anderson (1946), de Boer (1953), and Guggenheim (1966). It is widely used for equilibrium water-sorption isotherms in foods and other hygroscopic materials.

- Anderson, R. B. (1946), *Modifications of the Brunauer, Emmett and Teller Equation*, *Journal of the American Chemical Society*, **68** (4), 686–691. DOI: 10.1021/ja01208a049.
- de Boer, J. H. (1953), *The Dynamical Character of Adsorption*, Clarendon Press, Oxford.
- Guggenheim, E. A. (1966), *Applications of Statistical Mechanics*, Clarendon Press, Oxford.
- Roos, Y. H. (2024), *Water sorption modeling and monolayer of biological and food materials*, *LWT*, **201**, 116271. DOI: 10.1016/j.lwt.2024.116271.

The 2024 review discusses the interpretation of GAB parameters, including the estimated monolayer moisture content, and cautions that a physically distinct monolayer is not necessarily established by a good fit to complex biological materials. In the implementation below, the independent variable is water activity $a_w$, a dimensionless quantity defined as the equilibrium water-vapour partial pressure divided by its saturation pressure at the same temperature. The coefficient `C` is a GAB parameter, not a concentration.

MATHEMATICAL FORMULATION

$$q = \frac{q_m C K x}{(1 - Kx)(1 - Kx + CKx)}$$

The 3 adjustable parameters are: `q_m`, `C`, `K`. The independent variable is `water activity or relative pressure`.

0 <= x <= 1 water activity; 0 < K < 1 and C >= 1. These constraints ensure the implemented branch has no pole on [0,1] and is monotonic.

Parameters and uptake units are specific to moisture sorption, not directly exchangeable with gas adsorption concentration models.

The implementation requires the following admissible parameter and input domain:

$$q_m > 0, \quad C > 0, \quad K > 0, \quad C \geq 1, \quad K < 1, \quad 0 \leq a_w \leq 1.$$

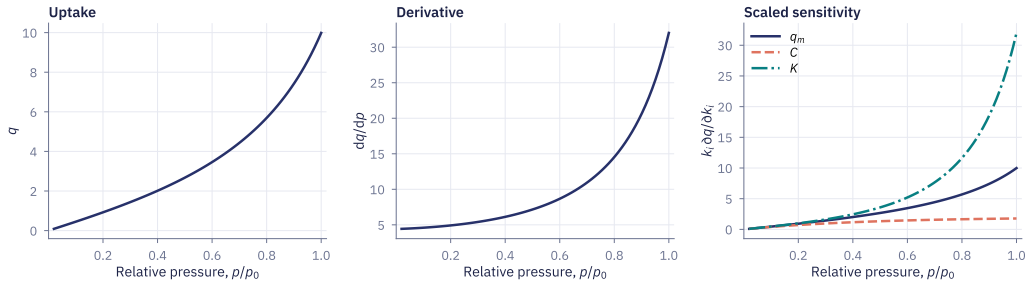



**Figure 23.** (a) GAB equilibrium moisture for $q_m = 3.4$, $C = 1.8$, $K = 0.72$; (b) Activity derivative of GAB equilibrium moisture for $q_m = 3.4$, $C = 1.8$, $K = 0.72$; (c) Scaled sensitivities $\lambda_{q_m}$, $\lambda_C$, $\lambda_K$ versus water activity.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::GAB model{3.4, 1.8, 0.72};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 2.6727484
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 7.1570725
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(x) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_m > 0, \quad C > 0, \quad K > 0, \quad C \geq 1, \quad K < 1, \quad 0 \leq a_w \leq 1.$$

# 24

# Henderson isotherm

THREE PARAMETERS / A, B, a_w / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Henderson, S. M. (1952), *A basic concept of equilibrium moisture*, *Agricultural Engineering*, **33** (1), 29–32. This is the primary publication of the Henderson moisture-sorption relation, as recorded in the FAO bibliography and the CiNii catalogue. No DOI has been verified for this article.

The original Henderson relation describes the equilibrium moisture content of hygroscopic materials, particularly agricultural products, as a function of water activity and temperature. It has **two empirical material coefficients**. Water activity is the dimensionless vapour-pressure ratio $a_w = p_w / p_w^{\mathrm{sat}}$ at the same temperature. The formulation used here requires absolute temperature in kelvin.

Later studies have compared the original relation with temperature-dependent modifications and other moisture-sorption models. Barrozo, Bego and Oliveira (2008), *Equilibrium moisture of kale seed (Brassica oleracea var. acephala): an alternative method for choice of the best model*, *International Journal of Food Science and Technology*, **43** (10), 1844–1849, DOI 10.1111/j.1365-2621.2008.01728.x, discuss how model selection and parameter behaviour depend on the moisture data. Iji, Inyang and Etuk (2025), *American Journal of Food Science and Technology*, **13** (2), 27–37, DOI 10.12691/ajfst-13-2-2, review empirical moisture-sorption models and their range-dependent applicability. These references do not establish that the Henderson equation is the best-fitting model for every material or water-activity interval.

### MATHEMATICAL FORMULATION

$$1 - a_w = \exp(-ATq_e^B), \qquad q_e(T; A, B, a_w) = \left(\frac{-\ln(1-a_w)}{AT}\right)^{1/B}.$$

The mathematical domain is

$$A > 0, \quad B > 0, \quad 0 \le a_w < 1, \quad T > 0 \text{ K}.$$

Here $a_w$ is the dimensionless water activity, and $T$ **must always be provided in kelvin (K)**. For consistent units, $A$ has units inverse to the product of temperature and equilibrium moisture content raised to the power $B$. The derivative produced by this class is $\mathrm{d}q_e/\mathrm{d}T$ at fixed $A$, $B$ and $a_w$, not a concentration derivative.

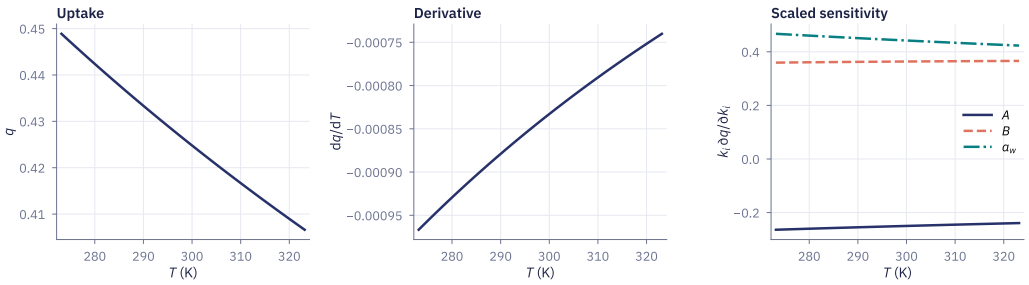



**Figure 24.** (a) Henderson equilibrium moisture for $A = 0.015$, $B = 1.7$, $a_w = 0.65$; (b) Temperature derivative of Henderson equilibrium moisture for $A = 0.015$, $B = 1.7$, $a_w = 0.65$; (c) Scaled sensitivities $\lambda_A$, $\lambda_B$, $\lambda_{a_w}$ versus absolute temperature.

### C++20 USAGE

```
#include <isotherm/isotherm.hpp>

const isotherm::models::Henderson model{0.015, 1.7, 0.65};                // A, B,
a_w
const isotherm::Real temperature_k = 298.15;                              // Kelvin
const isotherm::Real moisture = isotherm::evaluate(model, temperature_k);  //
0.42634249
const isotherm::Real slope = isotherm::derivative(model, temperature_k);   //
-0.00084115277
```

### NUMERICAL SENSITIVITY

For the three constructor parameters $k_i \in \{A, B, a_w\}$, scaled numerical sensitivity is

$$\lambda_i(T) = k_i \frac{\partial q_e(T)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$A > 0, \quad B > 0, \quad 0 \le a_w < 1, \quad T > 0 \text{ K}.$$

# 25

# Hill isotherm

●●● THREE PARAMETERS / q_max, K_d, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

The familiar Hill-type saturation law originates in A. V. Hill's 1910 analysis of cooperative oxygen binding by haemoglobin, not in a 1946 paper on gas adsorption. The relevant communication, *The possible effects of the aggregation of the molecules of haemoglobin on its dissociation curves*, appeared in *The Journal of Physiology*, **40** (supplement), iv–vii, within the 1910 Physiological Society proceedings. The linked DOI identifies the proceedings containing the communication.

A different researcher, T. L. Hill, published *Statistical Mechanics of Multimolecular Adsorption. II. Localized and Mobile Adsorption and Absorption*, *The Journal of Chemical Physics*, **14** (7), 441–453 (1946), DOI 10.1063/1.1724166. This is a foundational statistical-mechanical study of adsorption, but it is not the origin of the 1910 Hill binding equation.

As a contemporary adsorption application, Li et al. (2024), *A novel magnetic Ti-MOF/chitosan composite for efficient adsorption of Pb(II) from aqueous solutions: Synthesis and investigation*, *International Journal of Biological Macromolecules*, **258**, 129170, DOI 10.1016/j.ijbiomac.2023.129170, fitted the Hill model to equilibrium data and reported a model-derived capacity of 944.9 mg/g. Such a fit does not, by itself, establish molecular-scale cooperative interactions. The implemented $n$ is an empirical exponent whose interpretation depends on independent physical evidence.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\max} c_e^n}{K_d + c_e^n}$$

The three adjustable parameters are `q_max`, `K_d` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K_d > 0, \quad n > 0, \quad C_e \geq 0.$$

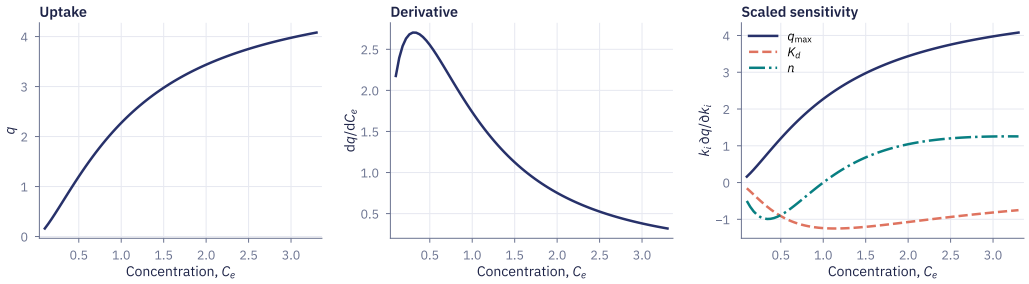



**Figure 25.** (a) Hill uptake for $q_{\max} = 5$, $K_d = 1.2$, $n = 1.4$; (b) Concentration derivative of Hill uptake for $q_{\max} = 5$, $K_d = 1.2$, $n = 1.4$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_{K_d}$, $\lambda_n$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Hill model{5, 1.2, 1.4};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.8939085
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 2.0589286
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K_d > 0, \quad n > 0, \quad C_e \geq 0.$$

26

# Hill–de Boer isotherm

THREE PARAMETERS / q_max, K, W / IMPLICIT / TEMPERATURE

## ORIGINAL PUBLICATION AND SCOPE

The Hill–de Boer isotherm describes an adsorbed phase with molecular mobility and lateral interactions. Its theoretical antecedent is T. L. Hill (1946), *Statistical Mechanics of Multimolecular Adsorption. II. Localized and Mobile Adsorption and Absorption*, *The Journal of Chemical Physics*, **14** (7), 441–453, DOI 10.1063/1.1724166. Hill considered a mobile adsorbed first layer through a two-dimensional equation of state. J. H. de Boer subsequently provided a wider treatment in *The Dynamical Character of Adsorption* (Clarendon Press, Oxford, 1953; bibliographic record).

A recent comparison by Chu, Hashim, Tran and Bollinger (2023), *S-shaped adsorption isotherms modeled by the Frumkin–Fowler–Guggenheim and Hill–de Boer equations*, *Monatshefte für Chemie*, **154**, 1127–1135, DOI 10.1007/s00706-023-03116-w, found that similar curve-fitting performance can conceal substantial differences in estimated saturation capacities. De Smedt et al. (2025) also applied the Hill–de Boer equation to CO2 adsorption by pelletised activated carbon in *Biomass and Bioenergy*, **107699**, DOI 10.1016/j.biombioe.2025.107699. These studies illustrate applications of the model without constituting validation of every fitted parameter as an independently measurable physical quantity.

## MATHEMATICAL FORMULATION

$$Kc_e = \frac{\vartheta}{1-\vartheta}\exp\left(\frac{\vartheta}{1-\vartheta} - \frac{W\vartheta}{RT}\right), \qquad q_e = q_{\max}\vartheta$$

The three adjustable parameters are `q_max` (capacity), `K` (affinity) and `W` (interaction energy). Capacity and affinity must be strictly positive; interaction energy must be non-negative. All values must be finite. Equilibrium concentration must be non-negative, while the absolute temperature is a separate, strictly positive input **always in kelvin (K)**. By default, the gas constant is 8.31446261815324 J mol$^{-1}$ K$^{-1}$, so the interaction energy uses joules per mole. The constructor can select kJ/(mol K) or L atm/(mol K) provided that the interaction energy is expressed in the corresponding units.

A bracketed, monotonic log-activity solver determines the surface coverage $0 \le \vartheta < 1$; derivatives use implicit differentiation, rather than numerical differentiation of iterates.

The accepted branch requires $W/(RT) \le 4$, a conservative sufficient condition for a unique root. The library deliberately rejects stronger interaction branches rather than selecting one equilibrium root arbitrarily.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad W \ge 0, \quad W/(RT) \le 4, \quad C_e \ge 0, \quad T > 0\ \mathrm{K}.$$

The temperature input $T$ **must always be supplied in kelvin (K)**. The interaction energy and selected value of $R$ must use compatible units.

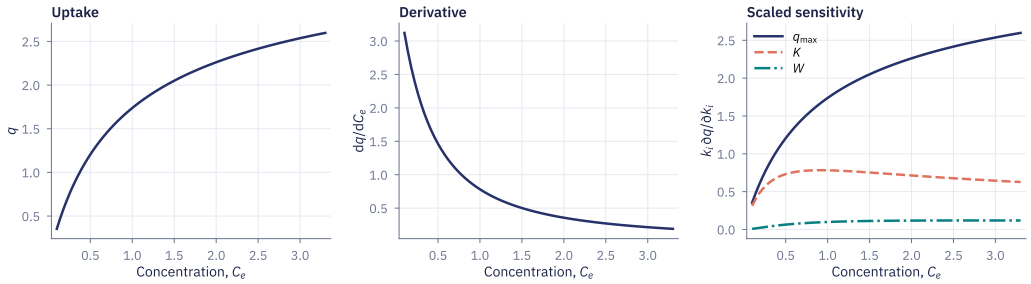



**Figure 26.** (a) Hill–de Boer uptake for $q_{max} = 5$, $K = 0.8$, $W = 900$ at $T = 298.15\,\mathrm{K}$; (b) Concentration derivative of Hill–de Boer uptake for $q_{max} = 5$, $K = 0.8$, $W = 900$ at $T = 298.15\,\mathrm{K}$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_W$ versus equilibrium concentration at $T = 298.15\,\mathrm{K}$.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::HillDeBoer model{5, 0.8, 900};
const isotherm::Real temperature_k = 298.15;                                    //
Always kelvin.
const isotherm::Real uptake = isotherm::evaluate(model, 0.8, temperature_k);    //
1.5633426
const isotherm::Real slope = isotherm::derivative(model, 0.8, temperature_k);   //
0.97551664

// Equivalent energy representation: W=0.9 kJ/mol.
const isotherm::models::HillDeBoer model_kj{
    5, 0.8, 0.9, isotherm::gas_constant_unit::kilojoule_per_mole_kelvin};
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e; T) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad W \geq 0, \quad W/(RT) \leq 4, \quad C_e \geq 0, \quad T > 0\ \mathrm{K}.$$

# 27

# Holl–Krich isotherm

●●● THREE PARAMETERS / q_max, K, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Holl–Krich is described in the adsorption literature as a three-parameter, Langmuir-type relation with a concentration exponent. An example is Aniagor and Menkiti (2020), who include the model in a comparative study of Cu(II) and Pb(II) adsorption on activated carbon.

Aniagor, C. O., and Menkiti, M. C. (2020), *Relational description of an adsorption system based on isotherm, adsorption density, adsorption potential, hopping number and surface coverage*, *Sigma Journal of Engineering and Natural Sciences*, **38** (3), 1073–1098. Full text.

The equation used here is algebraically identical to the Langmuir–Freundlich form with parameters $q_{max}$, $K$ and $n$ and is also equivalent to a reparametrised Sips isotherm. In particular, $n = 1$ gives the Langmuir equation. These names therefore do not denote distinct mathematical response functions in this parametrisation. A single, verifiable primary publication establishing the *Holl–Krich* name and its priority could not be established from the available sources; the 2020 publication is cited as a documented use, not as the origin.

### MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\max} K c_e^n}{1 + K c_e^n}$$

The three adjustable parameters are `q_max`, `K` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

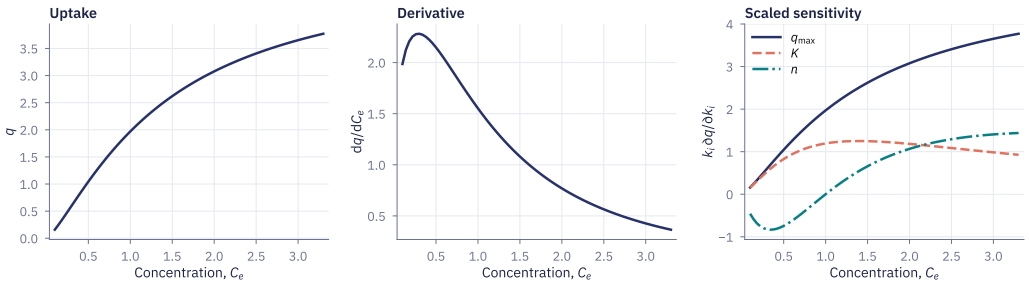



**Figure 27.** (a) Holl–Krich uptake for $q_{\max} = 5$, $K = 0.65$, $n = 1.3$; (b) Concentration derivative of Holl–Krich uptake for $q_{\max} = 5$, $K = 0.65$, $n = 1.3$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_K$, $\lambda_n$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::HollKrich model{5, 0.65, 1.3};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.6360077
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 1.7886431
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

# 28

# Jossens isotherm

THREE PARAMETERS / H, F, p / IMPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Jossens, L., Prausnitz, J. M., Fritz, W., Schlünder, E. U., and Myers, A. L. (1978), *Thermodynamics of multi-solute adsorption from dilute aqueous solutions*, *Chemical Engineering Science*, **33** (8), 1097–1106. DOI: 10.1016/0009-2509(78)85015-5. The paper introduces an adsorption relation in its treatment of adsorption from dilute solutions.

Chong, B. Y. K., Hashim, M. A., and Chu, K. H. (2024), *The Jossens isotherm: A critical look at water contaminant adsorption modeling*, *Separation Science and Technology*, **59** (10–14), 1147–1157. DOI: 10.1080/01496395.2024.2368236. The authors distinguish the original, loading-implicit Jossens relation from the Redlich–Peterson isotherm and discuss parameter identification.

The original Jossens relation is **loading-implicit**: the equilibrium concentration is an explicit function of the uptake. It must not be confused with the Redlich–Peterson form $q_e = AC_e/(1 + BC_e^n)$, which is frequently reported under the Jossens name (Chong, Hashim and Chu, 2024). This class implements the loading-implicit relation; earlier versions of the library exposed the Redlich–Peterson form under this name.

## MATHEMATICAL FORMULATION

$$C_e = \frac{q_e}{H}\exp\left(Fq_e^p\right)$$

The three adjustable parameters are `H` (Henry coefficient, uptake per concentration), `F` and `p`. All must be strictly positive and finite. Because

$$\ln C_e = \ln q_e - \ln H + Fq_e^p$$

is strictly increasing in $q_e$, every positive concentration has a unique uptake, bracketed by $0 < q_e < HC_e$; the C++ class finds it by bisection to full double resolution. An equivalent explicit form is $q_e = [W(Fp\,(HC_e)^p)/(Fp)]^{1/p}$, where $W$ is the principal Lambert function; it is used only to verify the implementation.

The analytical concentration derivative follows from implicit differentiation:

$$\frac{\mathrm{d}q_e}{\mathrm{d}C_e} = \frac{q_e}{C_e(1 + Fp\,q_e^p)}, \qquad \lim_{C_e \to 0^+} \frac{\mathrm{d}q_e}{\mathrm{d}C_e} = H.$$

The implementation returns the Henry-law limit $H$ at $C_e = 0$.

The implementation requires the following admissible parameter and input domain:

$$H > 0, \quad F > 0, \quad p > 0, \quad C_e \geq 0.$$

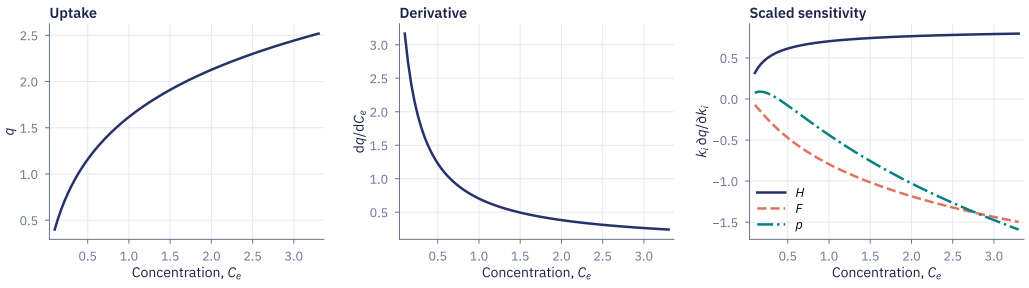



**Figure 28.** (a) Jossens uptake for $H = 5$, $F = 0.65$, $p = 1.15$; (b) Concentration derivative of Jossens uptake for $H = 5$, $F = 0.65$, $p = 1.15$; (c) Scaled sensitivities $\lambda_H$, $\lambda_F$, $\lambda_p$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Jossens model{5, 0.65, 1.15};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);   // 1.4623238
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 0.84734764
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$H > 0, \quad F > 0, \quad p > 0, \quad C_e \geq 0.$$

Chong, Hashim and Chu (2024) report strong correlation between the Jossens parameters when they are estimated from a single isotherm; the scaled sensitivities above should be inspected before interpreting fitted values.

29

# Khan isotherm

THREE PARAMETERS / q_max, K, n / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

Khan, A. R., Al-Waheab, I. R., and Al-Haddad, A. (1996), *A generalized equation for adsorption isotherms for multi-component organic pollutants in dilute aqueous solution*, *Environmental Technology*, **17** (1), 13–23. DOI: 10.1080/09593331708616356. The authors proposed a generalised equilibrium relation for organic pollutants adsorbed from dilute aqueous solutions, including multi-solute systems.

The single-component form implemented here is $q_e = q_{\max} K C_e / (1 + K C_e)^n$. It reduces to the Langmuir form when $n = 1$; at high concentration its response is proportional to $C_e^{1-n}$. Consequently, $n > 1$ produces a decreasing high-concentration branch rather than a finite, positive saturation plateau. The cited multi-component study provides the historical context, whereas the equations and parameter restrictions below specify the particular single-component C++ model.

MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\max} K c_e}{(1 + K c_e)^n}$$

The three adjustable parameters are `q_max`, `K` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

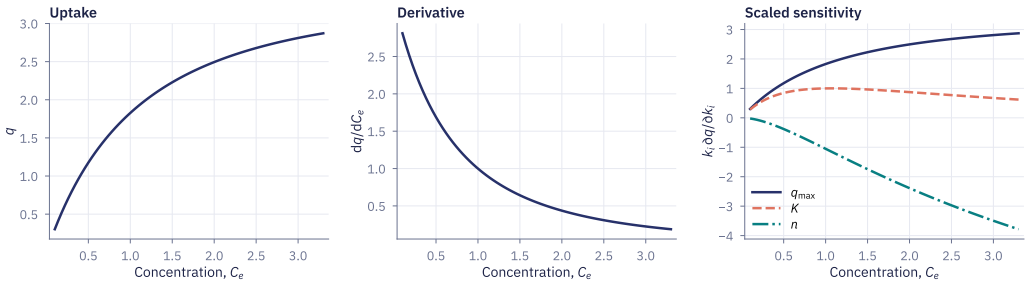



**Figure 29.** (a) Khan uptake for $q_{\max} = 5$, $K = 0.65$, $n = 1.15$; (b) Concentration derivative of Khan uptake for $q_{\max} = 5$, $K = 0.65$, $n = 1.15$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_K$, $\lambda_n$ versus equilibrium concentration.

C++20 USAGE

```
#include <isotherm/isotherm.hpp>

const isotherm::models::Khan model{5, 0.65, 1.15};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.6063982
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 1.2180092
```

NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

30

# Kiselev isotherm

THREE PARAMETERS / `q_max, K, a` / IMPLICIT

ORIGINAL PUBLICATION AND SCOPE

The Kiselev isotherm describes localised adsorption with an additional association term representing interactions between adsorbed molecules. The occupancy $\vartheta$ in the implemented relation is converted to uptake through $q_e = q_{\max}\vartheta$; the parameter convention is specified in the next section.

Kiselev, A. V. (1957), *Adsorption of vapours when complexes of adsorbate molecules are formed on the surface*, *Doklady Akademii Nauk SSSR*, **117** (6), 1023–1026. Primary bibliographic record. This earlier paper is directly identifiable in the journal archive.

Kiselev, A. V. (1958), *Vapor adsorption in the formation of adsorbate molecule complexes on the surface*, *Kolloidnyi Zhurnal*, **20**, 338–348. This citation is reproduced in subsequent adsorption literature, but an original-journal record and a DOI have not been independently confirmed here.

Chu, K. H. (2021), *The Kiselev isotherm for adsorption at the liquid–solid interface: Solving the mystery of negative equilibrium constants*, *Journal of Molecular Liquids*, **343**, 117674. DOI: 10.1016/j.molliq.2021.117674. Chu distinguishes sigmoidal (type V) uptake from type I data and explains why fitting the latter with an association-based model may produce negative estimates of the association coefficient. This is a warning about model selection, not a claim that every admissible parameter set produces a type V curve.

MATHEMATICAL FORMULATION

$$Kc_e = \frac{\vartheta}{(1 + a\vartheta)(1 - \vartheta)}, \qquad q_e = q_{\max}\vartheta$$

The three adjustable parameters are `q_max`, `K` and `a`. All must be positive and finite. Equilibrium concentration must be non-negative. The unique admissible coverage is determined by a bracketed log-activity solver; the analytical derivative follows from implicit differentiation.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad a > 0, \quad C_e \geq 0.$$

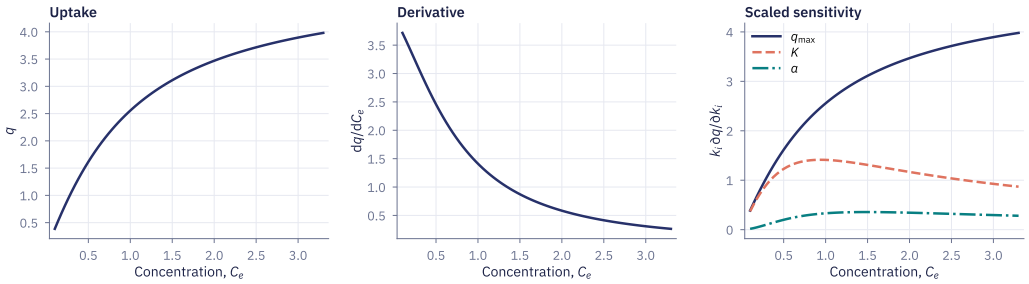



**Figure 30.** (a) Kiselev uptake for $q_{max} = 5$, $K = 0.8$, $a = 0.6$; (b) Concentration derivative of Kiselev uptake for $q_{max} = 5$, $K = 0.8$, $a = 0.6$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_a$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Kiselev model{5, 0.8, 0.6};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 2.2407551
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 1.750393
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad a > 0, \quad C_e \geq 0.$$

# 31

# Koble–Corrigan isotherm

THREE PARAMETERS / A, B, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Koble and Corrigan proposed an empirical adsorption relation for pure hydrocarbons. The three-parameter power-law form implemented here is saturating for positive coefficients and can be written as a Langmuir–Freundlich relation with $q_{max} = A/B$ and an affinity coefficient equal to $B$, using the same exponent $n$.

Koble, R. A., and Corrigan, T. E. (1952), *Adsorption Isotherms for Pure Hydrocarbons*, *Industrial & Engineering Chemistry*, **44** (2), 383–387. DOI: 10.1021/ie50506a049.

Chu, K. H., Debord, J., Harel, M., and Bollinger, J.-C. (2022), *Mirror, Mirror on the Wall, Which Is the Fairest of Them All? Comparing the Hill, Sips, Koble–Corrigan, and Liu Adsorption Isotherms*, *Industrial & Engineering Chemistry Research*, **61** (19), 6781–6790. DOI: 10.1021/acs.iecr.2c00507. The study demonstrates mathematical equivalence between these response families under suitable reparameterisation. Distinct names therefore do not establish distinct fitting flexibility.

### MATHEMATICAL FORMULATION

$$q_e = \frac{A c_e^n}{1 + B c_e^n}$$

The three adjustable parameters are `A`, `B` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$A > 0, \quad B > 0, \quad n > 0, \quad C_e \geq 0.$$

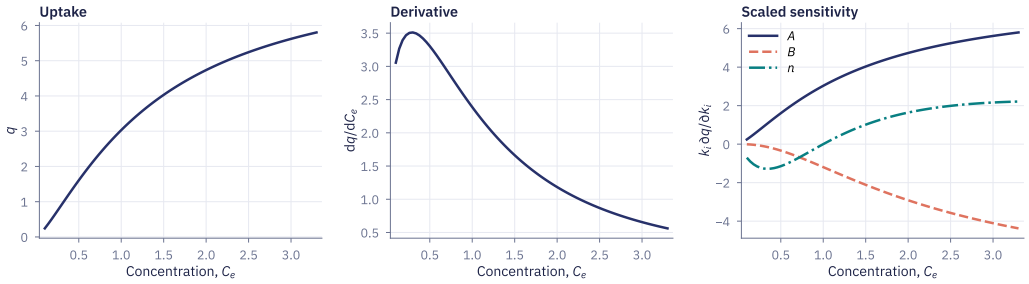



**Figure 31.** (a) Koble–Corrigan uptake for $A = 5$, $B = 0.65$, $n = 1.3$; (b) Concentration derivative of Koble–Corrigan uptake for $A = 5$, $B = 0.65$, $n = 1.3$; (c) Scaled sensitivities $\lambda_A$, $\lambda_B$, $\lambda_n$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::KobleCorrigan model{5, 0.65, 1.3};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 2.5169349
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 2.7517586
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$A > 0, \quad B > 0, \quad n > 0, \quad C_e \geq 0.$$

# 32

# Langmuir–Freundlich isotherm

THREE PARAMETERS / q_max, K, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

The Langmuir–Freundlich isotherm combines a power-law dependence at low concentration with finite saturation at high concentration. Its response family is also widely called the Sips isotherm, although the affinity and exponent may be parameterised differently in the literature. The equation implemented here uses $q_{max}$, $K$ and $n$.

Sips, R. (1948), *On the Structure of a Catalyst Surface*, *The Journal of Chemical Physics*, **16** (5), 490–495. DOI: 10.1063/1.1746922. Sips studied the distribution of adsorption-site energies and introduced a form with Freundlich-like low-pressure behaviour and saturation at high pressure.

Sips, R. (1950), *On the Structure of a Catalyst Surface. II*, *The Journal of Chemical Physics*, **18** (8), 1024–1026. DOI: 10.1063/1.1747848. This follow-up revises the earlier statistical treatment.

Jeppu, G. P., and Clement, T. P. (2012), *A modified Langmuir–Freundlich isotherm model for simulating pH-dependent adsorption effects*, *Journal of Contaminant Hydrology*, **129–130**, 46–53. DOI: 10.1016/j.jconhyd.2011.12.001. This is a pH-dependent extension, not the three-parameter equation implemented here.

Chu, K. H., Debord, J., Harel, M., and Bollinger, J.-C. (2022), *Mirror, Mirror on the Wall, Which Is the Fairest of Them All? Comparing the Hill, Sips, Koble–Corrigan, and Liu Adsorption Isotherms*, *Industrial & Engineering Chemistry Research*, **61** (19), 6781–6790. DOI: 10.1021/acs.iecr.2c00507. This comparison establishes that several named three-parameter isotherms belong to the same mathematical response family.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\max} K c_e^n}{1 + K c_e^n}$$

The three adjustable parameters are `q_max`, `K` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

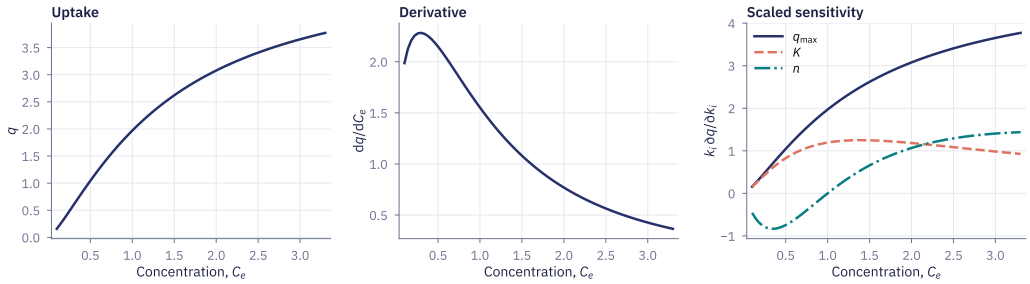



**Figure 32.** (a) Langmuir–Freundlich uptake for $q_{\max} = 5$, $K = 0.65$, $n = 1.3$; (b) Concentration derivative of Langmuir–Freundlich uptake for $q_{\max} = 5$, $K = 0.65$, $n = 1.3$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_K$, $\lambda_n$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::LangmuirFreundlich model{5, 0.65, 1.3};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);   // 1.6360077
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 1.7886431
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

33

# MacMillan–Teller isotherm

THREE PARAMETERS / q_max, K, c_s / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

The multilayer-adsorption treatment associated with this model was developed by W. G. **McMillan** and E. Teller in The Assumptions of the B.E.T. Theory (1951) (*Journal of Physical Chemistry*, **55**, 17–20) and The Role of Surface Tension in Multilayer Gas Adsorption (1951) (*Journal of Chemical Physics*, **19**, 25–32). The authors' surname is **McMillan**, whereas the C++ model identifier is `MacMillanTeller`.

The review by Foo and Hameed (2010) (*Chemical Engineering Journal*, **156**, 2–10) and the review by Benjelloun et al. (2021) (*Arabian Journal of Chemistry*) discuss adsorption-isotherm variants in aqueous systems. The 2021 review reports the concentration-based expression with exponent 1/3, which is the form implemented by the Isotherm++ model below. The original 1951 publications concern multilayer gas adsorption; the concentration-based expression is cited here from the later review.

## MATHEMATICAL FORMULATION

$$q_e = q_{\max}\left[\frac{K}{\ln(c_s/c_e)}\right]^{1/3}$$

The three adjustable parameters are `q_max`, `K` and `c_s`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The concentration must remain strictly below the saturation concentration $c_s$; the model diverges as that boundary is approached.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad c_s > 0, \quad 0 \le C_e < c_s.$$

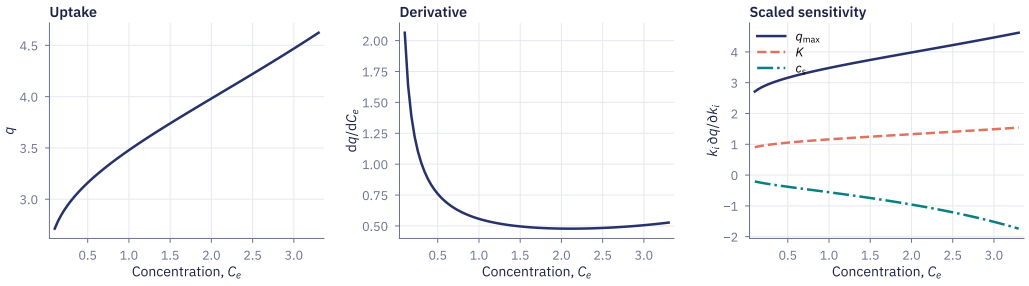



**Figure 33.** (a) MacMillan–Teller uptake for $q_{\max} = 5$, $K = 0.7$, $c_s = 8$; (b) Concentration derivative of MacMillan–Teller uptake for $q_{\max} = 5$, $K = 0.7$, $c_s = 8$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_K$, $\lambda_{c_s}$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::MacMillanTeller model{5, 0.7, 8};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);   // 3.36199803
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 0.60837383
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad c_s > 0, \quad 0 \le C_e < c_s.$$

# 34

# OBrienMyers isotherm

THREE PARAMETERS / q_max, b, sigma / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

The original O'Brien and Myers (1984) article, *Physical adsorption of gases on heterogeneous surfaces. Series expansion of isotherms using central moments of the adsorption energy distribution* (*Journal of the Chemical Society, Faraday Transactions 1*, **80**, 1467–1477), derives an approximation to the adsorption integral equation in terms of central moments of an adsorption energy distribution. The authors compare the truncated expansion with numerical integration for a Langmuir local isotherm and a normal energy distribution, and report failure of convergence for very broad distributions.

A subsequent explicit presentation of the two-term form used here is given in RUPTURA: simulation code for breakthrough, ideal adsorption solution theory computations, and fitting of isotherm models (2023), Eq. (20). The width parameter $\sigma$ appears through $\sigma^2$ in this convention. The approximation is a truncated heterogeneous-surface model, not a general exact integral for arbitrary energy distributions.

MATHEMATICAL FORMULATION

$$q = q_{\max}\left[\frac{bx}{1+bx} + \sigma^2 \frac{bx(1-bx)}{2(1+bx)^3}\right]$$

The 3 adjustable parameters are: `q_max`, `b`, `sigma`. The independent variable is `concentration`.

All parameters strictly positive and 0<sigma<1 in this implementation; sigma is a width measure, and the truncation is not justified for arbitrarily broad energy distributions.

The width term is the square of sigma, not an independently fitted sigma-squared parameter.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad b > 0, \quad \sigma > 0, \quad \sigma < 1, \quad C_e \ge 0.$$

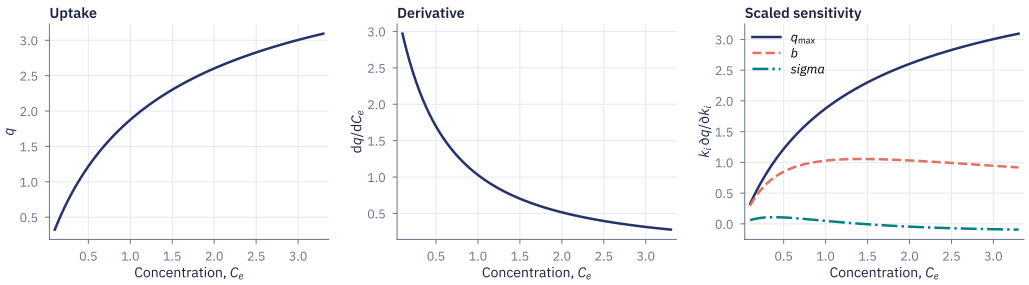



**Figure 34.** (a) O'Brien–Myers uptake for $q_{\max} = 4.5$, $b = 0.7$, $\sigma = 0.5$; (b) Concentration derivative of O'Brien–Myers uptake for $q_{\max} = 4.5$, $b = 0.7$, $\sigma = 0.5$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_b$, $\lambda_\sigma$ versus equilibrium concentration.

C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::OBrienMyers model{4.5, 0.7, 0.5};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 1.2186786
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 1.6954987
```

NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad b > 0, \quad \sigma > 0, \quad \sigma < 1, \quad C_e \geq 0.$$

35

# Quadratic isotherm

THREE PARAMETERS / q_max, K_a, K_b / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

The three-parameter quadratic site-interaction expression used here is explicitly documented by pyGAPS, Quadratic isotherm, which cites T. L. Hill's *An Introduction to Statistical Thermodynamics* as a statistical-mechanical reference. The quadratic denominator and the factor of two multiplying the $K_b$ term represent a particular site-interaction convention; the expression need not describe every model called a `quadratic isotherm`.

Xia, Nagrath and Cramer (2003), *Modeling of adsorption in hydrophobic interaction chromatography systems using a preferential interaction quadratic isotherm* (*Journal of Chromatography A*, **989**, 47–54), introduced a **preferential-interaction quadratic** model coupling salt-dependent interactions to nonlinear adsorption. It is an important related chromatographic application, but it should **not** be identified as the original derivation of the specific three-parameter equation below. A unique original research article for that precise parametrisation has not been established.

MATHEMATICAL FORMULATION

$$q = q_{\max} \frac{x(K_a + 2K_b x)}{1 + K_a x + K_b x^2}$$

The 3 adjustable parameters are: `q_max`, `K_a`, `K_b`. The independent variable is `concentration`.

All three parameters positive, concentration non-negative. Saturating limit is 2*q_max; the q_max parameter is a site-scale factor, not the total limiting uptake.

The factor 2 in the quadratic numerator means q_max is not the limiting capacity.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K_a > 0, \quad K_b > 0, \quad C_e \geq 0.$$

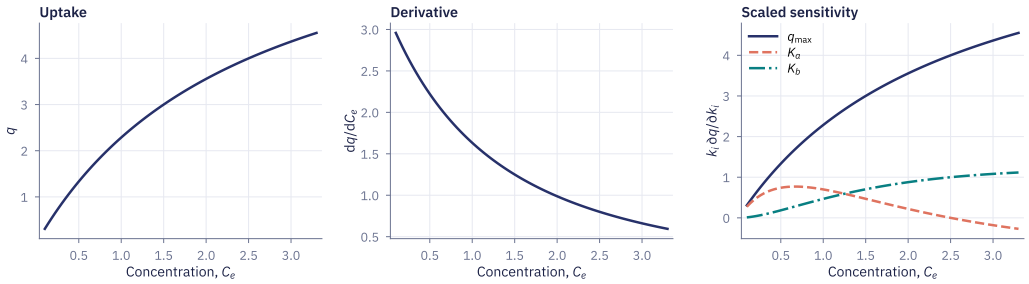



**Figure 35.** (a) Quadratic uptake for $q_{max} = 4$, $K_a = 0.8$, $K_b = 0.16$; (b) Concentration derivative of Quadratic uptake for $q_{max} = 4$, $K_a = 0.8$, $K_b = 0.16$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_{K_a}$, $\lambda_{K_b}$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Quadratic model{4, 0.8, 0.16};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 1.3333333
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 2.2222222
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K_a > 0, \quad K_b > 0, \quad C_e \geq 0.$$

# 36

# Radke–Prausnitz I isotherm

●●● THREE PARAMETERS / q_max, K, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

The original adsorption study is Radke and Prausnitz (1972), *Adsorption of Organic Solutes from Dilute Aqueous Solution of Activated Carbon*, *Industrial & Engineering Chemistry Fundamentals*, **11** (4), 445–451 (DOI). It concerns the equilibrium adsorption of organic solutes from dilute aqueous solutions onto activated carbon. The three expressions labelled I, II and III in this catalogue are specific mathematical parametrisations; their common historical attribution does not establish that they are interchangeable.

Tran, Bollinger, Salvestrini, Chu and Juang (2023), *Critical Review and Discussion of the Nonlinear Form of Radke–Prausnitz Model in Adsorption Solid–Liquid Phases*, *Journal of Environmental Engineering*, **149** (3), 03122006 (DOI: 10.1061/JOEEDU.EEENG-7074), examine the nonlinear expression, parameter dimensions and correct attribution to Radke and Prausnitz (1972). In particular, their analysis specifies the admissible exponent range for the correctly parametrised Radke–Prausnitz expression.

Tran, Bollinger, Lima and Juang (2023), *How to avoid mistakes in treating adsorption isotherm data (liquid and solid phases): Some comments about correctly using Radke-Prausnitz nonlinear model and Langmuir equilibrium constant*, *Journal of Environmental Management*, **325** (Part A), 116475 (DOI: 10.1016/j.jenvman.2022.116475), identify incorrect nonlinear forms and errors in the dimensional interpretation of adsorption equilibrium constants.

The implemented form I is $q_e = q_{max} K C_e/(1 + K C_e)^n$. It has a Henry-law limit at low concentration. At $n = 1$ it reduces to Langmuir; for $0 < n < 1$ its high-concentration asymptote is proportional to $C_e^{1-n}$. For $n > 1$, its derivative can become negative at sufficiently high concentration, so the numerical domain $n > 0$ accepted by the C++ class is broader than the range normally used for monotonic adsorption curves. The exponent convention is specific to this form and must not be transferred unchanged to forms II or III.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{max} K c_e}{(1 + K c_e)^n}$$

The three adjustable parameters are `q_max`, `K` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

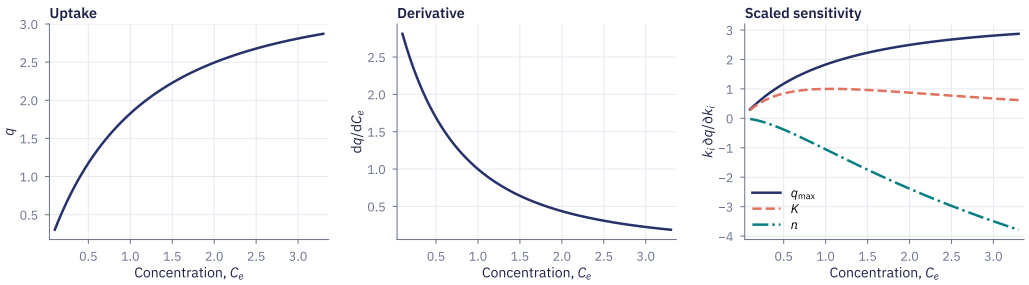



**Figure 36.** (a) Radke–Prausnitz I uptake for $q_{max} = 5$, $K = 0.65$, $n = 1.15$; (b) Concentration derivative of Radke–Prausnitz I uptake for $q_{max} = 5$, $K = 0.65$, $n = 1.15$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_n$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::RadkePrausnitsI model{5, 0.65, 1.15};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.6063982
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 1.2180092
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

37

# Radke–Prausnitz II isotherm

THREE PARAMETERS / q_max, K, n / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

The original adsorption study is Radke and Prausnitz (1972), *Adsorption of Organic Solutes from Dilute Aqueous Solution of Activated Carbon*, *Industrial & Engineering Chemistry Fundamentals*, **11** (4), 445–451 (DOI). It concerns the equilibrium adsorption of organic solutes from dilute aqueous solutions onto activated carbon. The three expressions labelled I, II and III in this catalogue are specific mathematical parametrisations; their common historical attribution does not establish that they are interchangeable.

Tran, Bollinger, Salvestrini, Chu and Juang (2023), *Critical Review and Discussion of the Nonlinear Form of Radke–Prausnitz Model in Adsorption Solid–Liquid Phases*, *Journal of Environmental Engineering*, **149** (3), 03122006 (DOI: 10.1061/JOEEDU.EEENG-7074), examine the nonlinear expression, parameter dimensions and correct attribution to Radke and Prausnitz (1972). In particular, their analysis specifies the admissible exponent range for the correctly parametrised Radke–Prausnitz expression.

Tran, Bollinger, Lima and Juang (2023), *How to avoid mistakes in treating adsorption isotherm data (liquid and solid phases): Some comments about correctly using Radke-Prausnitz nonlinear model and Langmuir equilibrium constant*, *Journal of Environmental Management*, **325** (Part A), 116475 (DOI: 10.1016/j.jenvman.2022.116475), identify incorrect nonlinear forms and errors in the dimensional interpretation of adsorption equilibrium constants.

The implemented form II is $q_e = q_{max} K C_e / (1 + K C_e^n)$. It has a Henry-law limit at low concentration; $n = 1$ reduces to Langmuir, whereas $0 < n < 1$ produces a Freundlich-type asymptote proportional to $C_e^{1-n}$. For $n > 1$ the response ultimately decreases with increasing concentration. If $C_e$ carries physical units, dimensional consistency requires $[K] = [C_e]^{-n}$ and $[q_{max}] = [q_e][C_e]^{n-1}$; consequently, q_max is not a saturation-capacity parameter for arbitrary $n$. This expression is not interchangeable with form I under unchanged parameters.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{max} K c_e}{1 + K c_e^n}$$

The three adjustable parameters are q_max, K and n. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

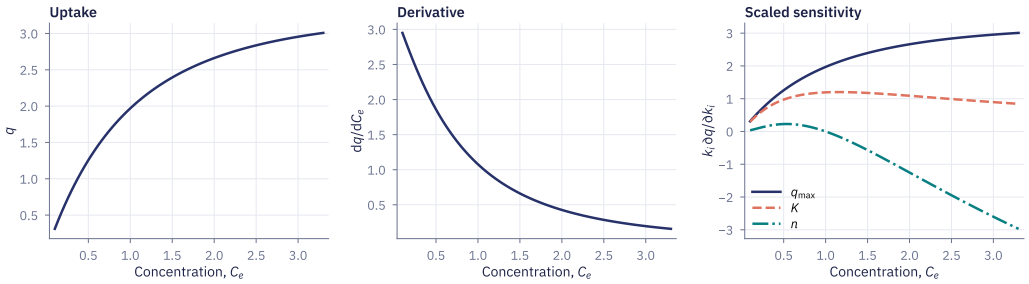



**Figure 37.** (a) Radke–Prausnitz II uptake for $q_{max} = 5$, $K = 0.65$, $n = 1.15$; (b) Concentration derivative of Radke–Prausnitz II uptake for $q_{max} = 5$, $K = 0.65$, $n = 1.15$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_n$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::RadkePrausnitsII model{5, 0.65, 1.15};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.7300084
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 1.3303679
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

# 38

# Radke–Prausnitz III isotherm

THREE PARAMETERS / q_max, K, n / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

The original adsorption study is Radke and Prausnitz (1972), *Adsorption of Organic Solutes from Dilute Aqueous Solution of Activated Carbon*, *Industrial & Engineering Chemistry Fundamentals*, **11** (4), 445–451 (DOI). It concerns the equilibrium adsorption of organic solutes from dilute aqueous solutions onto activated carbon. The three expressions labelled I, II and III in this catalogue are specific mathematical parametrisations; their common historical attribution does not establish that they are interchangeable.

Tran, Bollinger, Salvestrini, Chu and Juang (2023), *Critical Review and Discussion of the Nonlinear Form of Radke–Prausnitz Model in Adsorption Solid–Liquid Phases*, *Journal of Environmental Engineering*, **149** (3), 03122006 (DOI: 10.1061/JOEEDU.EEENG-7074), examine the nonlinear expression, parameter dimensions and correct attribution to Radke and Prausnitz (1972). In particular, their analysis specifies the admissible exponent range for the correctly parametrised Radke–Prausnitz expression.

Tran, Bollinger, Lima and Juang (2023), *How to avoid mistakes in treating adsorption isotherm data (liquid and solid phases): Some comments about correctly using Radke-Prausnitz nonlinear model and Langmuir equilibrium constant*, *Journal of Environmental Management*, **325** (Part A), 116475 (DOI: 10.1016/j.jenvman.2022.116475), identify incorrect nonlinear forms and errors in the dimensional interpretation of adsorption equilibrium constants.

The implemented form III is $q_e = q_{max} K C_e^n / (1 + K C_e^{n-1})$. For $0 < n < 1$, its low-concentration limit is $q_e \sim q_{max} C_e$ (Henry law), whereas the high-concentration limit is $q_e \sim q_{max} K C_e^n$ (Freundlich-type growth). At $n = 1$ it is linear, not Langmuir; the Langmuir expression arises in the limiting case $n \to 0$ at positive concentration, after identifying the corresponding saturation capacity and affinity. The C++ class admits $n > 0$, including values above unity; in that range the stated Henry-to-Freundlich asymptotes do not apply. In particular, the illustrated case $n = 1.3$ does not have them.

The algebraically equivalent alternative $q_e = A B C_e^\beta / (A + B C_e^{\beta-1})$ uses $q_{max} = A$, $K = B/A$ and $n = \beta$. This identity does not make `q_max` a saturation capacity. For dimensional concentration, consistency of the implemented expression requires $[K] = [C_e]^{1-n}$ and $[q_{max}] = [q_e]/[C_e]$.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{max} K c_e^n}{1 + K c_e^{n-1}}$$

The three adjustable parameters are `q_max`, `K` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

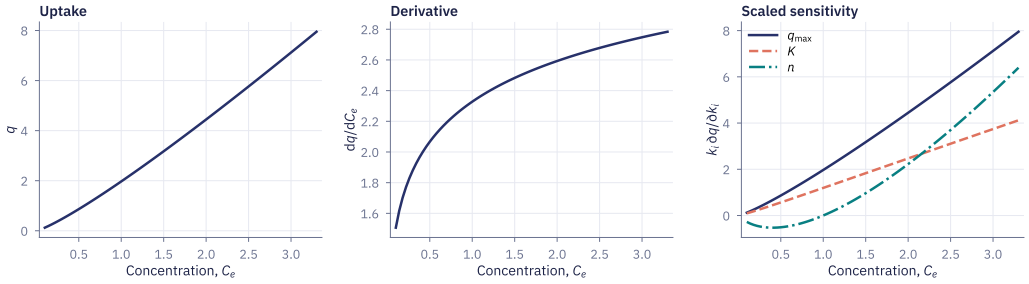



**Figure 38.** (a) Radke–Prausnitz III uptake for $q_{max} = 5$, $K = 0.65$, $n = 1.3$; (b) Concentration derivative of Radke–Prausnitz III uptake for $q_{max} = 5$, $K = 0.65$, $n = 1.3$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_n$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::RadkePrausnitsIII model{5, 0.65, 1.3};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.5123009
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 2.2430776
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

39

# Redlich–Peterson isotherm

THREE PARAMETERS / A, B, n / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Redlich–Peterson equation. Earlier versions of this catalogue also exposed this functional form under the name `Jossens`; that class now implements the original loading-implicit Jossens relation and is no longer equivalent to Redlich–Peterson.

The equation and parameter convention are specified below; the model name alone does not establish equivalence with every published formulation. For broader context see the 2022 adsorption-model review.

## MATHEMATICAL FORMULATION

$$q_e = \frac{A c_e}{1 + B c_e^n}$$

The three adjustable parameters are `A`, `B` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$A > 0, \quad B > 0, \quad n > 0, \quad C_e \geq 0.$$

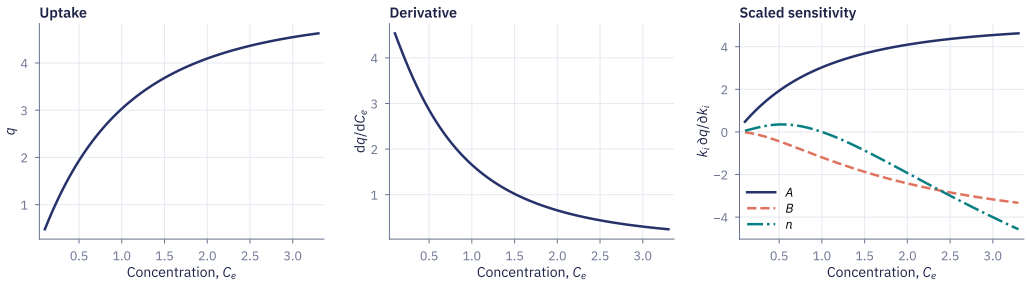



**Figure 39.** (a) Redlich–Peterson uptake for $A = 5$, $B = 0.65$, $n = 1.15$; (b) Concentration derivative of Redlich–Peterson uptake for $A = 5$, $B = 0.65$, $n = 1.15$; (c) Scaled sensitivities $\lambda_A$, $\lambda_B$, $\lambda_n$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::RedlichPeterson model{5, 0.65, 1.15};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 2.6615514
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 2.0467198
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$A > 0, \quad B > 0, \quad n > 0, \quad C_e \geq 0.$$

40

# Sips isotherm

THREE PARAMETERS / q_max, K, m / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Sips derived a saturating adsorption relation for heterogeneous surfaces, with a low-pressure limit resembling Freundlich behaviour. The class uses $q_{max}$, $K$ and $m$, with the concentration-dependent term $(KC_e)^{1/m}$. It is algebraically equivalent to the Langmuir–Freundlich form in this catalogue after setting the latter's exponent to $n = 1/m$ and its affinity coefficient to $K_{LF} = K^{1/m}$.

Sips, R. (1948), *On the Structure of a Catalyst Surface*, *The Journal of Chemical Physics*, **16** (5), 490–495. DOI: 10.1063/1.1746922.

Sips, R. (1950), *On the Structure of a Catalyst Surface. II*, *The Journal of Chemical Physics*, **18** (8), 1024–1026. DOI: 10.1063/1.1747848. The 1950 paper revisits the statistical basis of the earlier treatment.

Chu, K. H., Debord, J., Harel, M., and Bollinger, J.-C. (2022), *Mirror, Mirror on the Wall, Which Is the Fairest of Them All? Comparing the Hill, Sips, Koble–Corrigan, and Liu Adsorption Isotherms*, *Industrial & Engineering Chemistry Research*, **61** (19), 6781–6790. DOI: 10.1021/acs.iecr.2c00507. This analysis distinguishes differences in parameter definitions from genuinely different mathematical behaviour.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{max}(Kc_e)^{1/m}}{1 + (Kc_e)^{1/m}}$$

The three adjustable parameters are `q_max`, `K` and `m`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{max} > 0, \quad K > 0, \quad m > 0, \quad C_e \geq 0.$$

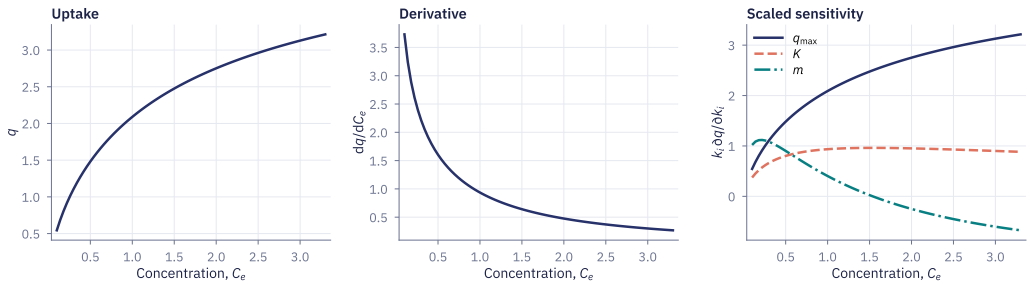



**Figure 40.** (a) Sips uptake for $q_{max} = 5$, $K = 0.65$, $m = 1.3$; (b) Concentration derivative of Sips uptake for $q_{max} = 5$, $K = 0.65$, $m = 1.3$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_m$ versus equilibrium concentration.

## C++20 USAGE

```
#include <isotherm/isotherm.hpp>

const isotherm::models::Sips model{5, 0.65, 1.3};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.8841557
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 1.1289876
```

NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad m > 0, \quad C_e \geq 0.$$

# 41

# Toth isotherm

THREE PARAMETERS / q_max, K, n / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

Tóth introduced an empirical three-parameter adsorption relation for heterogeneous solid–gas interfaces. The model retains a finite saturation capacity while providing a more flexible representation of the approach to saturation than the Langmuir equation. The exponent describes the shape of the isotherm; its fitted value should not, by itself, be taken as a unique measurement of the microscopic distribution of adsorption sites.

Tóth, J. (1971), *State Equations of the Solid-Gas Interface Layers*, *Acta Chimica Academiae Scientiarum Hungaricae*, **69** (3), 311–328. Digitised original journal volume (PDF); journal archive record; bibliographic record. The archive provides the complete volume rather than a separate article file. No DOI has been verified for the original publication.

Kumar, K. V., Gadipelli, S., Howard, C. A., Kwapinski, W., and Brett, D. J. L. (2021), *Probing adsorbent heterogeneity using Toth isotherms*, *Journal of Materials Chemistry A*, **9**, 944–962. DOI: 10.1039/D0TA08150G. The authors investigate the interpretation of adsorbent heterogeneity through Tóth-model parameters and affinity distributions.

Bbumba, S., Karume, I., Nsamba, H. K., Kigozi, M., and Kato, M. (2024), *An Insight into Isotherm Models in Physical Characterization of Adsorption Studies*, *European Journal of Applied Sciences*, **12** (2), 115–134. DOI: 10.14738/aivp.122.16738. This review discusses the Tóth equation alongside other two- and three-parameter models and compares fitting and error-assessment methods.

Musa, M. A., Chowdhury, S., Biswas, S., Alam, S. M. N., Parvin, S., and Sattar, M. A. (2024), *Removal of aqueous methylene blue dye over Vallisneria Natans biosorbent using artificial neural network and statistical response surface methodology analysis*, *Journal of Molecular Liquids*, **393**, 123624. DOI: 10.1016/j.molliq.2023.123624. The Tóth model was reported as the best-fitting isotherm in that study, with a reported capacity of 63.07 mg $g^{-1}$.

Zarei, S., Raanaei, H., Mohammad-Hosseini, V., and Kamali, S. (2025), *Efficient removal of zinc ion pollution by carbon-based magnetic alloy: Experimental, theoretical modeling and DFT studies*, *Inorganic Chemistry Communications*, **172**, 113735. DOI: 10.1016/j.inoche.2024.113735. The study compared ten adsorption isotherms for Zn(II) on a graphite–iron alloy and reported the best fit with Tóth, including a fitted maximum capacity of 729.7 mg $g^{-1}$.

Zuhara, S., and McKay, G. (2025), *Single and binary pollutant adsorption of strontium and barium on waste-derived activated carbons: Modelling, regeneration and mechanistic insights*, *Environmental Technology & Innovation*, **39**, 104220. DOI: 10.1016/j.eti.2025.104220. Tóth fitted the single-solute equilibrium data best in this study, whereas a Langmuir formulation was preferred for the binary systems.

Grossmann, Q., Saenz-Cavazos, P. A., Ferru, N., Williams, D. R., and Mazzotti, M. (2025), *Measuring and Modeling Water and Carbon Dioxide Adsorption on Amine Functionalized Alumina under Direct Air Capture Conditions*, *Industrial & Engineering Chemistry Research*, **64** (13), 7165–7175. DOI:

10.1021/acs.iecr.4c04581. The authors use a temperature-dependent Tóth relation for dry $CO_2$ adsorption and develop a separate collaborative, dual-site extension for humid conditions, named CATSO. Neither extension is implemented by this three-parameter class.

These publications illustrate applications and extensions, not evidence that the Tóth equation is universally the most suitable model. Applicability must be assessed for each adsorbate–adsorbent system and experimental range.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\max} c_e}{(1/K + c_e^n)^{1/n}}$$

This is the parameter convention implemented in Isotherm++. It is algebraically equivalent to the frequently used affinity-based form:

$$q_e = \frac{q_{\max} b C_e}{[1 + (bC_e)^n]^{1/n}}, \qquad b = K^{1/n}.$$

Accordingly, when concentration is dimensional, $K$ has units of $[C_e]^{-n}$, whereas $b$ has units of $[C_e]^{-1}$. The parameter $K$ must not be substituted numerically for the coefficient $b$ without this transformation. For $n = 1$, the equation reduces to the Langmuir isotherm. Its limiting behaviour is

$$q_e \sim q_{\max} K^{1/n} C_e \quad (C_e \to 0^+), \qquad \lim_{C_e \to \infty} q_e = q_{\max}.$$

The three adjustable parameters are `q_max`, `K` and `n`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

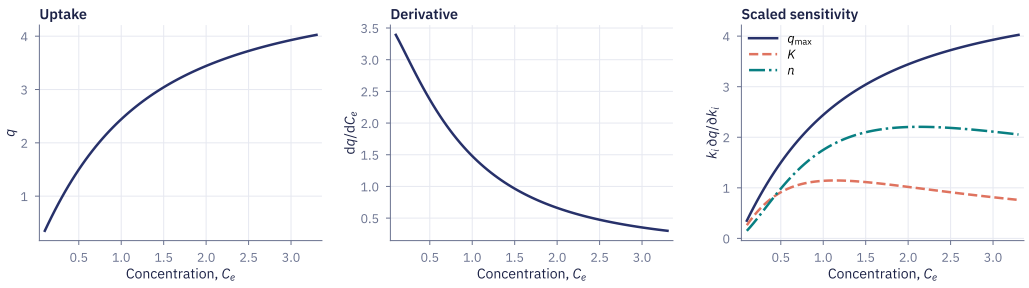



**Figure 41.** (a) Toth uptake for $q_{\max} = 5$, $K = 0.65$, $n = 1.3$; (b) Concentration derivative of Toth uptake for $q_{\max} = 5$, $K = 0.65$, $n = 1.3$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_K$, $\lambda_n$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Toth model{5, 0.65, 1.3};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 2.1171471
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 1.7805166
```

## NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad n > 0, \quad C_e \geq 0.$$

# 42

# Unilan isotherm

THREE PARAMETERS / q_max, K, a / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

The UNILAN (uniform-energy-distribution/Langmuir) equation represents energetic heterogeneity by averaging local Langmuir responses over a uniform distribution of adsorption energies. The resulting relation has three fitted coefficients, including a dimensionless heterogeneity spread. It is an equilibrium correlation; a fitted spread does not, by itself, establish a unique microscopic surface-energy distribution.

Valenzuela, D. P., and Myers, A. L. (1989), *Adsorption Equilibrium Data Handbook*, Prentice Hall, Englewood Cliffs, New Jersey. Bibliographic record. This handbook compiles adsorption-equilibrium data and UNILAN model applications. It is a standard reference for the equation, but should not automatically be described as its first derivation.

Tóth, J. (1997), *Some Consequences of the Application of Incorrect Gas/Solid Adsorption Isotherm Equations*, *Journal of Colloid and Interface Science*, **185** (1), 228–235. DOI: 10.1006/jcis.1996.4562. This critical discussion distinguishes useful equilibrium fitting from thermodynamically valid interpretation of heterogeneous-gas models, including the forms of UNILAN presented in the handbook.

Gürkan, E. H., Ilyas, B., and Tibet, Y. (2023), *Adsorption performance of heavy metal ions from aqueous solutions by a waste biomass based hydrogel: comparison of isotherm and kinetic models*, *International Journal of Environmental Analytical Chemistry*, **103** (6), 1343–1360. DOI: 10.1080/03067319.2021.1873314. UNILAN was one of the models compared for Cu(II) and Zn(II) adsorption on rice-husk biochar, calcium alginate and their composites; the article does not establish that UNILAN was universally the best-fitting model.

Sha'arani, S. A. W., Yasid, N. A., Halmi, M. I. E., Othman, A. R., and Shukor, M. Y. (2023), *Adsorption of Sodium Dodecyl Sulfate onto Activated Coconut Shell-Based Adsorbent: Isothermal Remodelling*, *Journal of Environmental Microbiology and Toxicology*, **11** (1), 52–61. DOI: 10.54987/jemat.v11i1.884. The authors compared 19 nonlinear isotherm models; UNILAN ranked second after Moreau under the error criteria considered.

Dan-Iya, B. I., Khayat, M. E., and Shukor, M. Y. (2023), *Isothermal Modelling of the Adsorption of Crystal Violet onto Modified Charred Rice Husk*, *Journal of Environmental Microbiology and Toxicology*, **11** (1), 1–9. DOI: 10.54987/jemat.v11i1.837. UNILAN was included in this model comparison, but the authors identified Jovanovic as the best-fitting model for the data analysed.

MATHEMATICAL FORMULATION

$$q_e = \frac{q_{max}}{2a} \ln\left[\frac{1 + Kc_e \exp(a)}{1 + Kc_e \exp(-a)}\right]$$

The coefficient $q_{max}$ is the limiting uptake of the heterogeneous Langmuir component, $K$ is the affinity coefficient (with dimensions of inverse concentration), and $a$ is a dimensionless half-width for the assumed uniform energy distribution. As $a \to 0^+$, this equation reduces continuously to the Langmuir relation:

$$\lim_{a \to 0^+} q_e = \frac{q_{max} K C_e}{1 + K C_e}.$$

The current C++ constructor excludes the exact value $a = 0$; this is an implementation restriction, not a failure of the mathematical limit. With positive parameters, the uptake tends to $q_{max}$ as $C_e \to \infty$.

The three adjustable parameters are `q_max`, `K` and `a`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

For numerical stability the implemented heterogeneity spread is restricted to $a \le 50$, avoiding unsafe intermediate exponentials. This is an explicit implementation restriction, not a physical upper bound.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad a > 0, \quad a \le 50, \quad C_e \ge 0.$$

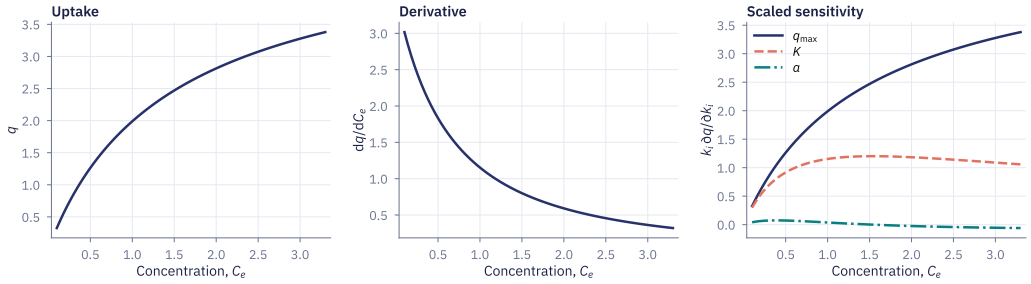



**Figure 42.** (a) Unilan uptake for $q_{\max} = 5$, $K = 0.65$, $a = 0.7$; (b) Concentration derivative of Unilan uptake for $q_{\max} = 5$, $K = 0.65$, $a = 0.7$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_K$, $\lambda_a$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Unilan model{5, 0.65, 0.7};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.7383849
const isotherm::Real slope = isotherm::derivative(model, 0.8);   // 1.3673534
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad a > 0, \quad a \le 50, \quad C_e \ge 0.$$

# 43

# Valenzuela–Myers isotherm

●●● THREE PARAMETERS / `q_max, K, a` / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

The name `ValenzuelaMyers` is retained as an Isotherm++ model identifier. It must not be interpreted as the name of a separately established adsorption equation. Valenzuela and Myers published a widely used handbook compiling adsorption-equilibrium data and heterogeneous-surface models, including UNILAN. The formula implemented here is mathematically the UNILAN relation with a *reciprocal affinity* parameter; it is not an additional independent adsorption model.

Valenzuela, D. P., and Myers, A. L. (1984), *Gas Adsorption Equilibria*, *Separation and Purification Methods*, **13** (2), 153–183. DOI: 10.1080/03602548408058530. This earlier review provides context on gas-adsorption equilibrium and separation processes; it does not establish a separate `ValenzuelaMyers` isotherm of the form used here.

Valenzuela, D. P., and Myers, A. L. (1989), *Adsorption Equilibrium Data Handbook*, Prentice Hall, Englewood Cliffs, New Jersey. Bibliographic record. The handbook is the appropriate historical reference for UNILAN and its use in gas-adsorption data correlation, not proof of a distinct Valenzuela–Myers equation.

Tóth, J. (1997), *Some Consequences of the Application of Incorrect Gas/Solid Adsorption Isotherm Equations*, *Journal of Colloid and Interface Science*, **185** (1), 228–235. DOI: 10.1006/jcis.1996.4562. This analysis cautions that good empirical fits of certain Tóth and UNILAN formulations do not guarantee thermodynamic consistency.

For a three-parameter UNILAN implementation using the conventional affinity parameter, see Unilan isotherm.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\max}}{2a} \ln\left[\frac{K + c_e \exp(a)}{K + c_e \exp(-a)}\right]$$

The parameter $K$ in this class has the dimensions of concentration and is *not* the affinity parameter $K_U$ stored by `isotherm::models::Unilan`. Their conversion is

$$K_U = \frac{1}{K_{VM}}, \qquad q_e^{VM}(C_e; q_{\max}, K_{VM}, a) = q_e^{UNILAN}(C_e; q_{\max}, 1/K_{VM}, a).$$

Thus the two Isotherm++ classes describe the same family of curves with different numerical meanings for `K`. This is an algebraic identity, not a new experimentally distinct isotherm. In the homogeneous limit $a \to 0^+$, the expression becomes $q_{\max} C_e/(K + C_e)$. The constructor excludes $a = 0$.

The three adjustable parameters are `q_max`, `K` and `a`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

For numerical stability the implemented heterogeneity spread is restricted to $a \le 50$, avoiding unsafe intermediate exponentials. This is an explicit implementation restriction, not a physical upper bound.

The implementation requires the following admissible parameter and input domain:

$$q_{\max} > 0, \quad K > 0, \quad a > 0, \quad a \le 50, \quad C_e \ge 0.$$

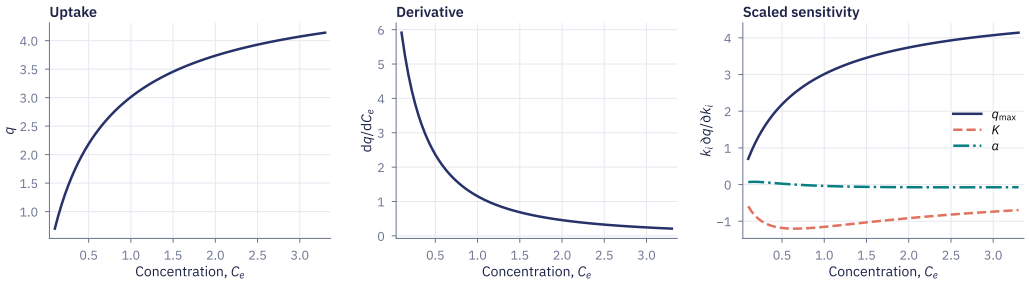



**Figure 43.** (a) Valenzuela–Myers uptake for $q_{\max} = 5$, $K = 0.65$, $a = 0.7$; (b) Concentration derivative of Valenzuela–Myers uptake for $q_{\max} = 5$, $K = 0.65$, $a = 0.7$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_K$, $\lambda_a$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::ValenzuelaMyers model{5, 0.65, 0.7};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);   // 2.7486538
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 1.4874073
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K > 0, \quad a > 0, \quad a \le 50, \quad C_e \ge 0.$$

# 44

# Vieth–Sladek isotherm

THREE PARAMETERS / q_max, K, H / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Vieth and Sladek used a dual-mode equilibrium-sorption relation as part of a model for transient gas transport through glassy polymers. The equilibrium uptake combines a Henry-law dissolution term and a finite-capacity Langmuir-type population associated with polymer microvoids. The corresponding diffusion model requires additional transport assumptions; the equilibrium equation alone does not predict diffusion rates.

Vieth, W. R., and Sladek, K. J. (1965), *A model for diffusion in a glassy polymer*, *Journal of Colloid Science*, **20** (9), 1014–1033. DOI: 10.1016/0095-8522(65)90071-1. The original study applies the equilibrium dual-mode description to transient gas sorption in polymer films. The Langmuir-like fraction is associated with immobilised molecules in polymer microvoids, while molecules dissolved in the amorphous phase provide the mobile fraction in the authors' transport interpretation.

Dan-Iya, B. I., Khayat, M. E., and Shukor, M. Y. (2023), *Isothermal Modelling of the Adsorption of Crystal Violet onto Modified Charred Rice Husk*, *Journal of Environmental Microbiology and Toxicology*, **11** (1), 1–9. DOI: 10.54987/jemat.v11i1.837. Vieth–Sladek was among the models used for nonlinear regression of crystal-violet sorption data; Jovanovic gave the best fit in that comparison.

Sha'arani, S. A. W., Yasid, N. A., Halmi, M. I. E., Othman, A. R., and Shukor, M. Y. (2023), *Adsorption of Sodium Dodecyl Sulfate onto Activated Coconut Shell-Based Adsorbent: Isothermal Remodelling*, *Journal of Environmental Microbiology and Toxicology*, **11** (1), 52–61. DOI: 10.54987/jemat.v11i1.884. Vieth–Sladek was included in the comparison of 19 empirical models; the best-ranked model was Moreau, rather than Vieth–Sladek.

### MATHEMATICAL FORMULATION

$$q_e = Hc_e + \frac{q_{\max} K c_e}{1 + K c_e}$$

Here $H$ is a Henry-law coefficient (uptake per concentration), $K$ is the Langmuir affinity (inverse concentration), and $q_{\max}$ is the capacity of the *saturable component only*. The total uptake is not bounded by $q_{\max}$ when $H > 0$:

$$q_e \sim HC_e + q_{\max} \quad (C_e \to \infty).$$

The implemented derivative is $H + q_{max} K/(1 + KC_e)^2$. These parameters describe equilibrium sorption; diffusion coefficients and transient transport mechanisms are not parameters of this class.

The three adjustable parameters are `q_max`, `K` and `H`. All must be strictly positive and finite. The model accepts finite, non-negative equilibrium concentrations, while the derivative requires strictly positive concentration.

An analytical concentration derivative is implemented and checked against an independently calculated centred difference.

The implementation requires the following admissible parameter and input domain:

$$q_{max} > 0, \quad K > 0, \quad H > 0, \quad C_e \geq 0.$$

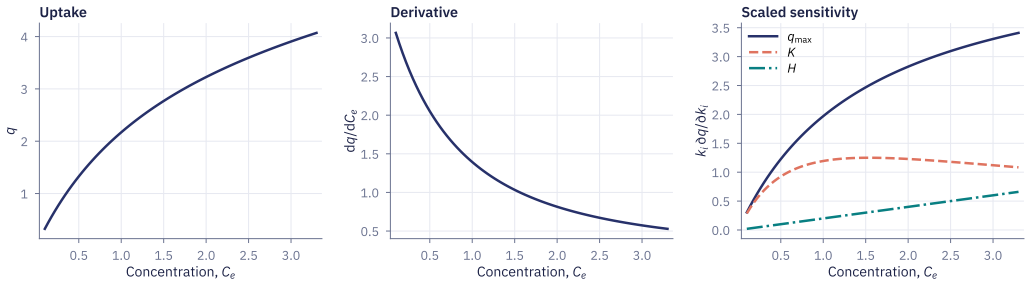



**Figure 44.** (a) Vieth–Sladek uptake for $q_{max} = 5$, $K = 0.65$, $H = 0.2$; (b) Concentration derivative of Vieth–Sladek uptake for $q_{max} = 5$, $K = 0.65$, $H = 0.2$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_K$, $\lambda_H$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::ViethSladek model{5, 0.65, 0.2};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.8705263
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 1.6066828
```

### NUMERICAL SENSITIVITY

For each coefficient $k_i$, the numerical scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K > 0, \quad H > 0, \quad C_e \geq 0.$$

# 45

# Baudu isotherm

●●●● FOUR PARAMETERS / `q_max, K1, K2, K3` / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Baudu's four-parameter equation is an empirical modification of the Langmuir adsorption relation, introduced to accommodate concentration- dependent behaviour that cannot be represented adequately by a single Langmuir affinity and capacity over an extended concentration range. The original work concerns solute interactions with activated-carbon fibres. The source identified below is a doctoral thesis rather than a journal article; its original derivation has not been independently checked against the thesis text in this documentation update.

Baudu, M. (1990), *Etude des interactions solutes-fibres de charbon actif: applications et regeneration*, doctoral thesis, Universite de Rennes 1, France. Thesis record.

Foo, K. Y., and Hameed, B. H. (2010), *Insights into the modeling of adsorption isotherm systems*, *Chemical Engineering Journal*, **156** (1), 2–10. DOI: 10.1016/j.cej.2009.09.013. This review provides context for multi-parameter adsorption-equilibrium correlations.

Essam, A., Eldek, S. I., and Shehata, N. (2024), *Management of caffeine in wastewater using MOF and perovskite materials: optimization, kinetics, and adsorption isotherm modelling*, *Journal of Environmental Health Science and Engineering*, **22**, 345–360. DOI: 10.1007/s40201-024-00904-2. The authors compared Baudu with several other adsorption models for caffeine uptake on UiO-66 and a lanthanum-strontium ferrite perovskite. Their perovskite fit is reported with $R^2 = 0.86$; this value is specific to that dataset and is not evidence of general superiority.

Ramadoss, R., Gunasekaran, D., and Subramanian, D. (2022), *Removal of Divalent Nickel from Aqueous Solution Using Blue Green Marine Algae: Adsorption Modelling and Applicability of Various Isotherm Models*, in *Progress in Microalgae Research - A Path for Shaping Sustainable Futures*, IntechOpen. DOI: 10.5772/intechopen.103940. This comparison evaluates Baudu among four-parameter models but cautions that goodness-of-fit statistics alone do not ensure physically interpretable parameters.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\max} K_1 c_e^{1-K_2+K_3}}{1 + K_1 c_e^{1-K_2}}$$

The coefficient named $q_{\max}$ in the implementation is a multiplicative scale. For the general fitted exponents it is **not** necessarily a finite saturation capacity. At low concentrations, the response has a Freundlich-type power-law asymptote; the leading power depends on whether $K_2$ lies below, at, or above unity. For $K_2 < 1$, in particular,

$$q_e \sim q_{\max} K_1 C_e^{1-K_2+K_3} \qquad (C_e \to 0^+).$$

For $K_2 > 1$, the leading low-concentration power is $q_e \sim q_{\max} C_e^{K_3}$. The parameter constraints below are those enforced by the current C++ implementation and should not be interpreted as restrictions verified against Baudu's original thesis.

The four independently adjustable parameters are `q_max`, `K1`, `K2` and `K3`. No temperature parameter is required by this implementation.

All parameters are strictly positive; 0 < K3 < 1 and K2 < 1 + K3. The latter condition ensures a continuous zero-concentration limit and increasing response, a deliberate tightening of the original implementation.

Finite and non-negative concentrations are accepted. The response at zero concentration is defined by its continuous zero limit. The derivative is analytical and is evaluated only for strictly positive concentration.

Parameter and concentration restrictions:

$$q_{\max} > 0, \quad K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad 0 < K_3 < 1, \quad K_2 < 1 + K_3, \quad C_e \geq 0.$$

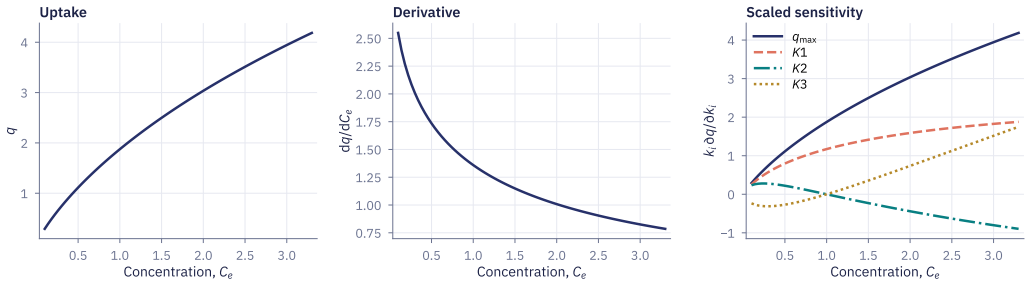



**Figure 45.** (a) Baudu uptake for $q_{max} = 5$, $K_1 = 0.6$, $K_2 = 0.4$, $K_3 = 0.35$; (b) Concentration derivative of Baudu uptake for $q_{max} = 5$, $K_1 = 0.6$, $K_2 = 0.4$, $K_3 = 0.35$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_{K_1}$, $\lambda_{K_2}$, $\lambda_{K_3}$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Baudu model{1.95325, 0.513664, 0.0435246, 0.00605498};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);   // 0.57203102
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 0.48768303
```

### NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad 0 < K_3 < 1, \quad K_2 < 1 + K_3, \quad C_e \geq 0.$$

# 46

# DualSiteLangmuir isotherm

●●●● FOUR PARAMETERS / q_1, K_1, q_2, K_2 / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

The dual-site Langmuir (DSL) model describes adsorption on two populations of sites, each represented by a Langmuir term with its own capacity and affinity. It extends the single-site relation established by Langmuir (1918). The four coefficients used here describe a *single* adsorbate at a fixed temperature. This C++ class does not implement multicomponent competition or the additional constraints required for a thermodynamically consistent mixture model.

Langmuir, I. (1918), *The Adsorption of Gases on Plane Surfaces of Glass, Mica and Platinum*, *Journal of the American Chemical Society*, **40** (9), 1361–1403. DOI: 10.1021/ja02242a004. This is the original single-site Langmuir foundation, not a publication proposing the four-parameter DSL model.

Mathias, P. M., Kumar, R., Moyer, J. D., Schork, J. M., Srinivasan, S. R., Auvil, S. R., and Talu, O. (1996), *Correlation of Multicomponent Gas Adsorption by the Dual-Site Langmuir Model. Application to Nitrogen/Oxygen Adsorption on 5A-Zeolite*, *Industrial & Engineering Chemistry Research*, **35** (7), 2477–2483. DOI: 10.1021/ie950291y. This study develops and examines a thermodynamically consistent multicomponent DSL treatment and illustrates it with nitrogen/oxygen adsorption on 5A zeolite.

Brandani, S. (2024), *On adsorption azeotropy and a classification based on the dual site Langmuir isotherm*, *Adsorption*, **30**, 1593–1601. DOI: 10.1007/s10450-023-00430-x. The analysis uses a thermodynamically consistent binary DSL formulation to classify adsorption azeotropy; its mixture-level conclusions cannot be obtained from the present single-component function alone.

Zhao, C., Yang, Y., Sun, H., Zhong, J., Zhang, K., Wang, K., Zhang, L., and Yao, J. (2024), *Pore-Scale Simulation of Deep Shale Gas Flow Considering the Dual-Site Langmuir Adsorption Model Based on the Pore Network Model*, *Langmuir*, **40** (43), 22844–22855. DOI: 10.1021/acs.langmuir.4c02889. This is a recent example of DSL use in shale-gas transport modelling.

## MATHEMATICAL FORMULATION

$$q = \frac{q_1 K_1 x}{1 + K_1 x} + \frac{q_2 K_2 x}{1 + K_2 x}$$

With positive parameters, $q_e(0) = 0$ and the limiting uptake is $q_1 + q_2$ as $C_e \to \infty$. Swapping the two site pairs $(q_1, K_1)$ and $(q_2, K_2)$ leaves the curve unchanged, so the pair labels are not independently identifiable from single-component equilibrium data. When $K_1 = K_2$, the expression reduces to one Langmuir term with combined capacity $q_1 + q_2$.

The 4 adjustable parameters are: `q_1`, `K_1`, `q_2`, `K_2`. The independent variable is `concentration`.

All parameters positive. Concentration-dependent, at fixed temperature; asymptotic uptake q_1+q_2.

Parameter values are illustrative and not estimates from experimental data. The mathematical representation does not establish experimental validity.

Parameter and concentration restrictions:

$$q_1 > 0, \quad K_1 > 0, \quad q_2 > 0, \quad K_2 > 0, \quad C_e \geq 0.$$

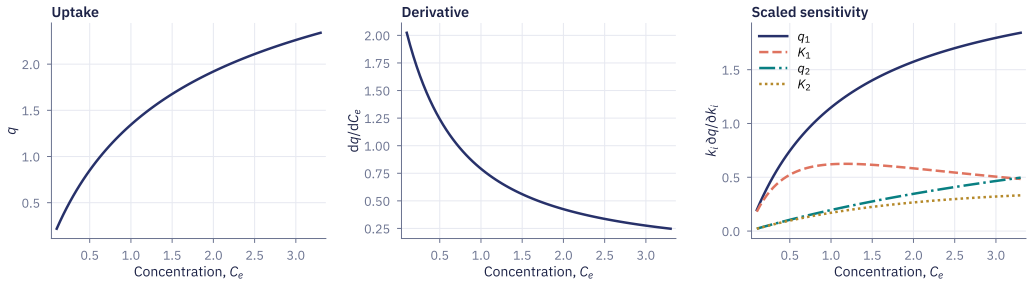



**Figure 46.** (a) Dual-Site Langmuir uptake for $q_1 = 2.5$, $K_1 = 0.85$, $q_2 = 1.5$, $K_2 = 0.15$; (b) Concentration derivative of Dual-Site Langmuir uptake for $q_1 = 2.5$, $K_1 = 0.85$, $q_2 = 1.5$, $K_2 = 0.15$; (c) Scaled sensitivities $\lambda_{q_1}$, $\lambda_{K_1}$, $\lambda_{q_2}$, $\lambda_{K_2}$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::DualSiteLangmuir model{2.5, 0.85, 1.5, 0.15};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);   // 0.8502652
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 1.2411757
```

## NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_1 > 0, \quad K_1 > 0, \quad q_2 > 0, \quad K_2 > 0, \quad C_e \geq 0.$$

47

# Fritz–Schlunder IV isotherm

FOUR PARAMETERS / q_max, K1, K2, K3 / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Fritz and Schlünder (1974) developed empirical relations for adsorption equilibria of organic solutes on activated carbon. The four-parameter expression implemented here is one member of the family; different publications use distinct coefficient and exponent conventions. The formula below, rather than the model name alone, defines the library implementation. A strong regression fit does not by itself identify the microscopic adsorption mechanism.

Fritz, W., and Schlünder, E.-U. (1974), *Simultaneous adsorption equilibria of organic solutes in dilute aqueous solutions on activated carbon*, *Chemical Engineering Science*, **29** (5), 1279–1282. DOI: 10.1016/0009-2509(74)80128-4.

Kumar, K. V., Monteiro de Castro, M. C., Martinez-Escandell, M., Molina-Sabio, M., and Rodriguez-Reinoso, F. (2010), *Adsorption on Heterogeneous Surfaces: Site Energy Distribution Functions from Fritz–Schlüender Isotherms*, *ChemPhysChem*, **11** (12), 2555–2560. DOI: 10.1002/cphc.201000307. This analysis examines site-energy distributions for Fritz–Schlünder models but does not imply a unique microscopic interpretation of an arbitrary four-parameter fit.

Kumar, M., Sujatha, S., Gokulan, R., Vijayakumar, A., Praveen, S., and Elayaraja, S. (2021), *Prediction of RSM and ANN in the decolorization of Reactive Orange 16 using biochar derived from Ulva lactuca*, *Desalination and Water Treatment*, **211**, 304–318. DOI: 10.5004/dwt.2021.26615. Fritz–Schlünder IV and Marczewski–Jaroniec were reported as the best-fitting models for these dye-adsorption data, with a reported $R^2 = 0.9999$. This paper was published in **2021**, not 2024.

Nouacer, I., Hammadi, S., Benalia, M., Djedid, M., and Bencheikh, S. E. (2023), *Biosorption of carcinogenic metal ions using the immobilized microalgae Auxenochlorella protothecoides on a lignocellulosic matrix: Optimization, kinetics, isotherm and mechanism*, *South African Journal of Botany*, **157**, 297–313. DOI: 10.1016/j.sajb.2023.02.023. The authors compared nonlinear equilibrium models for nickel biosorption and selected Fritz–Schlünder IV using several error functions. A model ranking for this dataset does not establish universal predictive performance.

## MATHEMATICAL FORMULATION

$$q_e = \frac{q_{\max} C_e^{K_2}}{1 + K_1 C_e^{K_3}}$$

The denominator and numerator exponents control the asymptotic response independently. For $C_e \to \infty$,

$$q_e \sim \frac{q_{\max}}{K_1} C_e^{K_2 - K_3}.$$

Thus the scale $q_{\max}$ does not in general represent a saturation capacity. In the special case $K_2 = K_3 = 1$, the expression becomes a Langmuir curve with affinity $K_1$ and limiting uptake $q_{\max}/K_1$, not $q_{\max}$.

The four independently adjustable parameters are `q_max`, `K1`, `K2` and `K3`. No temperature parameter is required by this implementation.

All four parameters are strictly positive; 0 < K3 <= 1. Depending on the exponents the response can increase without saturation or turn over; the library does not silently enforce a monotonic subfamily.

Finite and non-negative concentrations are accepted. The response at zero concentration is defined by its continuous zero limit. The derivative is analytical and is evaluated only for strictly positive concentration.

Parameter and concentration restrictions:

$$q_{\max} > 0, \quad K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad 0 < K_3 \le 1, \quad C_e \ge 0.$$

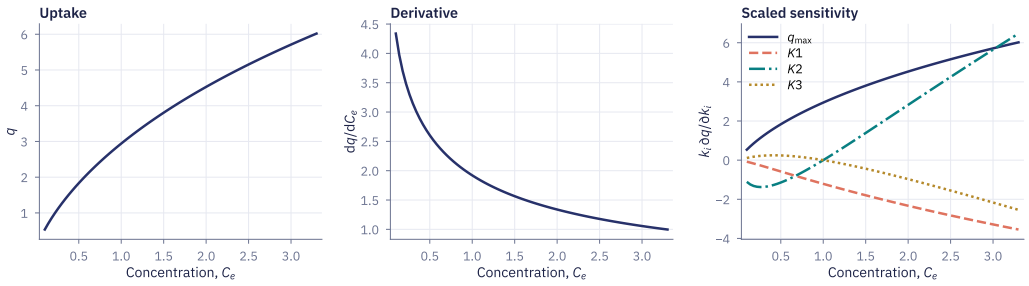



**Figure 47.** (a) Fritz–Schlunder IV uptake for $q_{\max} = 5$, $K_1 = 0.7$, $K_2 = 0.9$, $K_3 = 0.6$; (b) Concentration derivative of Fritz–Schlunder IV uptake for $q_{\max} = 5$, $K_1 = 0.7$, $K_2 = 0.9$, $K_3 = 0.6$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_{K_1}$, $\lambda_{K_2}$, $\lambda_{K_3}$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::FritzSchlunderIV model{71.3166, 0.357335, 0.779255, 0.669959};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 45.831032
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 35.611192
```

### NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{\max} > 0, \quad K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad 0 < K_3 \le 1, \quad C_e \ge 0.$$

# 48

# JensenSeaton isotherm

●●●● FOUR PARAMETERS / K, q_m, k, t / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Jensen, C. R. C., and Seaton, N. A. (1996), *An Isotherm Equation for Adsorption to High Pressures in Microporous Adsorbents*, *Langmuir*, **12** (11), 2866–2867. DOI: 10.1021/la9509460. This is the original source for the Jensen–Seaton relation. The equation interpolates between a Henry-law limit at low pressure and a high-pressure response that continues to rise, instead of enforcing a horizontal saturation plateau. The latter behaviour accounts empirically for compression of adsorbate within micropores.

The equation below is the four-parameter form also described in the pyGAPS Jensen–Seaton model documentation. The Isotherm++ interface names its independent argument equilibrium `concentration`. For comparison with gas-pressure measurements, the independent variable, its numerical value and the fitted coefficients must use mutually consistent units; the class does not calculate an equation of state or convert concentration into pressure automatically.

Rivera-Torrente, M., Kroon, D., Coulet, M.-V., et al. (2022), *Understanding the Effects of Binders in Gas Sorption and Acidity of Aluminium Fumarate Extrudates*, *Chemistry – A European Journal*, **28** (5), e202103420. DOI: 10.1002/chem.202103420. This application fitted single-component CO2 and CH4 measurements on aluminium fumarate materials using three-site Jensen–Seaton ex-

pressions and then estimated mixture selectivity using IAST. Its three-site construction and IAST calculation are additional operations, not features of the present four-parameter C++ class. The journal issue is dated **2022**, not 2023.

The cited papers do not establish that one fitted curve is preferable for all microporous materials. Model selection remains dependent on the pressure range, sorbent, temperature and observational uncertainty.

## MATHEMATICAL FORMULATION

$$q = Kx\left[1 + \left(\frac{Kx}{q_m(1+kx)}\right)^t\right]^{-1/t}$$

The 4 adjustable parameters are: `K`, `q_m`, `k`, `t`. The independent variable is `concentration`.

All four parameters strictly positive; q increases without a hard horizontal high-pressure plateau.

Parameter values are illustrative and not estimates from experimental data. The mathematical representation does not establish experimental validity.

Parameter and concentration restrictions:

$$K > 0, \quad q_m > 0, \quad k > 0, \quad t > 0, \quad C_e \geq 0.$$

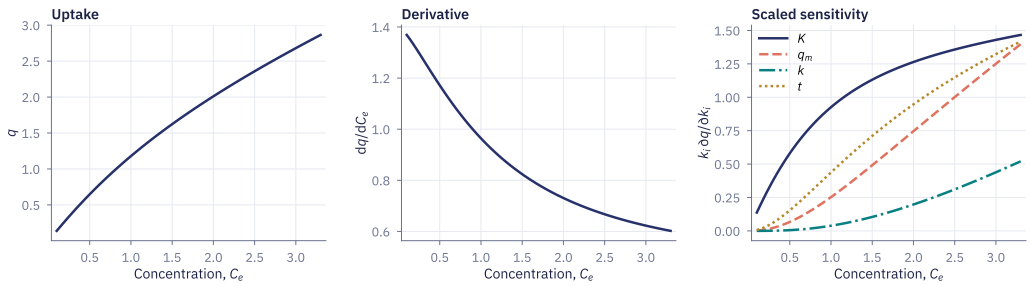



**Figure 48.** (a) Jensen–Seaton uptake for $K = 1.4$, $q_m = 3$, $k = 0.18$, $t = 1.4$; (b) Concentration derivative of Jensen–Seaton uptake for $K = 1.4$, $q_m = 3$, $k = 0.18$, $t = 1.4$; (c) Scaled sensitivities $\lambda_K$, $\lambda_{q_m}$, $\lambda_k$, $\lambda_t$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::JensenSeaton model{1.4, 3, 0.18, 1.4};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);   // 0.64740681
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 1.1717692
```

## NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$K > 0, \quad q_m > 0, \quad k > 0, \quad t > 0, \quad C_e \geq 0.$$

49

# Marczewski–Jaroniec isotherm

FOUR PARAMETERS / q_max, K1, K2, K3 / EXPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Marczewski, A. W., and Jaroniec, M. (1983), *A New Isotherm Equation for Single-Solute Adsorption from Dilute Solutions on Energetically Heterogeneous Solids*, *Monatshefte für Chemie*, **114** (6), 711–715. DOI: 10.1007/BF01134184. This is a primary source for generalised adsorption-isotherm relations based on local adsorption and distributions of adsorption energies over heterogeneous solids.

Marczewski, A. W., Derylo-Marczewska, A., and Jaroniec, M. (1986), *Energetic heterogeneity and molecular size effects in physical adsorption on solid surfaces*, *Journal of Colloid and Interface Science*, **109** (2), 310–324. DOI: 10.1016/0021-9797(86)90309-7. The later study examines the effects of site-energy heterogeneity and adsorbate size. Its correct volume and pages are **109**, 310–324, not 114, 319–327.

Dan-Iya, B. I., and Shukor, M. Y. (2023), *Isothermal Modelling of the Adsorption of Crystal Violet onto Modified Charred Rice Husk*, *Journal of Environmental Microbiology and Toxicology*, **11** (1), 1–9. DOI: 10.54987/jemat.v11i1.837. This nonlinear regression study assessed Marczewski–Jaroniec amongst a range of candidate models. For these data it identified **Jovanovic**, rather than Marczewski–Jaroniec, as the preferred model according to its reported statistical selection.

The four-parameter expression below defines the particular Marczewski–Jaroniec parametrisation used by Isotherm++. It should not be assumed to be numerically interchangeable with every equation labelled `Marczewski-Jaroniec` in the literature; the exponents, affinity convention and admissible parameter ranges must be compared first.

## MATHEMATICAL FORMULATION

$$q_e = q_{\max}\left[\frac{(K_1 c_e)^{K_2}}{1+(K_1 c_e)^{K_2}}\right]^{K_3/K_2}$$

The four independently adjustable parameters are `q_max`, `K1`, `K2` and `K3`. No temperature parameter is required by this implementation.

All parameters are strictly positive and 0 < K3 < 1. The model is saturating and approaches q_max as concentration tends to infinity.

Finite and non-negative concentrations are accepted. The response at zero concentration is defined by its continuous zero limit. The derivative is analytical and is evaluated only for strictly positive concentration.

Parameter and concentration restrictions:

$$q_{\max} > 0, \quad K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad 0 < K_3 < 1, \quad C_e \geq 0.$$

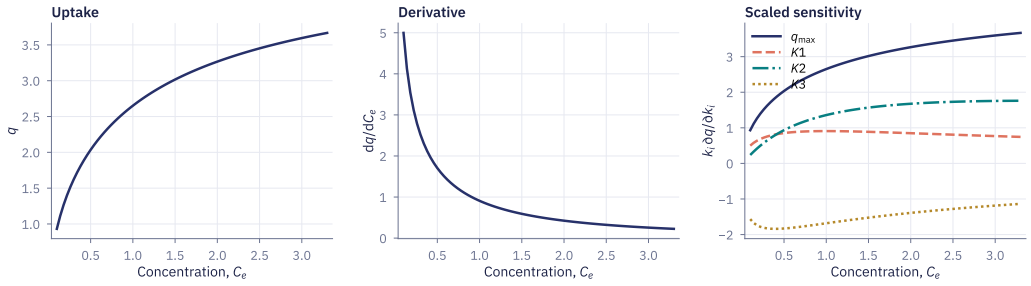



**Figure 49.** (a) Marczewski–Jaroniec uptake for $q_{max} = 5$, $K_1 = 0.7$, $K_2 = 0.8$, $K_3 = 0.6$; (b) Concentration derivative of Marczewski–Jaroniec uptake for $q_{max} = 5$, $K_1 = 0.7$, $K_2 = 0.8$, $K_3 = 0.6$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_{K_1}$, $\lambda_{K_2}$, $\lambda_{K_3}$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::MarczewskiJaroniec model{4.99159, 0.240965, 0.0686414,
0.0818727};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 2.0374566
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 0.11014187
```

### NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad 0 < K_3 < 1, \quad C_e \geq 0.$$

# 50

# Oswin isotherm

●●●● FOUR PARAMETERS / A, B, C, D / EXPLICIT / TEMPERATURE

### ORIGINAL PUBLICATION AND SCOPE

The original empirical **two-parameter** moisture-sorption equation was introduced by Oswin, C. R. (1946), *The kinetics of package life. III. The isotherm*, *Journal of the Society of Chemical Industry*, **65** (12), 419–421. DOI: 10.1002/jctb.5000651216. The original work was concerned with sorption by hygroscopic materials and the shelf life of packaged products.

The established **modified Oswin** equation introduces a linear variation of its amplitude with temperature, while retaining a constant exponent, $(A + BT)[a_w/(1 - a_w)]^{1/C}$. This three-coefficient form is described in Mathematical Modeling and Hysteresis of Sorption Isotherms for Paddy Rice Grains (2019) and was applied in Desorption Isotherms of Grain Sorghum Flour (2024).

Kimani, P. K., Gitau, A. N., Mbuge, D. O., Agullo, J. O., and Yang, L. (2023), *Modelling Moisture Sorption Isotherms of Superabsorbent Polymer Fabric for Desiccant Drying of Crops*. DOI: 10.1155/2023/9992832. The authors compared several sorption models for a superabsorbent fabric and reported the modified Oswin, modified Freundlich and modified Henderson equations amongst the better-performing correlations over the investigated temperature range.

Tryding, J., Askfelt, H., Alexandersson, M., and Ristinmaa, M. (2023), *A full-range moisture sorption model for cellulose-based materials yielding consistent net isosteric heat of sorption*, *Drying Technology*, **41** (1), 61–76. DOI: 10.1080/07373937.2022.2084104. This paper develops a dif-

ferent generalisation of classical Oswin and Henderson relations that explicitly incorporates the fibre saturation point and thermodynamic constraints. Its equation is not identical to the one implemented here.

The **four-coefficient** Isotherm++ expression below additionally permits the exponent denominator to depend linearly on temperature through $C + DT$. This is a library-specific extension and must not be attributed to Oswin (1946) or to either 2023 application as their published formulation. When $D = 0$, it reduces to the reported three-coefficient modified Oswin form; $B = D = 0$ recovers the original-type power law. At a single temperature, only $A + BT$ and $C + DT$ are identifiable, so data at several temperatures are required to estimate all four coefficients independently.

## MATHEMATICAL FORMULATION

$$q_e(a_w, T) = (A + BT)\left(\frac{a_w}{1 - a_w}\right)^{1/(C + DT)}.$$

The admissible parameter and input domain is

$$A > 0, \quad B, D \in \mathbb{R}, \quad C > 0, \quad T > 0\,\mathrm{K}, \quad A + BT > 0, \quad C + DT > 0, \quad 0 \le a_w < 1.$$

The implemented activity derivative requires $a_w > 0$:

$$\frac{\partial q_e}{\partial a_w} = \frac{q_e}{(C + DT)\, a_w (1 - a_w)}.$$

Here $a_w$ is water activity, a dimensionless ratio of water vapour partial pressure to saturation vapour pressure at the same temperature. The independent temperature $T$ must be supplied in kelvin. The four fitted coefficients are $A$, $B$, $C$, and $D$. $A$ has the units of $q_e$, $B$ has the units of $q_e\,\mathrm{K}^{-1}$, $C$ is dimensionless, and $D$ has units of $\mathrm{K}^{-1}$. The sums $A + BT$ and $C + DT$ must remain positive at each evaluated temperature.

Coefficients fitted with temperature $t$ in degrees Celsius cannot be used unchanged. Because $t = T - 273.15$, their kelvin intercepts satisfy $A_K = A_{°C} - 273.15B$ and $C_K = C_{°C} - 273.15D$.

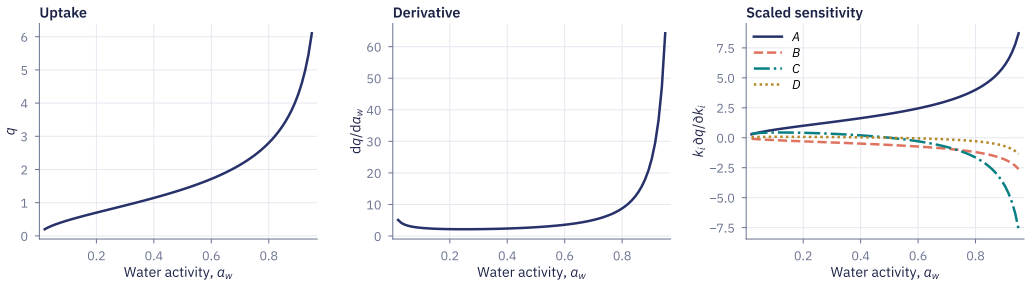



**Figure 50.** (a) Oswin equilibrium moisture for $A = 2.0$, $B = -0.002$, $C = 1.7$, $D = 0.001$ at $T = 300\,\mathrm{K}$; (b) Activity derivative of Oswin moisture for $A = 2.0$, $B = -0.002$, $C = 1.7$, $D = 0.001$ at $T = 300\,\mathrm{K}$; (c) Scaled sensitivities to $A$, $B$, $C$, and $D$ versus water activity at $T = 300\,\mathrm{K}$.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Oswin model{2.0, -0.002, 1.7, 0.001};
constexpr isotherm::Real temperature_k = 300.0;             // K
const isotherm::Real uptake = isotherm::evaluate(
    model, 0.5, temperature_k);                              // 1.4
const isotherm::Real slope = isotherm::derivative(
    model, 0.5, temperature_k);                              // 2.8
```

## NUMERICAL SENSITIVITY

For each model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(a_w) = k_i \frac{\partial q_e(a_w, T)}{\partial k_i}, \qquad T = 300\,\mathrm{K}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$A > 0, \quad B, D \in \mathbb{R}, \quad C > 0, \quad T > 0\ \mathrm{K}, \quad A + BT > 0, \quad C + DT > 0, \quad 0 \le a_w < 1.$$

The implemented activity derivative requires $a_w > 0$.

51

# Smith isotherm

FOUR PARAMETERS / A, B, C, a_w / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

Smith, S. E. (1947), *The Sorption of Water Vapor by High Polymers*, *Journal of the American Chemical Society*, **69** (3), 646–651. DOI: 10.1021/ja01195a053.

Smith's original equation (6) describes a two-coefficient moisture-sorption relation at a fixed temperature. The modified three-coefficient expression below allows its intercept to depend linearly on absolute temperature. A temperature-dependent Smith relationship has been reported, for example, in the study of moisture sorption in soybean grains. The temperature term is not part of Smith's 1947 equation.

The C++ class stores three empirical coefficients and a *fixed water activity* for its temperature section. Its four stored parameters do not represent four independently fitted material coefficients.

MATHEMATICAL FORMULATION

$$q_e(T; A, B, C, a_w) = (A + BT) - C\ln(1 - a_w).$$

The admissible parameter and input domain is

$$A > 0, \quad B \in \mathbb{R}, \quad C > 0, \quad 0 \le a_w < 1, \quad T > 0\ \mathrm{K}, \quad A + BT > 0.$$

Here $a_w = p_w / p_w^{\mathrm{sat}}$ is dimensionless water activity, where the vapour pressure and saturation pressure refer to the same temperature. $T$ is the **absolute temperature in kelvin**. $A$ and $C$ have units of equilibrium moisture content, while $B$ has units of equilibrium moisture content per kelvin. Literature expressions written as $q_e = a - bt_C - c\ln(1 - a_w)$, with $t_C$ in degrees Celsius, correspond to the kelvin-based form through $A = a + 273.15b$, $B = -b$, $C = c$.

The temperature derivative at fixed stored parameters is

$$\frac{\partial q_e}{\partial T} = B.$$

The derivative with respect to stored water activity is distinct:

$$\frac{\partial q_e}{\partial a_w} = \frac{C}{1 - a_w}.$$

The C++ `derivative(model, T)` function returns $B$, not the water-activity derivative. The latter can be calculated by comparing new instances constructed at different activities.

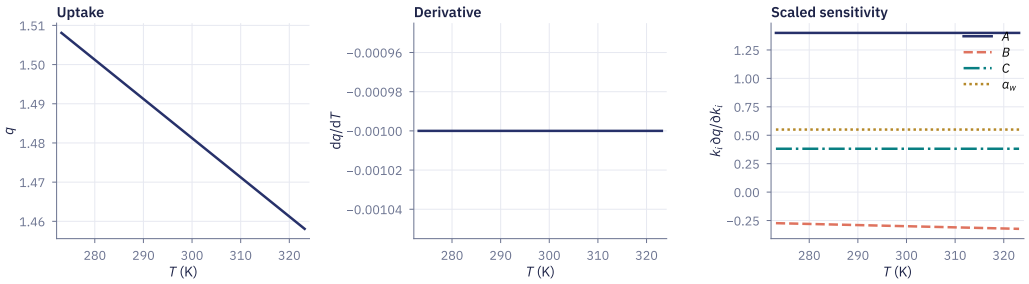



**Figure 51.** (a) Smith equilibrium moisture for $A = 1.4$, $B = -0.001$, $C = 0.55$ and fixed $a_w = 0.5$; (b) Temperature derivative of Smith equilibrium moisture with $A = 1.4$, $B = -0.001$, $C = 0.55$, $a_w = 0.5$; (c) Scaled sensitivities $\lambda_A$, $\lambda_B$, $\lambda_C$ and $\lambda_{a_w}$ versus absolute temperature at fixed $a_w = 0.5$.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Smith model{1.4, -0.001, 0.55, 0.5}; // A, B, C, a_w
constexpr isotherm::Real temperature_k = 300.0;              // K
const isotherm::Real uptake = isotherm::evaluate(model, temperature_k);    // 1.4812309
const isotherm::Real slope = isotherm::derivative(model, temperature_k);  // -0.001
```

### NUMERICAL SENSITIVITY

For each model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(T) = k_i \frac{\partial q_e(T)}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$A > 0, \quad B \in \mathbb{R}, \quad C > 0, \quad 0 \le a_w < 1, \quad T > 0\ \mathrm{K}, \quad A + BT > 0.$$

At a single fixed water activity, varying temperature identifies the slope $B$ and only the combined intercept $A - C\ln(1 - a_w)$. Determining $A$ and $C$ separately therefore requires additional activity conditions or independent information. The original Smith relation was developed for a restricted moisture-sorption regime; mathematical validity at $a_w = 0$ is not evidence of physical applicability there.

# 52

# Virial isotherm

●●●● FOUR PARAMETERS / K_H, A, B, C / IMPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Virial adsorption equations form a family of expansions, rather than a single universal isotherm. A standard treatment is Ruthven, D. M. (1984), *Principles of Adsorption and Adsorption Processes*, Wiley, New York. Publisher record. This book is a reference for adsorption-equilibrium theory and does not establish that the present four-parameter truncated expression was originally proposed by Ruthven.

The Isotherm++ class uses an *implicit* cubic virial expansion in adsorbed loading, where the independent input is equilibrium concentration and the numerical root provides equilibrium uptake. Its enforced strictly positive virial coefficients select a monotonic, uniquely invertible branch; they are restrictions of this implementation, not universal physical constraints on virial coefficients.

Chu, K. H., Hashim, M. A., Mudhoo, A., and Debord, J. (2023), *Beyond Freundlich and Langmuir: the Ruthven–virial equilibrium isotherm for aqueous-solid adsorption systems*, *Chemical Papers*, **77** (3), 1593–1600. DOI: 10.1007/s11696-022-02576-4. This study tests a **two-parameter** Ruthven–virial variant for aqueous-solid equilibrium; its exponential-integral formulation is not the same mathematical expression as the cubic four-parameter series implemented below.

*High Pressure Excess Isotherms for Adsorption of Oxygen and Nitrogen in Zeolites* (2011), *Langmuir*, **27** (17), 10648–10656. DOI: 10.1021/la201690x. This work reported that a virial correlation described high-pressure oxygen and nitrogen excess adsorption data on NaX and NaY zeolites better than a Toth correlation with the same number of parameters. It was published in **2011**, not 2023; its temperature-dependent and excess-adsorption context differs from the present concentration-based, fixed-coefficient model.

## MATHEMATICAL FORMULATION

$$x = \frac{q}{K_H}\exp(Aq + Bq^2 + Cq^3)$$

The 4 adjustable parameters are: `K_H`, `A`, `B`, `C`. The independent variable is `concentration`.

K_H positive, A,B,C strictly positive in this release (restricted monotonic region). High-pressure inversion has a unique positive root.

Unlike the pressure-explicit literature equation, the public evaluate(x) returns uptake, so an inverse solution is necessary.

Parameter and concentration restrictions:

$$K_H > 0, \quad A > 0, \quad B > 0, \quad C > 0, \quad C_e \geq 0.$$

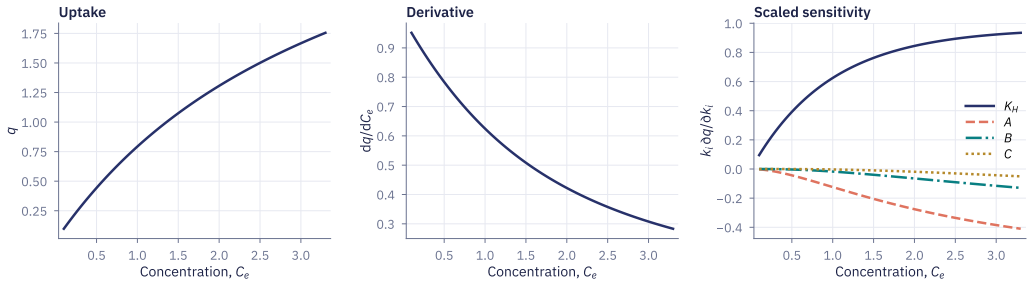



**Figure 52.** (a) Virial uptake for $K_H = 1$, $A = 0.25$, $B = 0.045$, $C = 0.01$; (b) Concentration derivative of Virial uptake for $K_H = 1$, $A = 0.25$, $B = 0.045$, $C = 0.01$; (c) Scaled sensitivities $\lambda_{K_H}$, $\lambda_A$, $\lambda_B$, $\lambda_C$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::Virial model{1, 0.25, 0.045, 0.01};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 0.44323026
const isotherm::Real slope = isotherm::derivative(model, 0.5);   // 0.78371504
```

## NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$K_H > 0, \quad A > 0, \quad B > 0, \quad C > 0, \quad C_e \geq 0.$$

53

# Weber–Van Vliet isotherm

FOUR PARAMETERS / K1, K2, K3, K4 / IMPLICIT

## ORIGINAL PUBLICATION AND SCOPE

Van Vliet, B. M., Weber, W. J., Jr., and Hozumi, H. (1980), *Modeling and prediction of specific compound adsorption by activated carbon and synthetic adsorbents*, *Water Research*, **14** (12), 1719–1728. DOI: 10.1016/0043-1354(80)90107-4. The original study compared activated carbons and synthetic adsorbents, using an empirical isotherm alongside a model for adsorption-bed performance. It is the historical reference for the Weber–Van Vliet family of correlations.

The four-parameter Isotherm++ expression below is written with **equilibrium concentration on the left-hand side** and loading as the unknown. Unlike explicit uptake equations, the C++ function `evaluate(C_e)` computes the positive inverse numerically. Its additional monotonicity condition $K_4 > K_2\exp(-2)$ selects a unique positive solution over the supported domain; this is a numerical admissibility condition specific to the implementation.

Dan-Iya, B. I., and Shukor, M. Y. (2023), *Isothermal Modelling of the Adsorption of Crystal Violet onto Modified Charred Rice Husk*, *Journal of Environmental Microbiology and Toxicology*, **11** (1), 1–9. DOI: 10.54987/jemat.v11i1.837. Weber–Van Vliet was among the models included in the nonlinear reanalysis. The authors reported **Jovanovic** as the preferred model for those data. Thus the study supports the *application* of Weber–Van Vliet, but not a claim that it was universally or specifically the best-fitting model.

The present formulation and admissible domain must be checked when transferring fitted parameters from other published conventions. Good statistical agreement does not alone establish mechanistic validity.

## MATHEMATICAL FORMULATION

$$c_e = K_1 q_e^{K_2 q_e^{K_3} + K_4}$$

$$\ln(c_e/K_1) = (K_2\exp(K_3 t) + K_4)t, \qquad t = \ln(q_e)$$

The positive root is found by bisection in `t`. For positive `K2` and `K3`, the derivative of the right-hand side with respect to `t` has lower bound `K4 - K2 exp(-2)`. Requiring this bound to be strictly positive guarantees uniqueness of the positive equilibrium uptake. The parameter bound above establishes this property.

The four independently adjustable parameters are `K1`, `K2`, `K3` and `K4`. No temperature parameter is required by this implementation.

All parameters are strictly positive and K4 > K2 exp(-2). This is a sufficient condition for uniqueness for every positive concentration; outside this region the equation can have multiple positive solutions and this implementation rejects it rather than returning an arbitrary root.

Finite and non-negative concentrations are accepted. The response at zero concentration is defined by its continuous zero limit. The derivative is analytical and is evaluated only for strictly positive concentration.

Parameter and concentration restrictions:

$$K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad K_4 > 0, \quad K_4 > K_2\exp(-2), \quad C_e \geq 0.$$

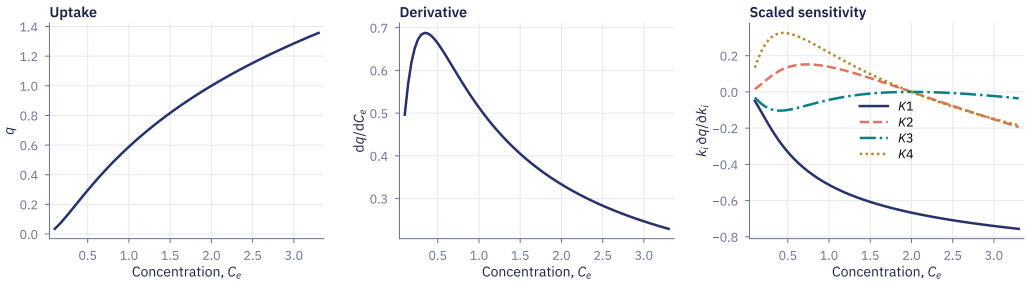



**Figure 53.** (a) Weber–Van Vliet uptake for $K_1 = 2$, $K_2 = 0.7$, $K_3 = 0.6$, $K_4 = 0.8$; (b) Concentration derivative of Weber–Van Vliet uptake for $K_1 = 2$, $K_2 = 0.7$, $K_3 = 0.6$, $K_4 = 0.8$; (c) Scaled sensitivities $\lambda_{K_1}$, $\lambda_{K_2}$, $\lambda_{K_3}$, $\lambda_{K_4}$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::WeberVanVliet model{0.087093, 3.19814, 0.0647897, 0.769393};
const isotherm::Real uptake = isotherm::evaluate(model, 0.8);    // 1.7216065
const isotherm::Real slope = isotherm::derivative(model, 0.8);  // 0.51254114
```

### NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad K_4 > 0, \quad K_4 > K_2 \exp(-2), \quad C_e \geq 0.$$

# 54

# WilsonVST isotherm

●●●● FOUR PARAMETERS / q_max, K_H, L1v, Lv1 / IMPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Suwanayuen, S., and Danner, R. P. (1980), *A gas adsorption isotherm equation based on vacancy solution theory*, *AIChE Journal*, **26** (1), 68–76. DOI: 10.1002/aic.690260112. This is the primary paper for the pure-component vacancy solution theory (VST) isotherm. It treats adsorbate molecules and hypothetical vacancies as solution components and represents non-ideality using Wilson activity coefficients.

Suwanayuen, S., and Danner, R. P. (1980), *Vacancy solution theory of adsorption from gas mixtures*, *AIChE Journal*, **26** (1), 76–83. DOI: 10.1002/aic.690260113. The companion paper extends the treatment to adsorption from gas mixtures. A single component's C++ isotherm is not by itself a full multicomponent thermodynamic calculation.

Cochran, T. W., Kabel, R. L., and Danner, R. P. (1985), *Vacancy solution theory of adsorption using Flory–Huggins activity coefficient equations*, *AIChE Journal*, **31** (2), 268–277. DOI: 10.1002/aic.690310214. This is a related **Flory–Huggins VST** variant, not the primary source for the Wilson-activity formulation.

*Understanding the vacancy solution theory* (2026), *Adsorption*. DOI: 10.1007/s10450-026-00678-z. This later analysis reviews the reference states and thermodynamic-consistency conditions in VST. Those mixture-level considerations must not be assumed to follow automatically from evaluation of the present single-component class.

The explicit pressure–coverage relation below follows the Wilson VST form also documented by pyGAPS. Isotherm++ inverts that relation for uptake by bisection. Its $0.8 \le L_{1v}, L_{v1} \le 1.2$ restriction is a conservative numerical monotonicity condition imposed by this implementation, not a general thermodynamic range required by Wilson's activity model.

## MATHEMATICAL FORMULATION

$$x = \frac{q_{\max}}{K_H} \frac{\vartheta}{1-\vartheta} f_W(\vartheta)$$

The Wilson activity correction is defined by

$$\ln f_W = \ln L_{1v} + \ln u - \ln v - \frac{L_{v1}(1-L_{v1})\vartheta}{u} - \frac{(1-L_{1v})\vartheta}{v}$$

where

$$u = 1 - (1-L_{v1})\vartheta, \quad v = L_{1v} + (1-L_{1v})\vartheta$$

The 4 adjustable parameters are: `q_max`, `K_H`, `L1v`, `Lv1`. The independent variable is `concentration`.

q_max,K_H strictly positive. To guarantee a single root, both Wilson interaction factors L1v and Lv1 are restricted to [0.8,1.2], a conservative sufficient region; other literature states are not claimed to be covered.

f_W(theta) is the Wilson vacancy activity correction. The derivative is obtained from the explicit log-pressure relation, and bisection in surface coverage selects its unique root in the restricted parameter domain.

Parameter and concentration restrictions:

$$q_{\max} > 0, \quad K_H > 0, \quad L_{1v} > 0, \quad L_{v1} > 0, \quad 0.8 \le L_{1v} \le 1.2, \quad 0.8 \le L_{v1} \le 1.2, \quad C_e \ge 0.$$

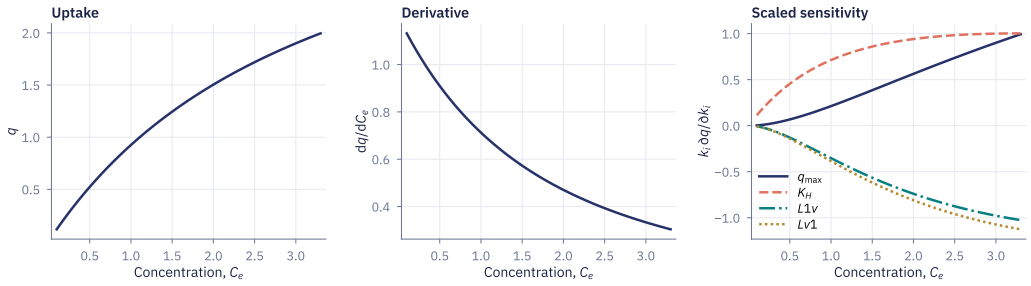



**Figure 54.** (a) Wilson VST uptake for $q_{\max} = 4$, $K_H = 1.2$, $L_{1v} = 0.9$, $L_{v1} = 1.1$; (b) Concentration derivative of Wilson VST uptake for $q_{\max} = 4$, $K_H = 1.2$, $L_{1v} = 0.9$, $L_{v1} = 1.1$; (c) Scaled sensitivities $\lambda_{q_{\max}}$, $\lambda_{K_H}$, $\lambda_{L_{1v}}$, $\lambda_{L_{v1}}$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::WilsonVST model{4, 1.2, 0.9, 1.1};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);    // 0.52244468
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 0.90961164
```

## NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K_H > 0, \quad L_{1v} > 0, \quad L_{v1} > 0, \quad 0.8 \le L_{1v} \le 1.2, \quad 0.8 \le L_{v1} \le 1.2, \quad C_e \ge 0.$$

55

# FritzSchlunderV isotherm

FIVE PARAMETERS / q_max, K1, K2, K3, K4 / EXPLICIT

ORIGINAL PUBLICATION AND SCOPE

Five-parameter Fritz–Schlunder equations extend the empirical family associated with Fritz and Schlünder (1974). They provide independently adjustable amplitudes and concentration powers. The five-parameter parametrisation adopted by a particular software implementation need not coincide term by term with every published expression bearing this name.

- Fritz, W., and Schlünder, E.-U. (1974), *Simultaneous adsorption equilibria of organic solutes in dilute aqueous solutions on activated carbon*, *Chemical Engineering Science*, **29** (5), 1279–1282. DOI: 10.1016/0009-2509(74)80128-4.
- Kumar, K. V., Monteiro de Castro, M. C., Martinez-Escandell, M., Molina-Sabio, M., and Rodriguez-Reinoso, F. (2010), *Adsorption on Heterogeneous Surfaces: Site Energy Distribution Functions from Fritz–Schlüender Isotherms*, *ChemPhysChem*, **11** (12), 2555–2560. DOI: 10.1002/cphc.201000307.

The 2010 study explicitly discusses a five-parameter Fritz–Schlüender representation and related adsorption-site energy distributions. This does not establish that fitted coefficients can be interpreted independently as physical site properties. The `FritzSchlunderV` class implements the rational power-law equation presented below. In contrast, the `FrenkelHalseyHill` class implements a logarithmic multilayer-adsorption model. These formulations are mathematically distinct and represent different adsorption model families.

MATHEMATICAL FORMULATION

$$q = \frac{q_{max} x^{K_3}}{K_1 + K_2 x^{K_4}}$$

The 5 adjustable parameters are: `q_max`, `K1`, `K2`, `K3`, `K4`. The independent variable is `concentration`.

All five parameters strictly positive. Saturation and monotonicity depend on the relative powers; no universally bounded uptake is implied.

This equation is also implemented by the other five-parameter class; these two identifiers should not be counted as distinct mathematical families.

Parameter and concentration restrictions:

$$q_{max} > 0, \quad K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad K_4 > 0, \quad C_e \ge 0.$$

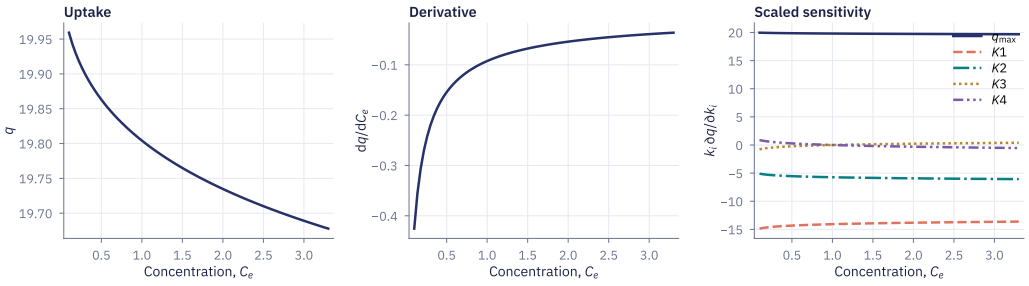



**Figure 55.** (a) Fritz–Schlunder V uptake for $q_{max} = 6.05758$, $K_1 = 0.217337$, $K_2 = 0.0885359$, $K_3 = 0.0169304$, $K_4 = 0.0746286$; (b) Concentration derivative of Fritz–Schlunder V uptake for $q_{max} = 6.05758$, $K_1 = 0.217337$, $K_2 = 0.0885359$, $K_3 = 0.0169304$, $K_4 = 0.0746286$; (c) Scaled sensitivities $\lambda_{q_{max}}$, $\lambda_{K_1}$, $\lambda_{K_2}$, $\lambda_{K_3}$, $\lambda_{K_4}$ versus equilibrium concentration.

### C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::FritzSchlunderV model{6.05758, 0.217337, 0.0885359,
0.0169304, 0.0746286};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);   // 19.863038
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // -0.15436923
```

### NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

### VALIDATION AND LIMITATIONS

$$q_{max} > 0, \quad K_1 > 0, \quad K_2 > 0, \quad K_3 > 0, \quad K_4 > 0, \quad C_e \geq 0.$$

# 56

# TripleSiteLangmuir isotherm

●●●●●● SIX PARAMETERS / q_1, K_1, q_2, K_2, q_3, K_3 / EXPLICIT

### ORIGINAL PUBLICATION AND SCOPE

Triple-site Langmuir as specified by https://pygaps.readthedocs.io/en/develop/reference/fitting.html , with six independent adjustable parameters.

### MATHEMATICAL FORMULATION

$$q = \sum_{i=1}^{3} \frac{q_i K_i x}{1 + K_i x}$$

The 6 adjustable parameters are: `q_1`, `K_1`, `q_2`, `K_2`, `q_3`, `K_3`. The independent variable is `concentration`.

All six coefficients strictly positive. Requires a SIX-parameter category, not a forced five-parameter approximation.

The model assumes three distinct, non-interacting populations of adsorption sites, each characterised by its own saturation capacity $q_i$ and affinity constant $K_i$. The equilibrium uptake is the sum of the three independent contributions. As $C_e \to \infty$, each contribution approaches its respective $q_i$, so the total saturation capacity is

$$q_{sat} = q_1 + q_2 + q_3.$$

This is an asymptotic limit, not necessarily attained at any finite concentration. The total saturation capacity is determined by the three $q_i$ values and is not a seventh independently adjustable parameter.

Parameter and concentration restrictions:

$$q_1 > 0, \quad K_1 > 0, \quad q_2 > 0, \quad K_2 > 0, \quad q_3 > 0, \quad K_3 > 0, \quad C_e \geq 0.$$

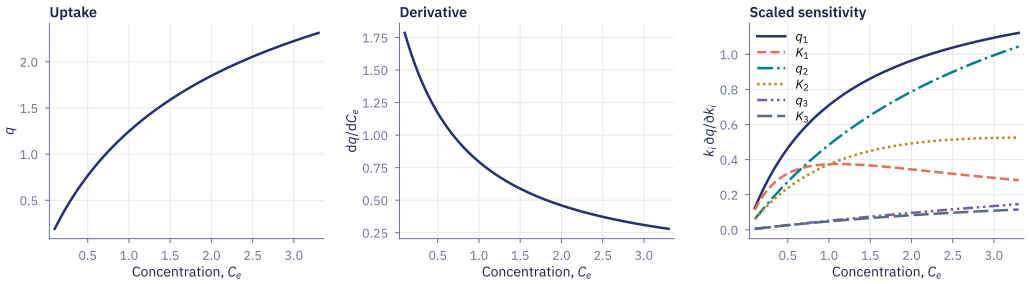



**Figure 56.** (a) Triple-Site Langmuir uptake for $q_1 = 1.5$, $K_1 = 0.9$, $q_2 = 2.1$, $K_2 = 0.3$, $q_3 = 0.7$, $K_3 = 0.08$; (b) Concentration derivative of Triple-Site Langmuir uptake for $q_1 = 1.5$, $K_1 = 0.9$, $q_2 = 2.1$, $K_2 = 0.3$, $q_3 = 0.7$, $K_3 = 0.08$; (c) Scaled sensitivities $\lambda_{q_1}$, $\lambda_{K_1}$, $\lambda_{q_2}$, $\lambda_{K_2}$, $\lambda_{q_3}$, $\lambda_{K_3}$ versus equilibrium concentration.

## C++20 USAGE

```cpp
#include <isotherm/isotherm.hpp>

const isotherm::models::TripleSiteLangmuir model{1.5, 0.9, 2.1, 0.3, 0.7, 0.08};
const isotherm::Real uptake = isotherm::evaluate(model, 0.5);   // 0.76635336
const isotherm::Real slope = isotherm::derivative(model, 0.5);  // 1.1702384
```

## NUMERICAL SENSITIVITY

For every model parameter $k_i$, the scaled sensitivity is

$$\lambda_i(C_e) = k_i \frac{\partial q_e}{\partial k_i}.$$

Numerical values are obtained by perturbing individual parameters and evaluating new instances of the C++ model. The accompanying C++20 example performs this procedure.

## VALIDATION AND LIMITATIONS

$$q_1 > 0, \quad K_1 > 0, \quad q_2 > 0, \quad K_2 > 0, \quad q_3 > 0, \quad K_3 > 0, \quad C_e \geq 0.$$

# User guide

## INSTALLING THE C++20 INTERFACE

```
cmake -S . -B build/release -DISOTHERM_BUILD_TESTS=OFF
cmake --build build/release
cmake --install build/release --prefix install
```

A consuming CMake application uses `find_package(IsothermPP CONFIG REQUIRED)` and links `Isotherm::Isotherm`. The stable public header is `isotherm/isotherm.hpp`.

The first implemented scientific models are `isotherm::models::Henry` and `isotherm::models::Langmuir`. See the scientific catalogue for a mathematical reference and the accompanying runnable examples. The independent GoogleTest suites verify the principal identities and domains of both models.

## BUILDING THE MANUALS

Activate the WSL virtual environment where you installed Sphinx and run:

```
python -m sphinx --version
cmake -S . -B build -DISOTHERM_BUILD_DOCS=ON \
  -DISOTHERM_SPHINX_PYTHON="$(command -v python)"
cmake --build build --target docs-check
cmake --build build --target docs-html
cmake --build build --target docs-pdf
```

PDF generation uses HTML/CSS and WeasyPrint; both formats share the same RST pages.

# C++20 architecture

## CORE PRINCIPLES

The library is header-only and never uses inheritance or virtual functions in its model architecture. A scientific model is a value type whose constructor validates the parameters and whose member functions `evaluate` and `derivative` return the adsorbed amount and its derivative with respect to the independent variable. Each model has an `isotherm::model_traits` specialisation that gives the number and names of its parameters, its independent variable, whether the equation is explicit or implicit and whether the model depends on temperature.

`scalar_isotherm`, `thermal_isotherm`, their `differentiable_` variants and `parameterised_model` are compile-time concepts. The core provides the free functions `evaluate`, `derivative` and `evaluate_into`, which validate the independent variable and reject non-finite results, and `parameters`, `parameter`, `with_parameter` and `with_parameters`, which read parameters or return a new, validated model with replaced values. Individual model implementations remain responsible for their physical domain and parameter restrictions.

## DERIVATIVES AND IMPLICIT MODELS

Every model returns the derivative from its exact expression. Implicit models compute the adsorbed amount numerically, in most cases by bisection on an interval that contains the root, to the resolution of double precision; the derivative then follows from implicit differentiation at that value.

## DATA AND UNITS

The scalar independent variable must be interpreted using the model's scientific metadata: concentration, pressure, relative pressure or, for some moisture-sorption sections, temperature. The library does not impose a dimensional-analysis system. Temperature-dependent models accept an absolute temperature in kelvin as an additional argument.

## ERROR HANDLING

Errors are raised as `isotherm_error` exceptions with explicit codes. No library operation calls `exit` or terminates the host application. Bulk evaluations write incrementally and do not provide rollback if a later point fails.

## CURRENT SCOPE

The library covers single-component equilibrium. It does not include a parameter-fitting module; parameters are estimated with an optimiser chosen by the user, through the parameter functions described above.

# Developer guide

## AUTOMATIC CMAKE DISCOVERY

To add an example, create `Examples/TwoParameters/ModelName/main.cpp`. CMake automatically finds it at the next build. Additional `*.cpp` files in the same folder are linked into that programme. No `CMakeLists.txt` changes are required.

For GoogleTest, create `IsothermTest/TwoParameters/ModelName/test_model.cpp`. All `test_*.cpp` in that folder form one GoogleTest executable, with each test registered in CTest. `ISOTHERM_REQUIRE_GTEST=ON` makes the GoogleTest dependency mandatory.

The `Core` group is reserved for independent architectural tests. Standalone smoke tests may use `Core/Area/smoke_main.cpp`.

Documentation must follow Australian English, including graph labels. Each model must be validated against independent mathematical references before it can be listed as complete.

# Verification and validation

The CTest suite combines a GoogleTest test file for every model, smoke programmes that check reference values without GoogleTest, tests of the parameter interface and of the declared algebraic equivalences, residual checks of the implicit equations, and checks of the documentation sources. It is run with `cmake --build build --target run_all_tests`. The library compiles without warnings under GCC 13 and 15 and Clang 18 with strict warning flags, and the test suite runs cleanly under AddressSanitizer and UndefinedBehaviorSanitizer.

An independent audit compared every equation and derivative with reference implementations written from the published equations and evaluated with 60-digit arithmetic, and checked the derivatives symbolically. The audit included extreme inputs, from about 1e-300 to 1e300, near poles and in the limits of each domain.

Model-specific conventions that differ from some published forms, such as the decreasing Jovanovic branch, the singular Halsey form, the dimensionless input of Dubinin–Radushkevich and the constrained implicit Elovich coverage branch, are documented on the corresponding model pages. The tests verify the implementation of the stated equations; they are not a substitute for experimental validation.

The documentation target `docs-figures` regenerates the model plots from the compiled C++20 examples rather than from independent Python model equations. `docs-html` builds the HTML manual with Sphinx, and `docs-pdf` renders the A4 PDF from HTML and CSS with WeasyPrint, redrawing each model's plots from the same example data.